

# Important Information

Thank you for choosing Freenove products!

## Getting Started

If you have not yet downloaded the zip file, associated with this kit, please do so now and unzip it.

## Get Support and Offer Input

Freenove provides free and responsive product and technical support, including but not limited to:

- Product quality issues
- Product use and build issues
- Questions regarding the technology employed in our products for learning and education
- Your input and opinions are always welcome
- We also encourage your ideas and suggestions for new products and product improvements

For any of the above, you may send us an email to:

[support@freenove.com](mailto:support@freenove.com)

## Safety and Precautions

Please follow the following safety precautions when using or storing this product:

- Keep this product out of the reach of children under 6 years old.
- This product should be **used only when there is adult supervision present** as young children lack necessary judgment regarding safety and the consequences of product misuse.
- This product contains small parts and parts, which are sharp. This product contains electrically conductive parts. **Use caution with electrically conductive parts near or around power supplies, batteries and powered (live) circuits.**
- When the product is turned ON, activated or tested, some parts will move or rotate. **To avoid injuries to hands and fingers keep them away from any moving parts!**
- It is possible that an improperly connected or shorted circuit may cause overheating. **Should this happen, immediately disconnect the power supply or remove the batteries and do not touch anything until it cools down!** When everything is safe and cool, review the product tutorial to identify the cause.
- Only operate the product in accordance with the instructions and guidelines of this tutorial, otherwise parts may be damaged or you could be injured.
- Store the product in a cool dry place and avoid exposing the product to direct sunlight.
- After use, always turn the power OFF and remove or unplug the batteries before storing.

## About Freenove

Freenove provides open source electronic products and services worldwide.

Freenove is committed to assist customers in their education of robotics, programming and electronic circuits so that they may transform their creative ideas into prototypes and new and innovative products. To this end, our services include but are not limited to:

- Educational and Entertaining Project Kits for Robots, Smart Cars and Drones
- Educational Kits to Learn Robotic Software Systems for Arduino, Raspberry Pi and micro: bit
- Electronic Component Assortments, Electronic Modules and Specialized Tools
- **Product Development and Customization Services**

You can find more about Freenove and get our latest news and updates through our website:

<http://www.freenove.com>

[sale@freenove.com](mailto:sale@freenove.com)

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## List

Freenove ESP32 Mini TV is currently available in four color options: **yellow, black, white, and blue.**

It is important to note that the differences between these versions are strictly limited to the external case color. All internal components—including the hardware architecture, circuit design, and software interfaces—remain identical across all models. Therefore, regardless of which color version you use, this tutorial is fully applicable for all learning and development activities.



If you have any concerns, please feel free to contact us via [support@freenove.com](mailto:support@freenove.com)

Freenove ESP32 Mini TV x 1



USB data cable x 1

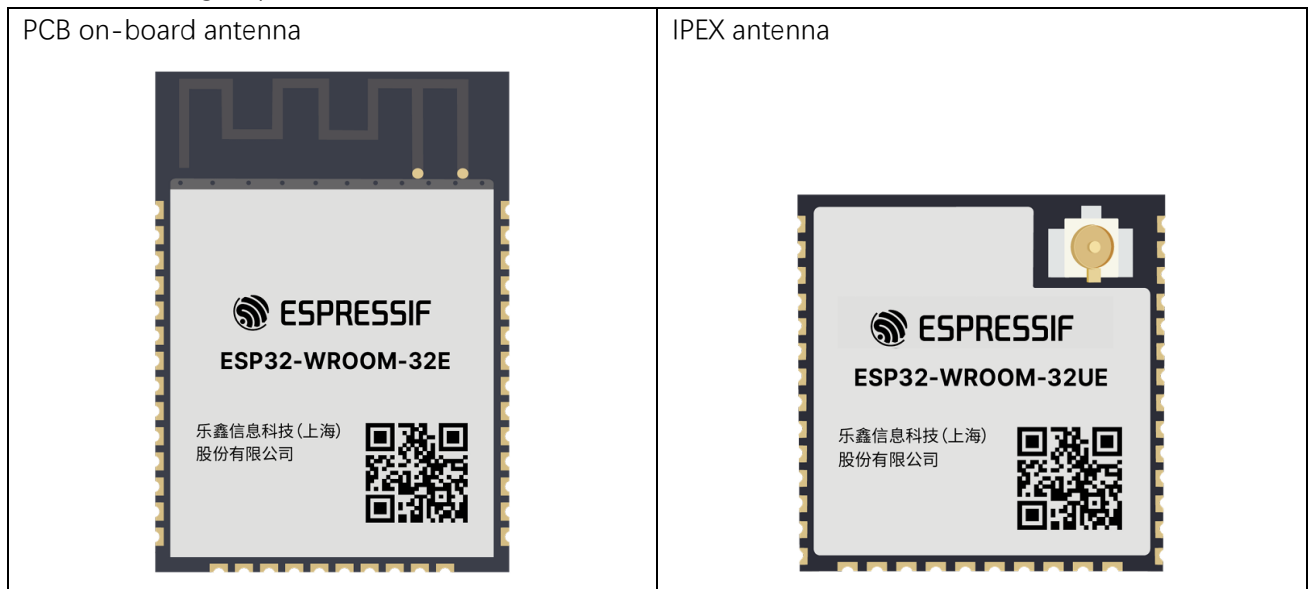


# Preface

## ESP32-WROOM

The ESP32-WROOM offers two antenna options: the PCB on-board antenna and the IPEX antenna.

- The PCB on-board antenna is an integrated antenna within the chip module itself, making it compact and convenient for both portability and design.
- The IPEX antenna is an external metal antenna connected to the module's integrated antenna, providing enhanced signal performance.



The ESP32-WROOM of this product is based on the ESP32-WROOM-32E module with built-in PCB on-board antenna.

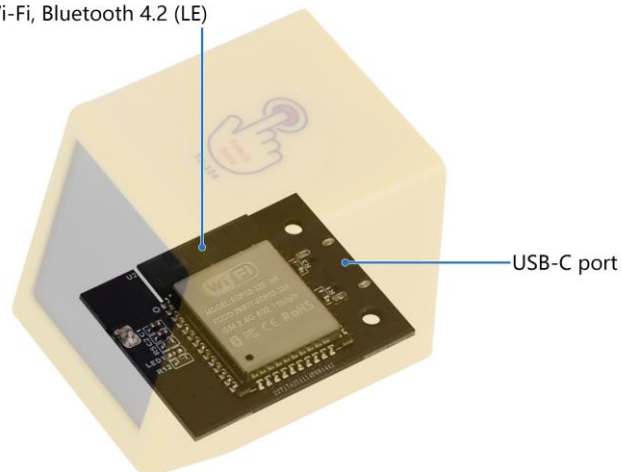
## Freenove ESP32 Mini TV

### ESP32 Module

A powerful, generic and small-sized controller with onboard wireless.

|                                 |             |
|---------------------------------|-------------|
| Dual-core 32-bit microprocessor | 520 KB SRAM |
| Up to 240 MHz clock frequency   | 448 KB ROM  |
| Compatible with Arduino IDE     | 4 MB flash  |

2.4 GHz Wi-Fi, Bluetooth 4.2 (LE)



### Color Screen

A delicate screen integrated on the mini TV.

|                        |                          |
|------------------------|--------------------------|
| 1.54 inch              | TFT screen               |
| 240x240 pixel          | ST7789 driver            |
| IPS wide viewing angle | 4-wire SPI communication |



## CH340 (Required)

ESP32 uses CH340 to download codes. So before using it, we need to install CH340 driver in our computers.

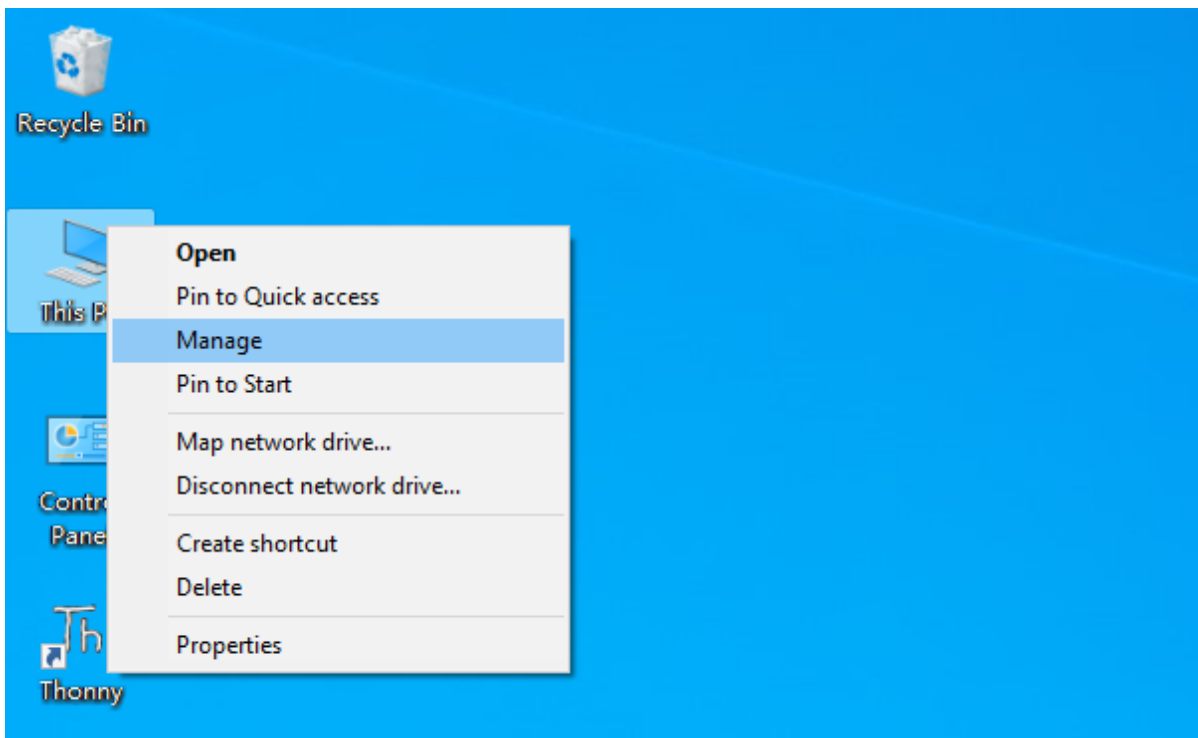
### Windows

Check whether CH340 has been installed

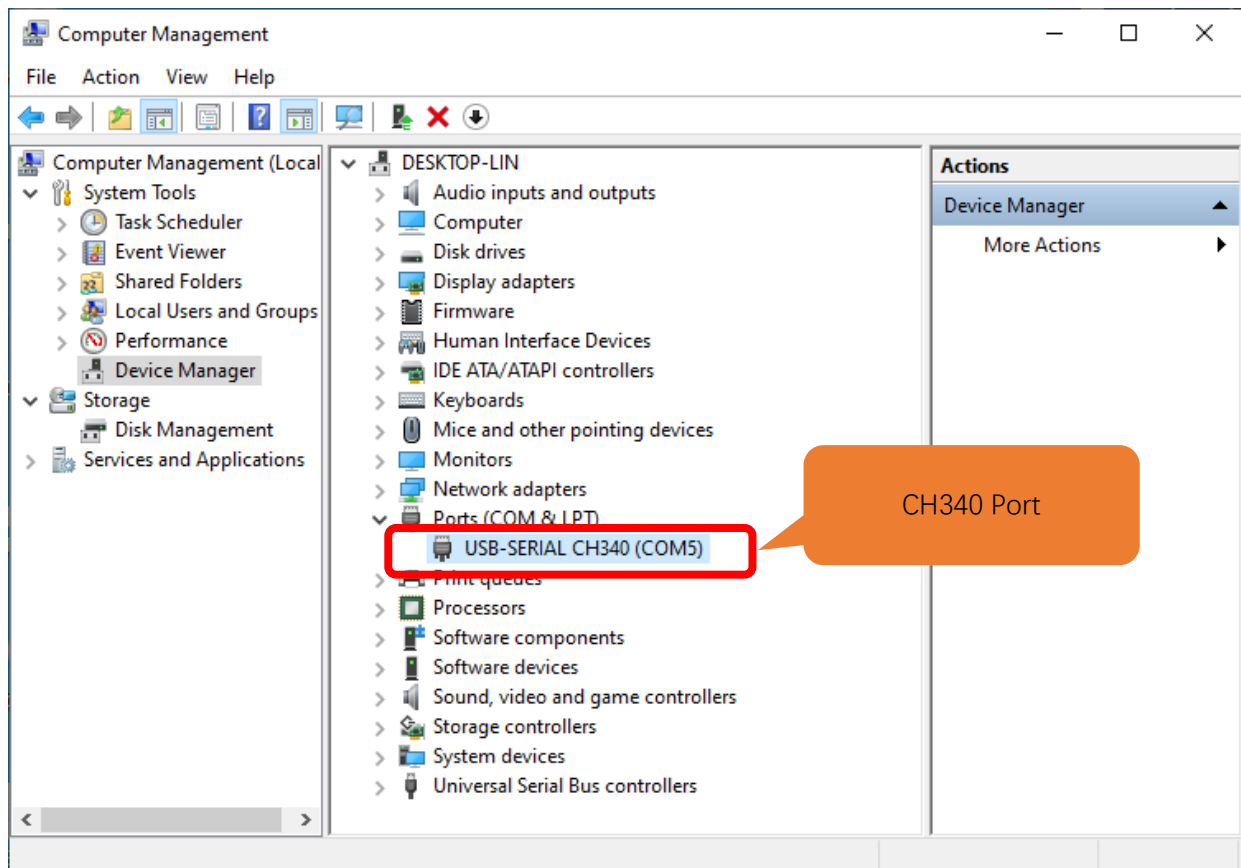
1. Connect your computer and ESP32-WROOM with a USB cable.



2. Turn to the main interface of your computer, select “**This PC**” and right-click to select “**Manage**”.



3. Click “Device Manager”. If your computer has installed CH340, you can see “USB-SERIAL CH340 (COMx)”. And you can click [here](#) to move to the next step.





## Installing CH340

- First, download CH340 driver, click <http://www.wch-ic.com/search?q=CH340&t=downloads> to download the appropriate one based on your operating system.

index / search / search CH340

All (14)

Downloads (7)

Products (4)

Application (2)

Video (1)

News (0)

keyword CH340

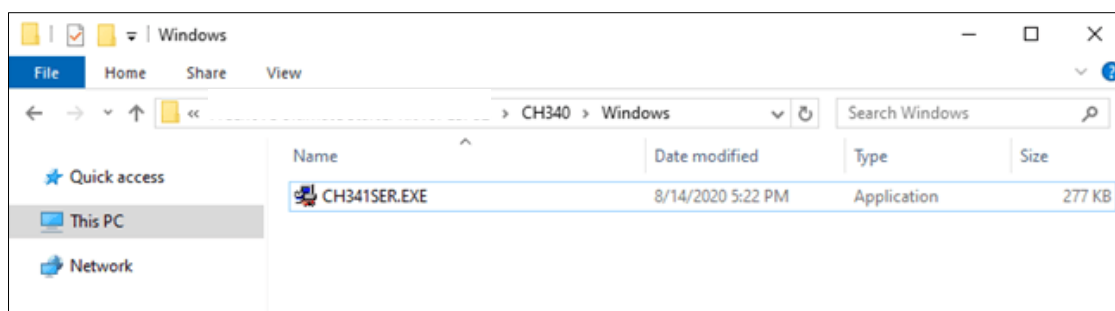
Downloads( 7 )

| file category       | file content                                                                                                                                                                                                                                                             | version | upload time |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|
| Driver&Tools        |                                                                                                                                                                                                                                                                          |         |             |
| CH341SER.EXE        | CH340/CH341 USB to serial port Windows driver, supports 32/64-bit Windows 10/8.1/8/7/VISTA/XP, Server 2016/2012/2008/2003, 2000/ME/98                                                                                                                                    | 3.5     | 2019-03-18  |
| CH341SER.ZIP        | CH340/CH341 USB to serial port Windows driver, includes DLL dynamic library and non-standard baud rate settings and other instructions. Supports 32/64-bit Windows 10/8.1/8/7/VISTA/XP, Server 2016/2012/2008/2003, 2000/ME/98                                           | 3.5     | 2019-03-05  |
| CH341SER_ANDROID... | CH340/CH341 USB to serial port Android free drive application library, for Android OS 3.1 and above version which supports USB Host mode already, no need to load Android kernel driver, no root privileges. Contains apk, lib (library), App Demo Example (USB to UART) | 1.6     | 2019-04-19  |
| CH341SER_LINUX...   | CH340/CH341 USB to serial port LINUX driver                                                                                                                                                                                                                              | 1.5     | 2018-03-18  |
| CH341SER_MAC.ZI...  | CH340/CH341 USB to serial port MAC OS driver                                                                                                                                                                                                                             | 1.5     | 2018-07-05  |
| Others              |                                                                                                                                                                                                                                                                          |         |             |
| PRODUCT_GUIDE.P...  | Electronic selection of product selection manual, please refer to related product technical manual for more technical information.                                                                                                                                       | 1.4     | 2018-12-29  |
| InstallNoteOn64...  | Instructions for the driver after 18 years of August cannot be installed under some 64-bit WIN7 (English)                                                                                                                                                                | 1.0     | 2019-01-10  |

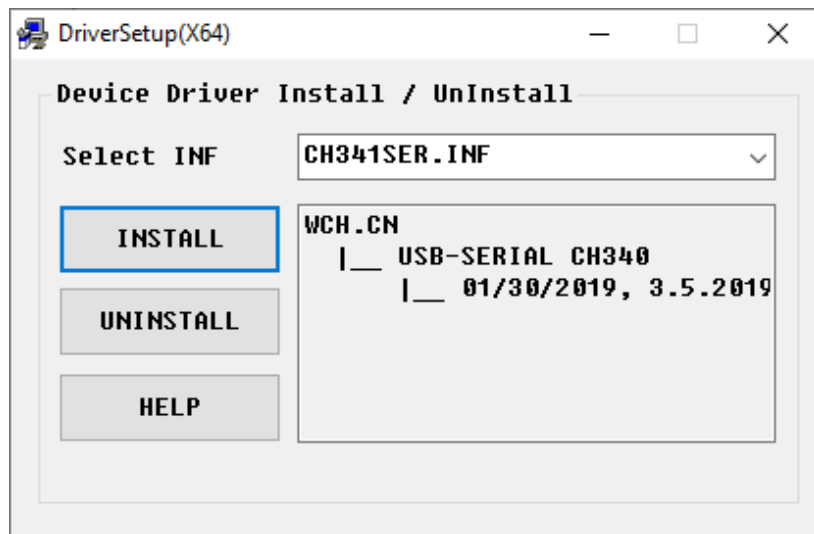
If you would not like to download the installation package, you can open “Freenove\_ESP32\_Mini\_TV/CH340”, we have prepared the installation package.

|         |                    |             |
|---------|--------------------|-------------|
| Linux   | 10/17/2022 1:30 PM | File folder |
| MAC     | 10/17/2022 1:30 PM | File folder |
| Windows | 10/17/2022 1:30 PM | File folder |

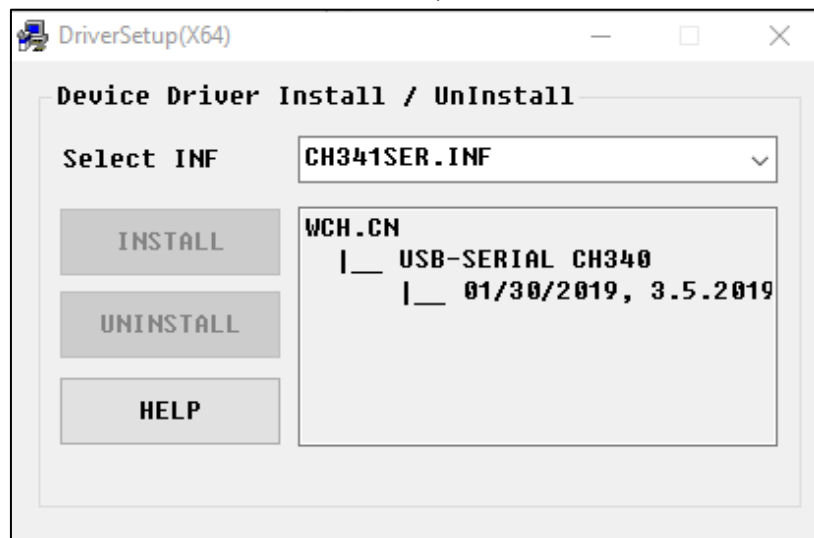
- Open the folder “Freenove\_ESP32\_Mini\_TV/CH340/Windows/”



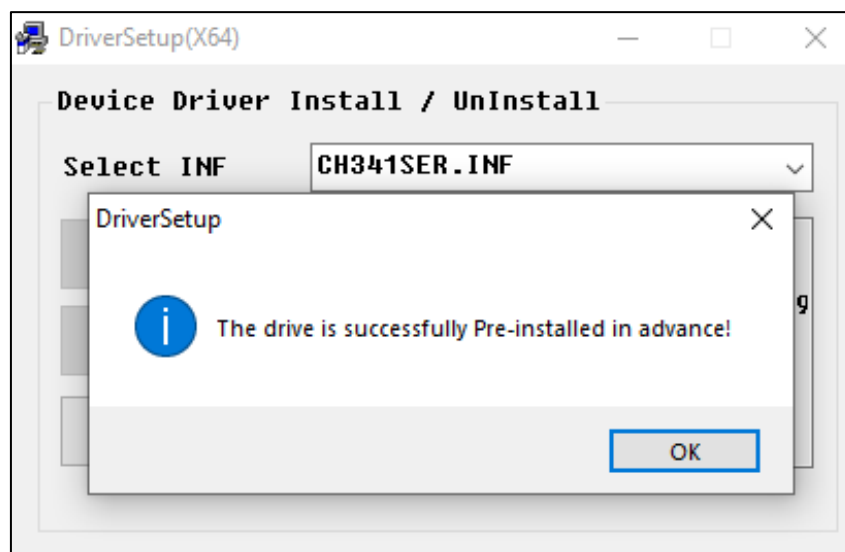
- Double click "CH341SER.EXE".



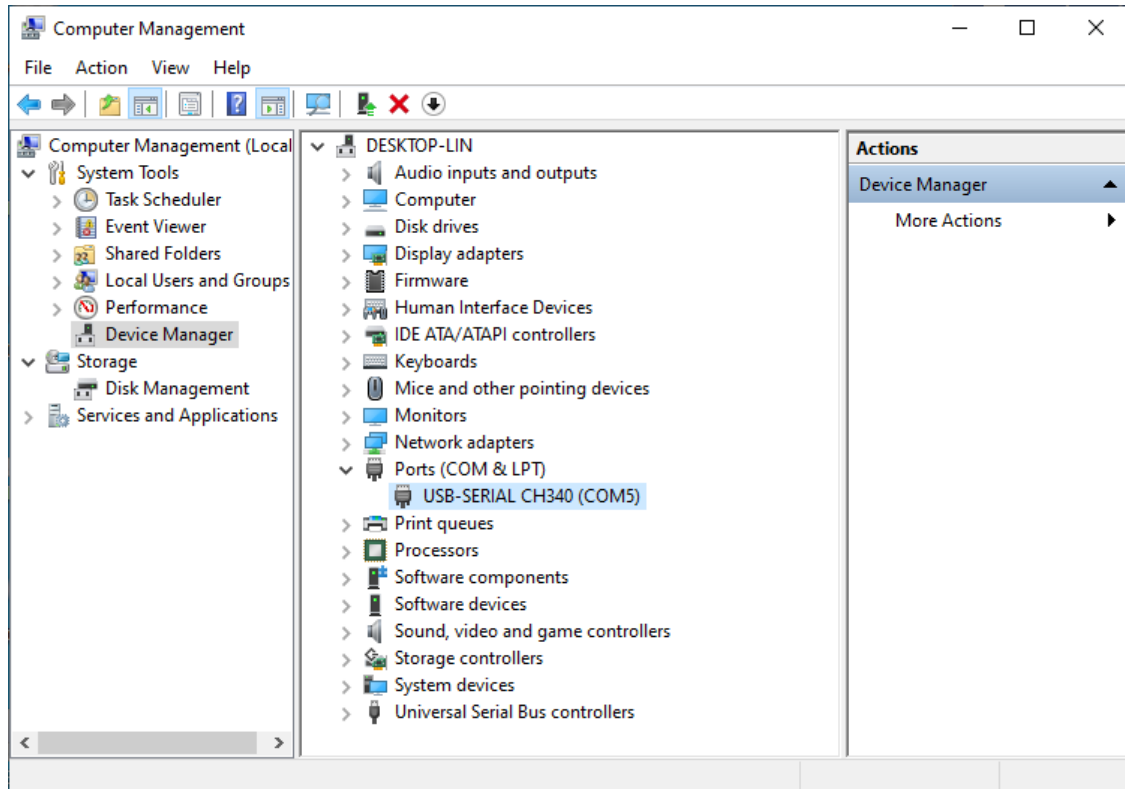
- Click "INSTALL" and wait for the installation to complete.



- Install successfully. Close all interfaces.



6. When ESP32 is connected to computer, select “This PC”, right-click to select “Manage” and click “Device Manager” in the newly pop-up dialog box, and you can see the following interface.



7. So far, CH340 has been installed successfully. Close all dialog boxes.

## MAC

First, download CH340 driver, click <http://www.wch-ic.com/search?q=CH340&t=downloads> to download the appropriate one based on your operating system.

The screenshot shows the WCH website's search results for 'ch340'. The left sidebar lists categories: All (14), Downloads (7), Products (4), Application (2), Video (1), and News (0). The main content area shows 'keyword ch340' and a list of downloads. The table below summarizes the listed files:

| file category       | file content                                                                                                                                                                                                                                                                                          | version | upload time |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|
| Driver&Tools        |                                                                                                                                                                                                                                                                                                       |         |             |
| CH341SER.EXE        | CH340/CH341 USB to serial port Windows driver, supports 32/64-bit Windows 10/8.1/8/7/VISTA/XP, Server 2016/2012/2008/2003, 2000/ME/98                                                                                                                                                                 | 3.5     | 2019-03-18  |
| CH341SER.ZIP        | CH340/CH341 USB to serial port Windows driver, includes DLL dynamic library and non-standard baud rate settings and other instructions. Supports 32/64-bit Windows 10/8.1/8/7/VISTA/XP, Server 2016/2012/2008/2003, 2000/ME/98                                                                        | 3.5     | 2019-03-05  |
| CH341SER_ANDROID... | CH340/CH341 USB to serial port Android free drive application library, for Android OS 3.1 and above version which supports USB Host mode already, no need to load Android kernel driver, no root privileges. Contains apk, lib library file (Java Driver), App Demo Example (USB to serial port SDK). | 1.6     | 2019-04-19  |
| CH341SER_LINUX....  | CH340/CH341 USB to serial port LINUX driver                                                                                                                                                                                                                                                           | 1.5     | 2018-03-18  |
| CH341SER_MAC.ZI...  | CH340/CH341 USB to serial port MAC OS driver                                                                                                                                                                                                                                                          | 1.5     | 2018-07-05  |
| Others              |                                                                                                                                                                                                                                                                                                       |         |             |

If you would not like to download the installation package, you can open

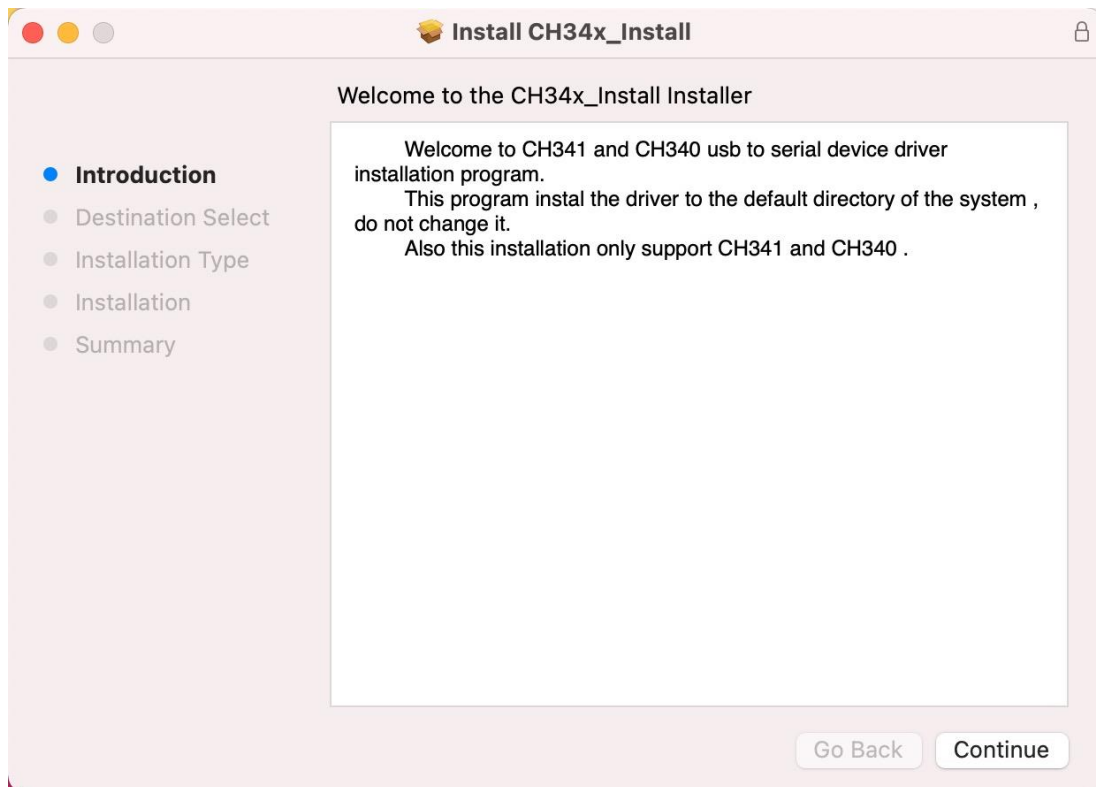
“Freenove\_ESP32\_Mini\_TV/CH340”, we have prepared the installation package.

Second, open the folder “Freenove\_ESP32\_Mini\_TV /CH340/MAC/”

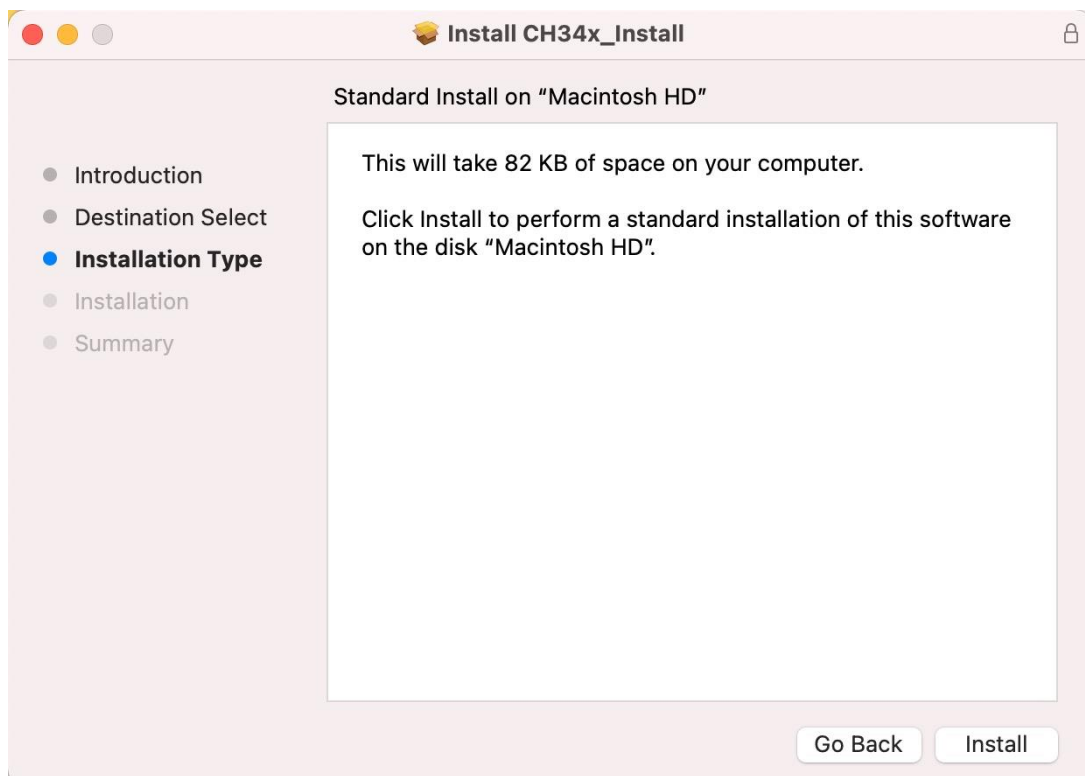
The screenshot shows a Mac file explorer window titled 'MAC'. The left sidebar shows the 'Favorites' list. The main pane displays the contents of the 'MAC' folder:

| Name                   | Modified               | Size   | Kind              |
|------------------------|------------------------|--------|-------------------|
| CH34x_Install_V1.5.pkg | Dec 8, 2018 at 2:00 PM | 26 KB  | Installer package |
| ReadMe.pdf             | Dec 8, 2016 at 4:09 PM | 146 KB | Adobe...ocument   |

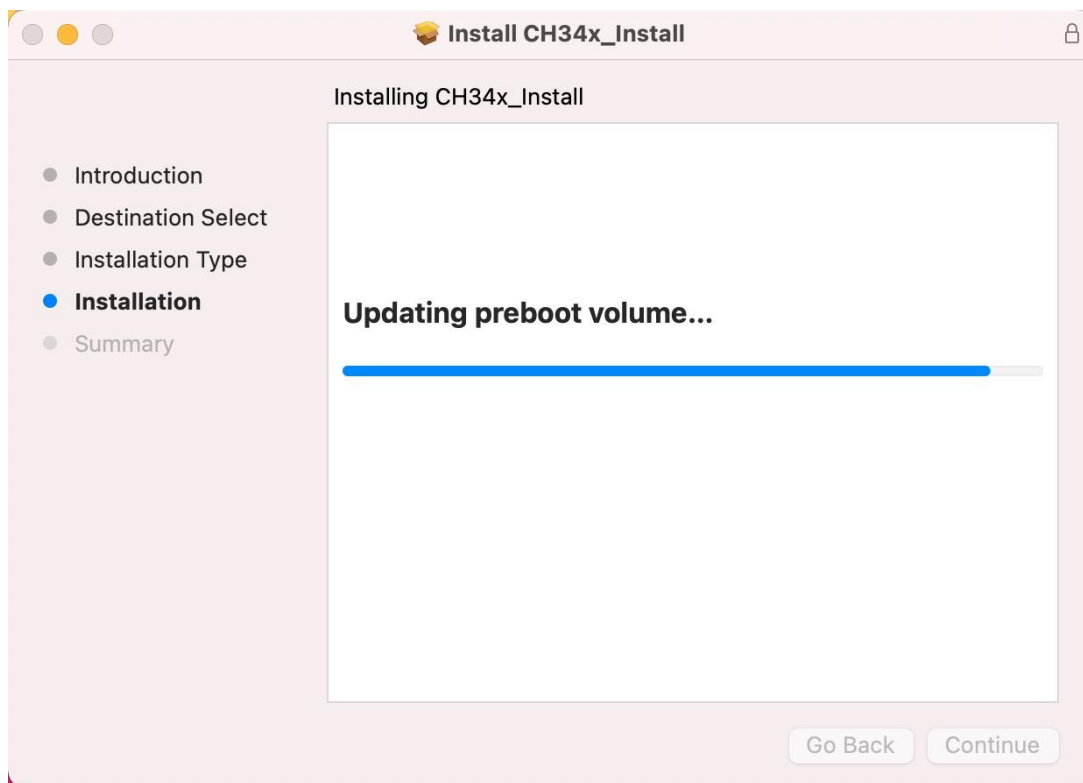
Third, click Continue.



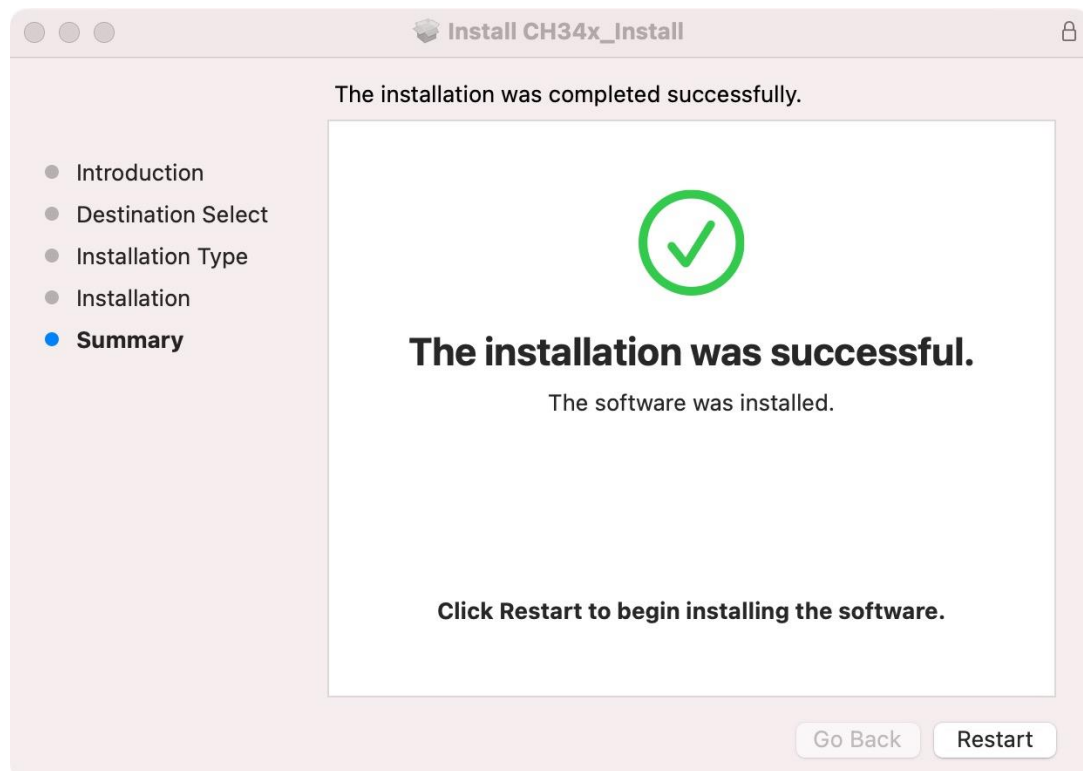
Fourth, click Install.



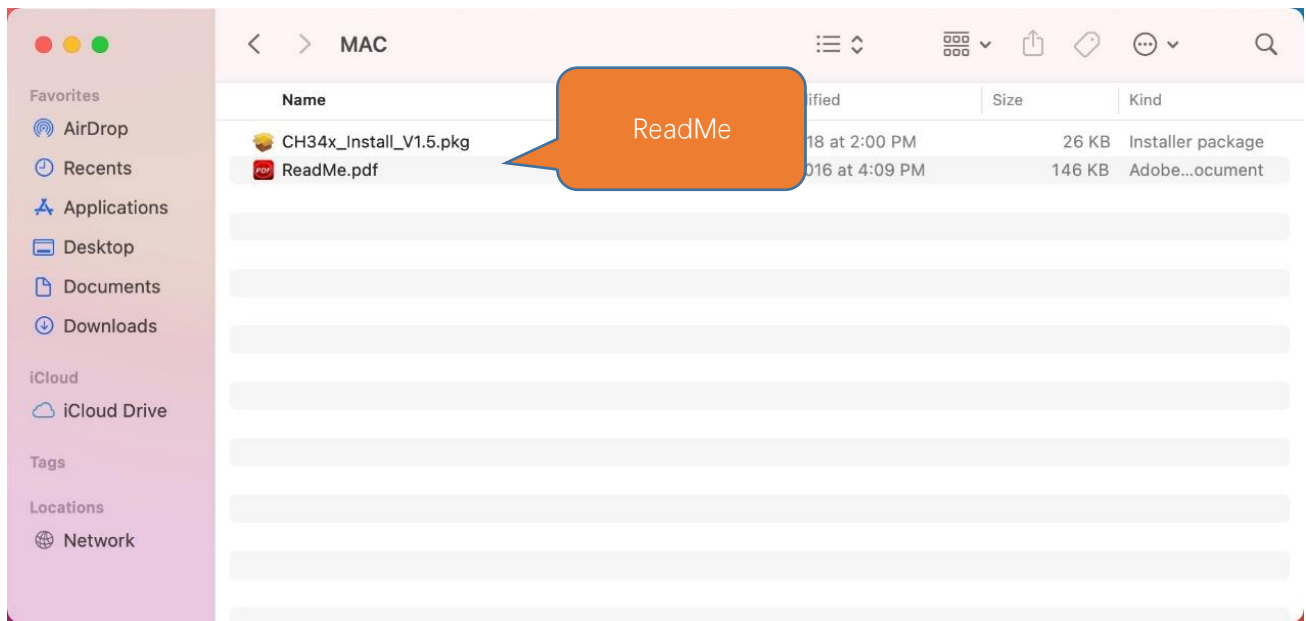
Then, waiting Finish.



Finally, restart your PC.



If you still haven't installed the CH340 by following the steps above, you can view readme.pdf to install it.

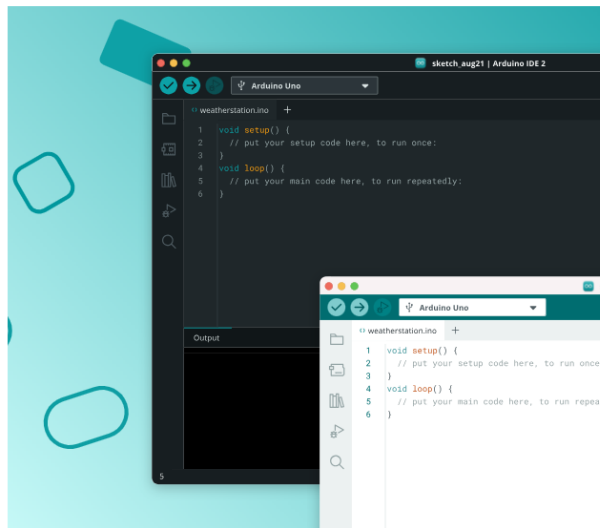


## Programming Software

Arduino Software (IDE) is used to write and upload the code for Arduino Board.

First, install Arduino Software (IDE): visit <https://www.arduino.cc/en/software/>

### Bring Your Projects to Life with Arduino Software



#### Arduino IDE 2.3.6 Release notes

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

Windows Win 10 or newer (64-bit)

DOWNLOAD



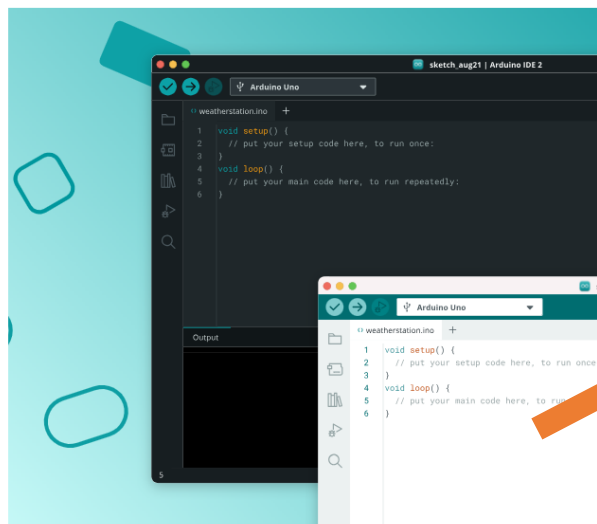
#### Nightly Builds

Download a preview of the incoming release with the most updated features and bugfixes.

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

Select and download corresponding installer based on your operating system. If you are a Windows user, please select the "Windows" to download and install the driver correctly.

### Bring Your Projects to Life with Arduino Software



#### Arduino IDE 2.3.6 Release notes

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

Windows Win 10 or newer (64-bit)

DOWNLOAD

Windows Win 10 or newer (64-bit)

Windows MSI installer

Windows ZIP file

Linux Appliance (64-bit X86-64)

Linux ZIP file (64-bit X86-64)

macOS Intel 10.15 Catalina or newer (64-bit)

macOS Apple Silicon 11 Big Sur or newer (64-bit)



#### Legacy IDE (1.8.19)

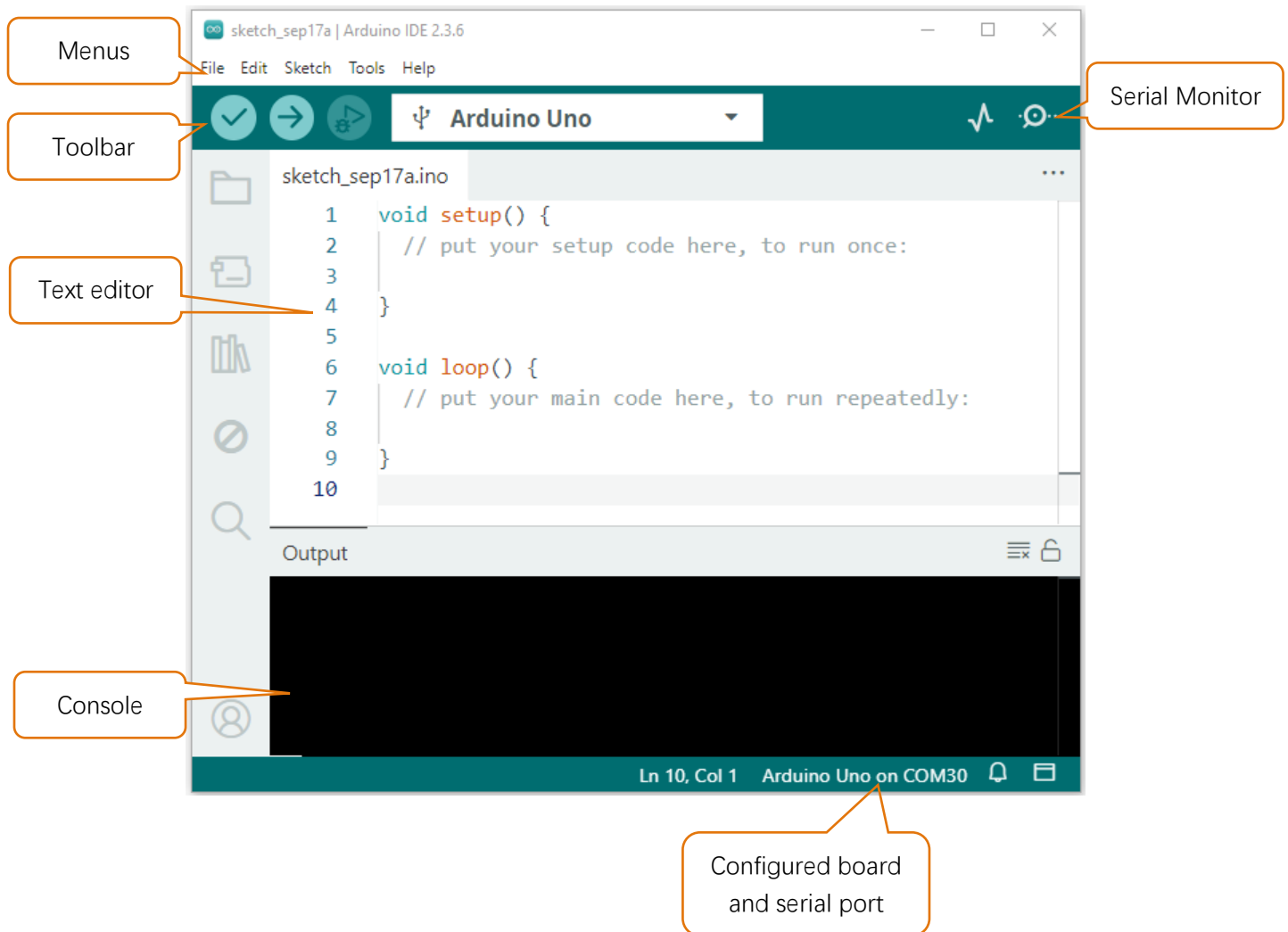
Download a legacy version of the Arduino IDE.



After the downloading completes, run the installer. For Windows users, there may pop up an installation dialog box of driver during the installation process. When it is popped up, please allow the installation. After installation is completed, an shortcut will be generated in the desktop.



Run it. The interface of the software is as follows:

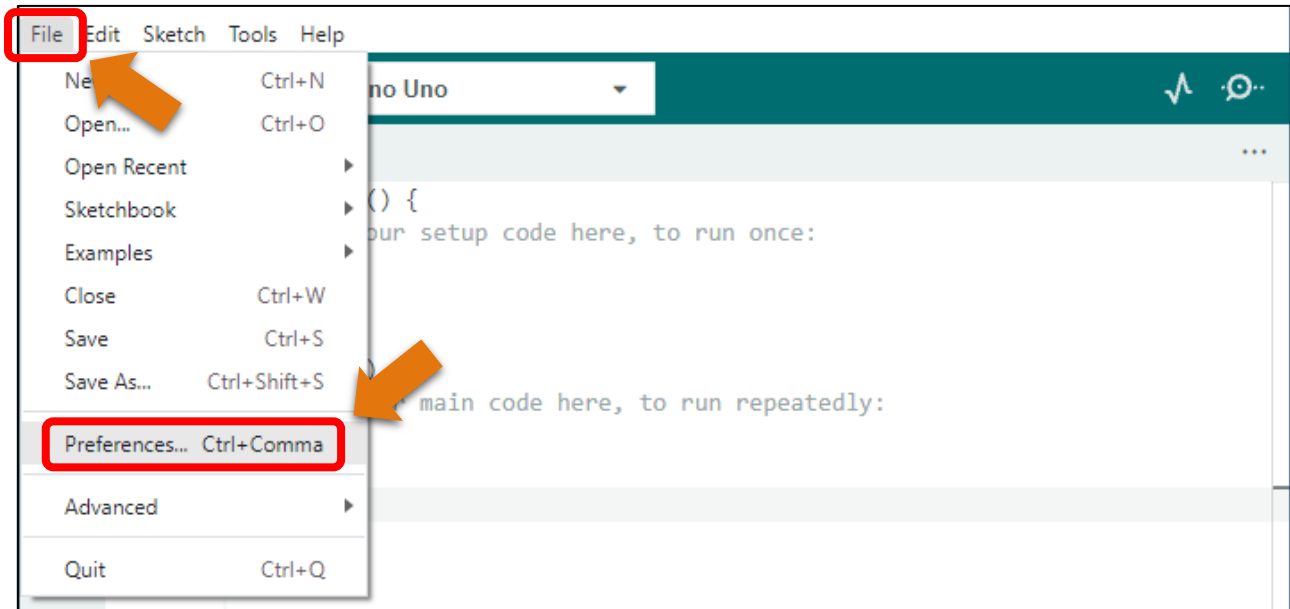


Programs written with Arduino IDE are called sketches. These sketches are written in a text editor and are saved with the file extension.ino. The editor has features for cutting/pasting and for searching/replacing text. The console displays text output by the Arduino IDE, including complete error messages and other information. The bottom right-hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, open the serial monitor, and access the serial plotter.

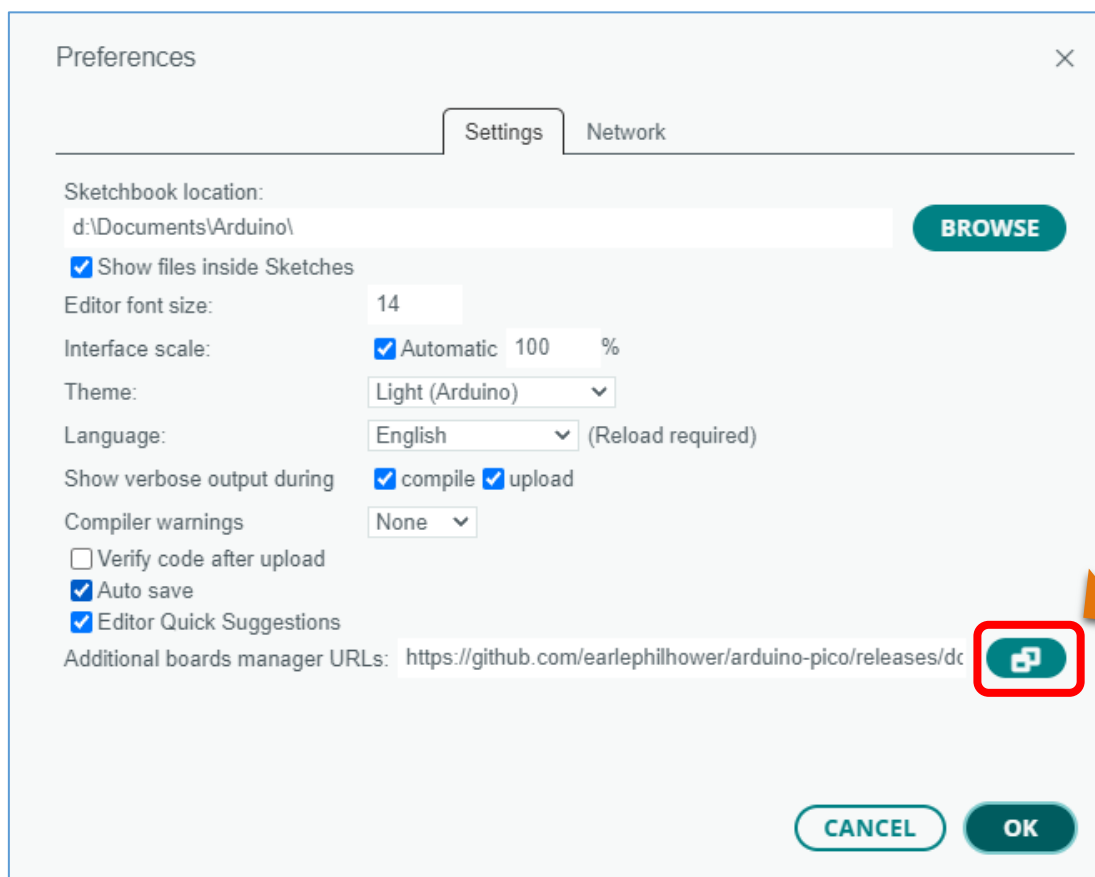
|                                                                                   |                                                                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|  | <b>Verify</b><br>Checks your code for errors compiling it.                                   |
|  | <b>Upload</b><br>Compiles your code and uploads it to the configured board.                  |
|  | <b>Debug</b><br>Troubleshoot code errors and monitor program running status.                 |
|  | <b>Serial Plotter</b><br>Real-time plotting of serial port data charts.                      |
|  | <b>Serial Monitor</b><br>Used for debugging and communication between devices and computers. |

## Environment Configuration

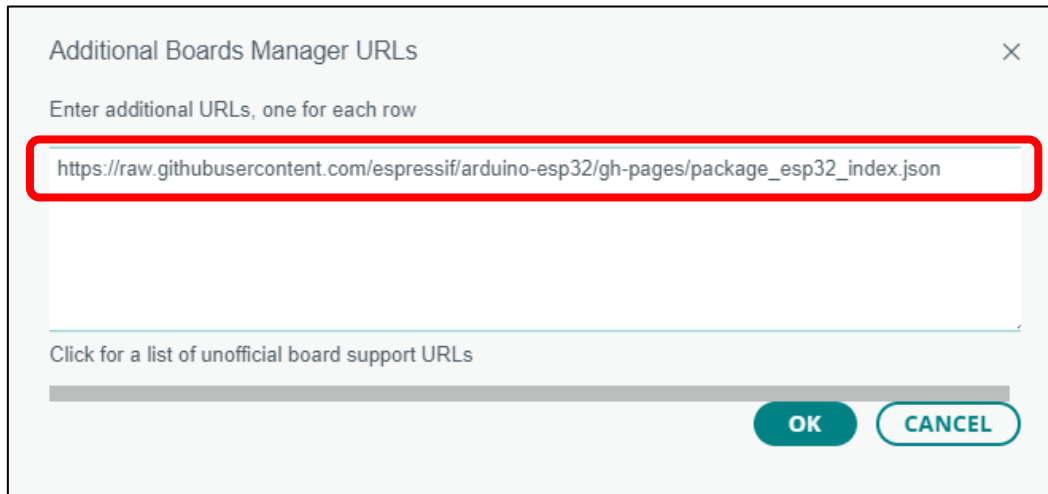
First, open the software platform arduino, and then click File in Menus and select Preferences.



Second, click on the symbol behind "Additional Boards Manager URLs"



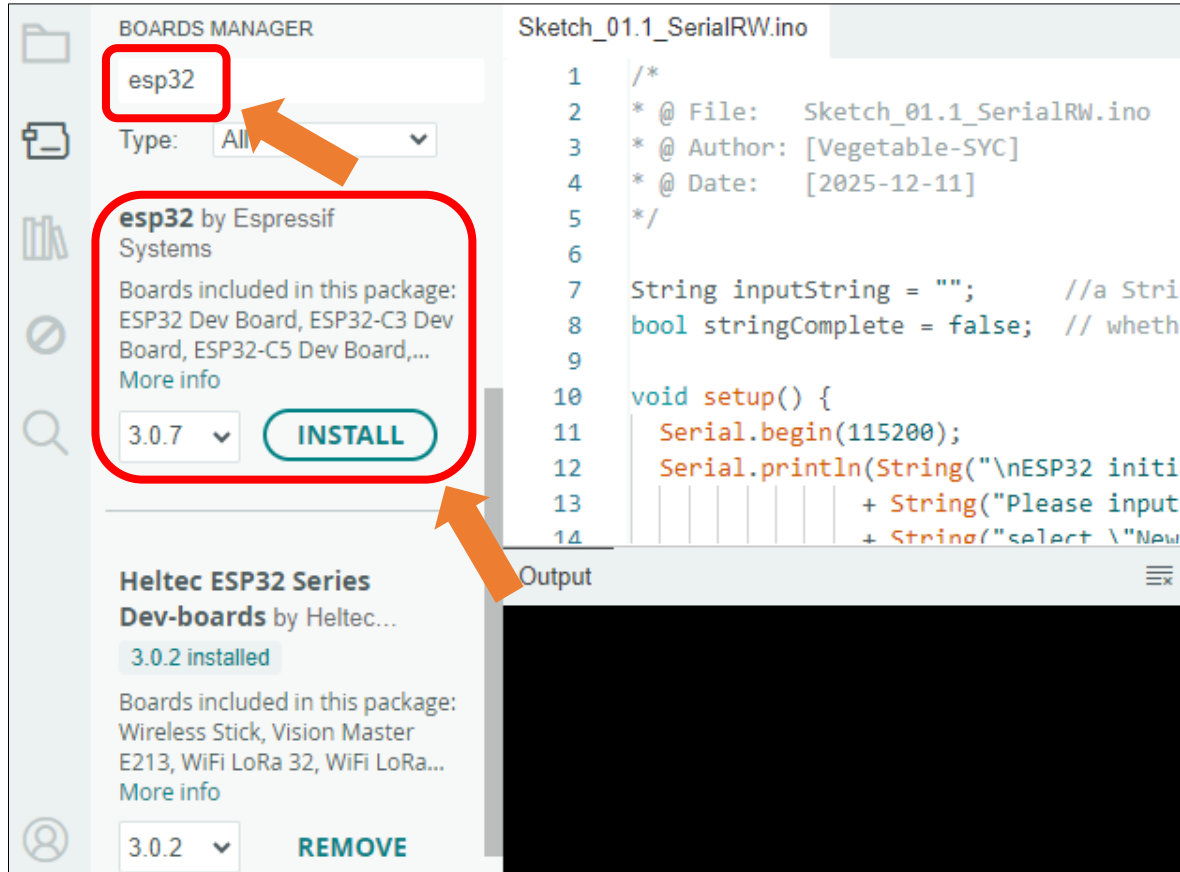
Third, fill in [https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package\\_esp32\\_index.json](https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json) in the new window, click OK, and click OK on the Preferences window again.



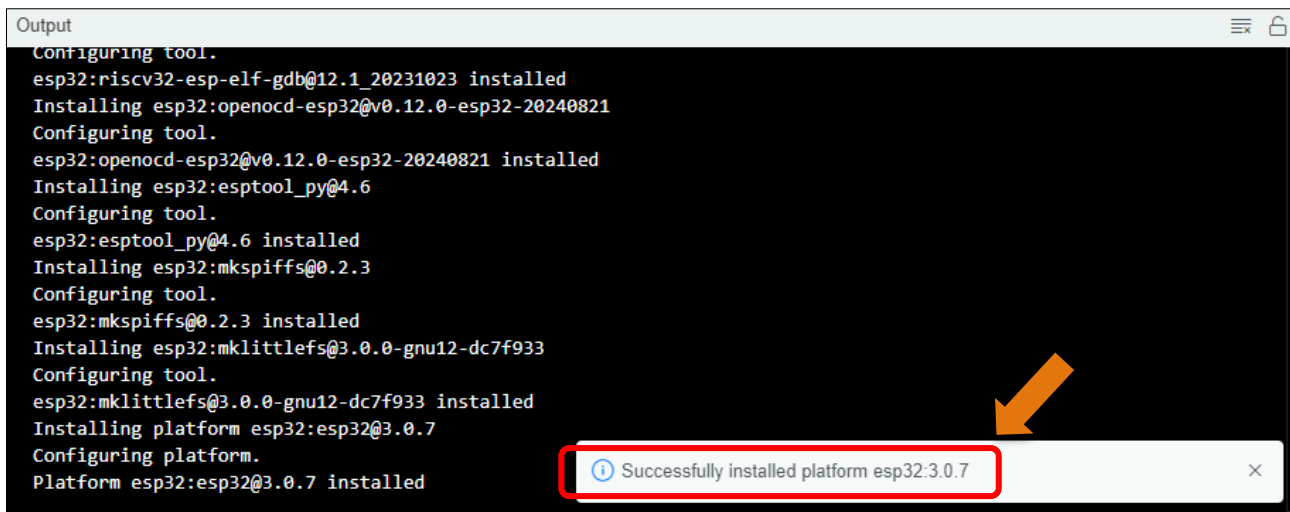
**Note:** if you copy and paste the URL directly, you may lose the "-". Please check carefully to make sure the link is correct.

Fourth, click "Boards Manager". Enter "esp32" in Boards manager, select 3.0.7, and click "INSTALL".

**Please note:** Repeated testing has revealed that higher versions of the board support package may cause compatibility issues with certain peripherals. It is recommended to use the more stable version 3.0.7.

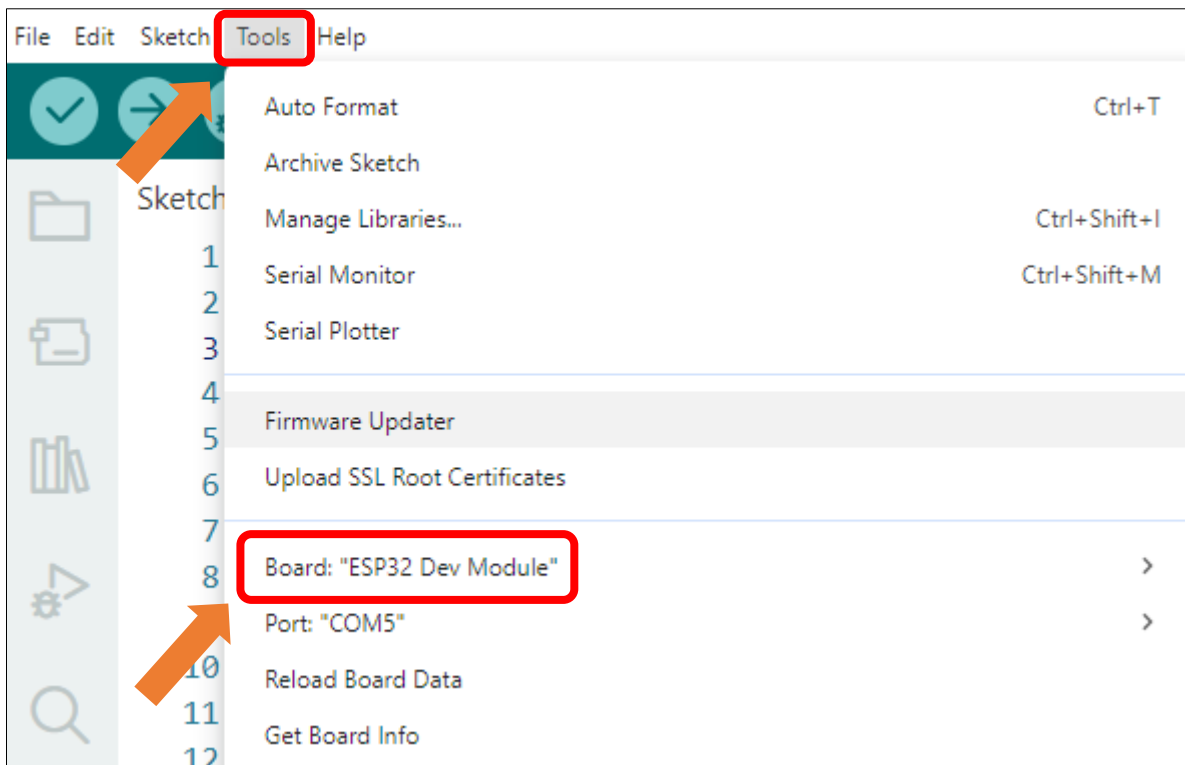


Arduino will download these files automatically. Wait for the installation to complete.



```
Output
Configuring tool.
esp32:riscv32-esp-elf-gdb@12.1_20231023 installed
Installing esp32:openocd-esp32@v0.12.0-esp32-20240821
Configuring tool.
esp32:openocd-esp32@v0.12.0-esp32-20240821 installed
Installing esp32:esptool_py@4.6
Configuring tool.
esp32:esptool_py@4.6 installed
Installing esp32:mkspiffs@0.2.3
Configuring tool.
esp32:mkspiffs@0.2.3 installed
Installing esp32:mklittlefs@3.0.0-gnu12-dc7f933
Configuring tool.
esp32:mklittlefs@3.0.0-gnu12-dc7f933 installed
Installing platform esp32:esp32@3.0.7
Configuring platform.
Platform esp32:esp32@3.0.7 installed
Successfully installed platform esp32:3.0.7
```

When finishing installation, click Tools in the Menus again and select Board: "ESP32 Dev Module", and then you can see information of ESP32.



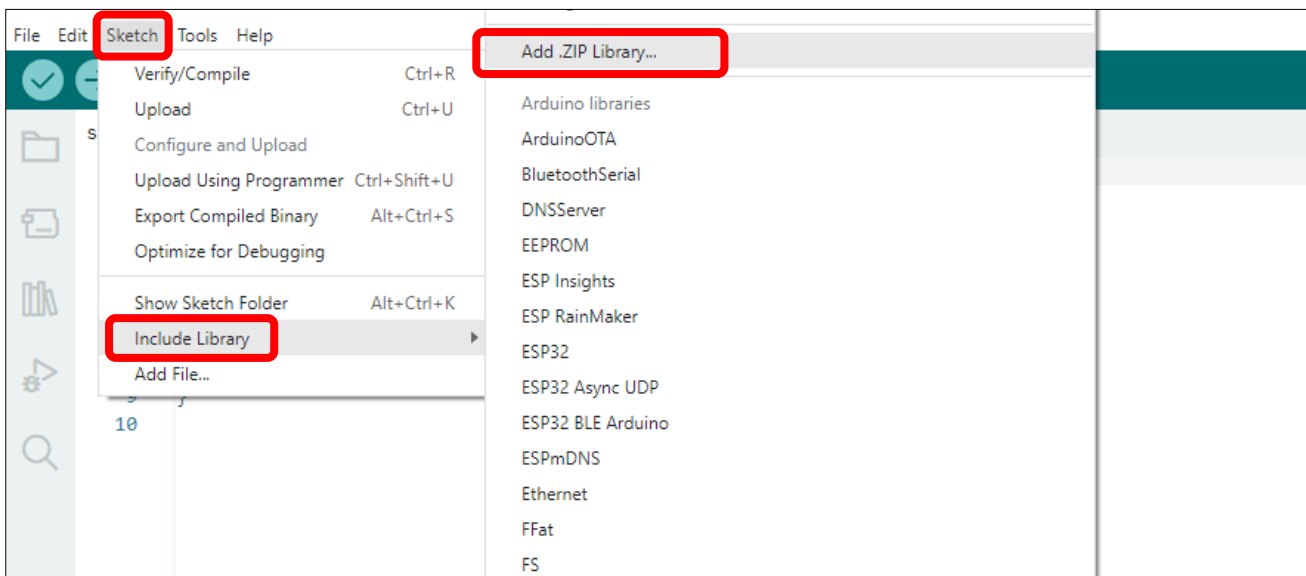
## Library Installation

Before starting the learning process, it is necessary to install some libraries in advance to enable the code to be compiled properly. For convenience, we have already packaged these libraries and placed them in the **Freenove\_ESP32\_Mini\_TV/Libraries** folder. Please refer to the following steps to install these libraries into the Arduino IDE.

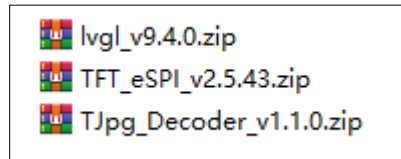
1. Open Arduino IDE.



2. Select Sketch->Include Library->Add .ZIP Library...

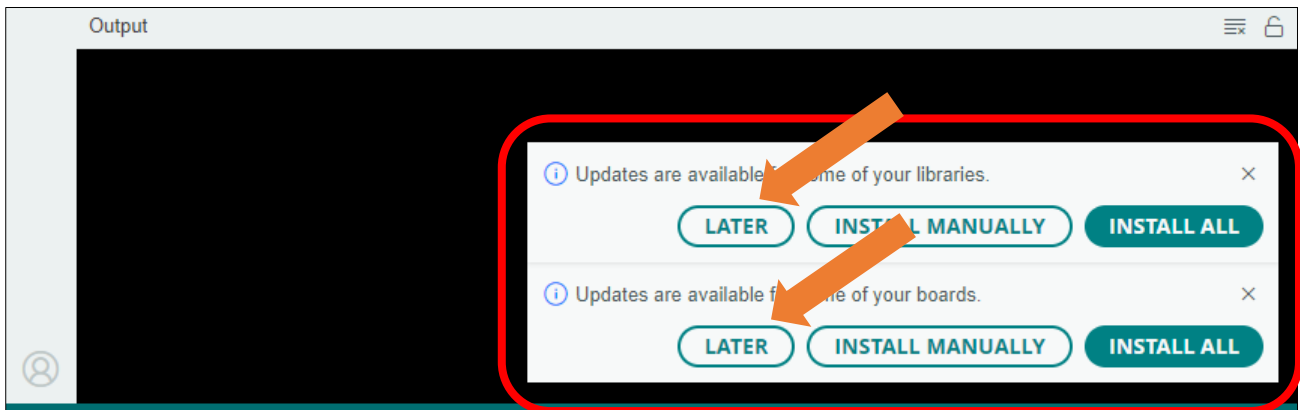


- On the newly pop-up window, select the files from the **Freenove\_ESP32\_Mini\_TV/Libraries**. Click Open to install the library.



- Repeat the above steps until all the three libraries are installed to Arduino.** So far, all libraries have been installed.

**Note:** Some libraries are not the latest version. Please do not update them even if it prompts every time you open the IDE. Just click LATER. Otherwise, it may lead to compilation failure.



# Chapter 1 Serial

## Project 1.1 SerialRW

### Related Knowledge

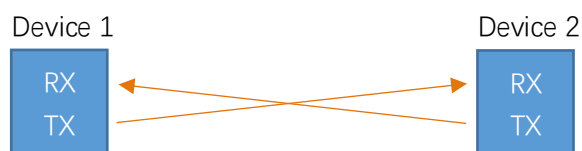
#### Serial and parallel communication

Serial communication uses one data cable to transfer data one bit by another in turn, while parallel communication means that the data is transmitted simultaneously on multiple cables. Serial communication takes only a few cables to exchange information between systems, which is especially suitable for computers to computer, long distance communication between computers and peripherals. Parallel communication is faster, but it requires more cables and higher cost, so it is not appropriate for long distance communication.



#### Serial communication

Serial communication generally refers to the Universal Asynchronous Receiver/Transmitter (UART), which is commonly used in electronic circuit communication. It has two communication lines, one is responsible for sending data (TX line) and the other for receiving data (RX line). The serial communication connections of two devices use is as follows:



Before serial communication starts, the baud rate in both sides must be the same. Only use the same baud rate can the communication between devices be normal. The baud rates commonly used are 9600 and 115200.



## Component List

Freenove ESP32 Mini TV



USB cable x1



## Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “**Sketch\_01.1\_SerialRW**” folder under “**Freenove\_ESP32\_Mini\_TV\Sketch**” and double-click “**Sketch\_01.1\_SerialRW.ino**”.

Sketch\_01.1\_SerialRW

```
1 String inputString = "";    //a String to hold incoming data
2 bool stringComplete = false; // whether the string is complete
3
4 void setup() {
5     Serial.begin(115200);
6     Serial.println(String("\nESP32 initialization completed!\n"));
7     + String("Please input some characters,\n")
```

```

8         + String("select \"Newline\" below and click send button. \n"));
9     }
10
11 void loop() {
12     if (Serial.available()) {          // judge whether data has been received
13         char inChar = Serial.read();    // read one character
14         inputString += inChar;
15         if (inChar == '\n') {
16             stringComplete = true;
17         }
18     }
19     if (stringComplete) {
20         Serial.printf("InputString: %s", inputString);
21         inputString = "";
22         stringComplete = false;
23     }
24 }

```

### Code Explanation

Set the baud rate to 115200.

```
5 Serial.begin(115200);
```

Determine whether there is data in the serial port buffer.

```
8 if (Serial.available())
```

Receive serial port data and save it in the inputString string.

```

13 char inChar = Serial.read();          // read one character
14 inputString += inChar;

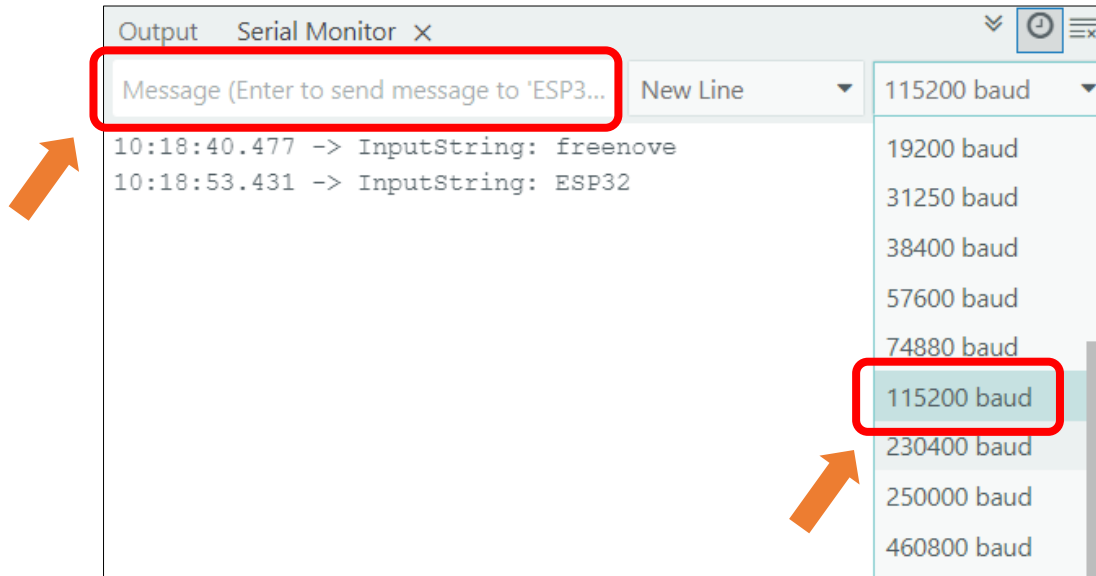
```

This code achieves the function of printing data on serial monitor. Click "**Upload**" to upload the code to Freenove ESP32 Mini TV.



After downloading the code, open the serial port monitor, and set the baud rate to **115200**, input any data in the messages bard and press Enter key, Freenove ESP32 Mini TV will print the received data.

**Please note: This chapter does not involve the use of the screen. After the code for this chapter, the screen may not light up, which is normal and not a hardware malfunction. If you need to verify whether the screen is functioning properly, please refer to [Chapter 6](#) for testing.**



#### Reference

##### `int available()`

`Serial.available()` checks the number of bytes currently available to read in the Serial receive buffer. It returns the number of bytes available (int type), or 0 if the buffer is empty.

##### `int read ()`

`Serial.read()` reads one byte of data from the Serial receive buffer and returns it as an int. If no data is available to read, it returns -1.

## Chapter 2 Touch

### Project 2.1 Touch Test

#### Component List

|                                                                                                             |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Freenove ESP32 Mini TV<br> | USB cable x1<br> |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|

#### Component knowledge

The fundamental principle of the ESP32's capacitive touch sensing is to monitor variations in the capacitance of its GPIO pins. Physically, a touch-capable pin and its associated sensing pad constitute an equivalent capacitor. The proximity of a human finger, which possesses inherent capacitance, increases the overall capacitance at the pin. For detection, the ESP32 employs an internal circuit to cycle through charging and discharging the pin at a high frequency. It records the number of pulses (the count of completed cycles) in a fixed duration. As a larger capacitance extends the charge/discharge time, the recorded pulse count within that period is correspondingly reduced.

Freenove ESP32 Mini TV utilizes capacitive touch sensing technology in its design. Internally, a wire connect the touch-sensitive pins to a copper foil pad affixed to the inside of the casing. This configuration transforms specific areas on the exterior shell into an active touch sensor, allowing the device to be controlled by simply tapping the designated spots on the casing.

Important Note: The internal disassembly diagram shown below is provided for illustrative purposes only. **Disassembling the device yourself is strongly discouraged. The unit is secured with extremely small screws that are highly susceptible to being lost or damaged during removal, which can compromise the structural integrity of the device.**



## Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “Sketch\_02.1\_Touch\_Test” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_02.1\_Touch\_Test.ino”.

### Sketch\_02.1\_Touch\_Test

```
1  const int TOUCH_PIN = 32;
2
3  void setup() {
4      Serial.begin(115200);
5      delay(1000);
6      Serial.println("ESP32 Touch Test");
7      touchSetCycles(0xf000, 0x1000);
8  }
9
10 void loop() {
11     Serial.println(touchRead(TOUCH_PIN));
12     delay(100);
13 }
```

### Code Explanation

Set the baud rate to 115200.

```
4  Serial.begin(115200);
```

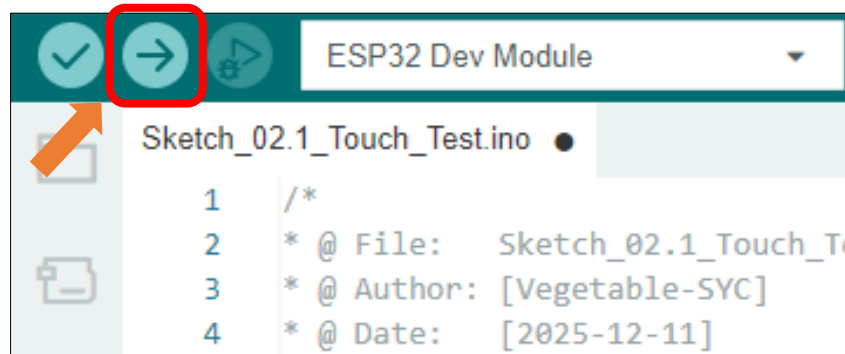
Configure touch detection parameters

```
7  touchSetCycles(0xf000, 0x1000);
```

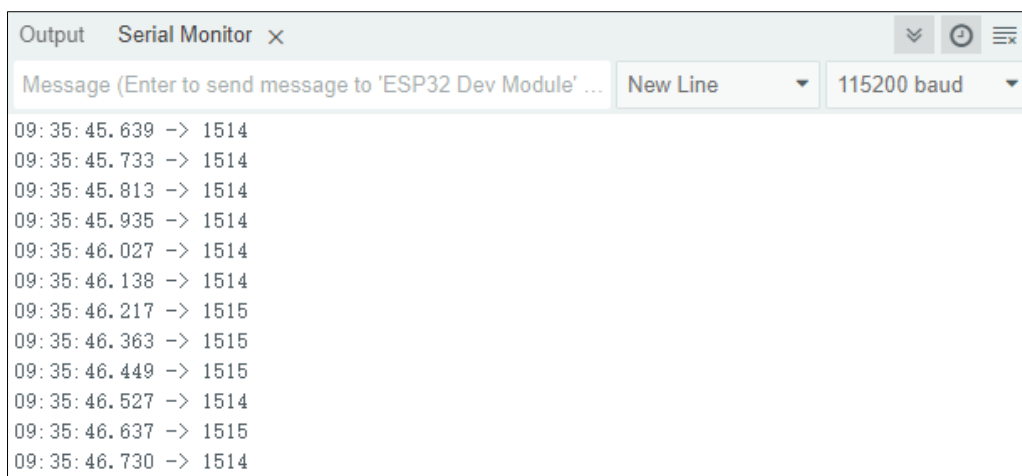
Print capacitance values on the serial monitor

```
7  Serial.println(touchRead(TOUCH_PIN));
```

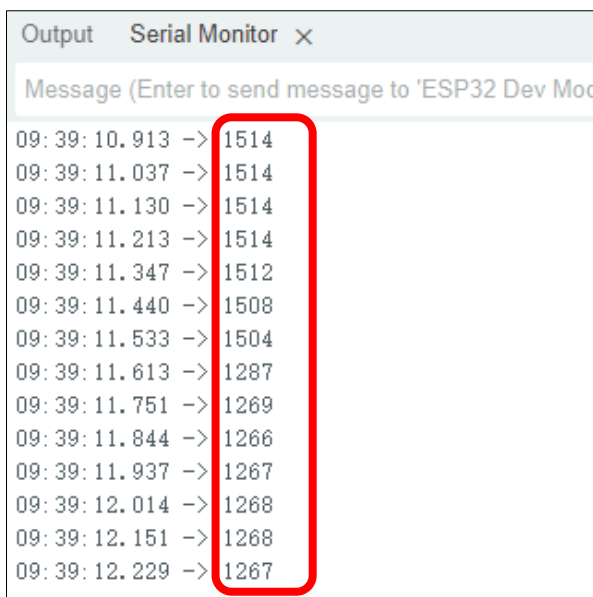
The purpose of this code is to display data on the serial monitor. Click **“Upload”** to upload the code to Freenove ESP32 Mini TV.



After downloading the code, open the serial port monitor, and set the baud rate to **115200**



Touching the capacitive touch area of the Freenove ESP32 Mini TV with your finger will significantly reduce the capacitance reading.



**Please note:** Repeated testing has revealed that newer versions of the board support package may cause compatibility issues with certain peripherals. If abnormal serial port data transmission occurs, please use the more stable version 3.0.7.

## Project 2.2 Touch Button

Building on the understanding of capacitive touch operation principles from the previous section, this part will focus on accurately distinguishing between short and long presses using these principles.

### Component List

|                                                                                                             |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Freenove ESP32 Mini TV<br> | USB cable x1<br> |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|

### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



### Sketch

Open “**Sketch\_02.2\_Touch\_Button**” folder under “**Freenove\_ESP32\_Mini\_TV\Sketch**” and double-click “**Sketch\_02.2\_Touch\_Button.ino**”.

#### Sketch\_02.2\_Touch\_Button

```
1  /*
2  * @ File:   Sketch_02.2_Touch_Button.ino
3  * @ Author: [Vegetable-SYC]
4  * @ Date:   [2025-12-19]
```

```
5  */
6
7  const int TOUCH_PIN = 32;
8  const int TOUCH_THRESHOLD = 80; // Touch sensitivity threshold
9  const int PRESS_Time = 3000;    // Threshold duration for long press (ms)
10 uint32_t Touch_Time = 0;        // Timestamp of the initial touch
11 int Touch_data = 0, Touch_old_data = 0;
12 int short_press_num = 0;
13 int long_press_num = 0;
14 bool LONG_PRESS_FLAG = 0;
15
16 void setup() {
17     Serial.begin(115200);
18     delay(1000); // Wait for serial monitor to initialize
19
20     /* Configure touch sensor cycles */
21     touchSetCycles(0xf000, 0x1000);
22
23     /* Initial calibration and baseline data reading */
24     Touch_old_data = touchRead(TOUCH_PIN);
25     Serial.println(Touch_old_data);
26 }
27
28 void loop() {
29     // Read touch sensor data
30     Touch_data = touchRead(TOUCH_PIN);
31
32     // Track and update the maximum baseline value
33     if(Touch_old_data < Touch_data)
34         Touch_old_data = Touch_data;
35
36     // Press Detection Logic
37     if((Touch_old_data - Touch_data) > TOUCH_THRESHOLD)
38     {
39         // Start timer on initial touch detection
40         if(Touch_Time == 0)
41             Touch_Time = millis();
42
43         // Evaluate if the press duration meets Long Press requirement
44         if(millis() - Touch_Time > PRESS_Time)
45         {
46             Touch_Time = 0;
47             LONG_PRESS_FLAG = 1;
48         }
```



```

49     Serial.print("Long Press Detected: ");
50     long_press_num++;
51     Serial.println(long_press_num);
52 }
53 }
54 else {
55     // Release Detection Logic
56     if(Touch_Time != 0)
57     {
58         Touch_Time = 0;
59
60         if(LONG_PRESS_FLAG != 0)
61         {
62             // Event: Long press released
63             LONG_PRESS_FLAG = 0;
64         }
65         else
66         {
67             // Event: Short press released
68             Serial.print("Short Press Detected: ");
69             short_press_num++;
70             Serial.println(short_press_num);
71         }
72     }
73 }
74 }

```

### Code Explanation

Preset basic parameters

```

7  const int TOUCH_PIN = 32;
8  const int TOUCH_THRESHOLD = 80; // Touch sensitivity threshold
9  const int PRESS_Time = 3000;    // Threshold duration for long press (ms)

```

Set the baud rate to 115200.

```

17  Serial.begin(115200);

```

Configure touch detection parameters.

```

21  touchSetCycles(0xf000, 0x1000);

```

Obtain capacitive touch reference value.

```

24  Touch_old_data = touchRead(TOUCH_PIN);

```

Continuously reading the capacitive touch values.

```

30  Touch_data = touchRead(TOUCH_PIN);

```

Determine if the difference between the current capacitive touch value and the reference value exceeds the preset touch threshold.

```

37  if((Touch_old_data - Touch_data) > TOUCH_THRESHOLD)

```

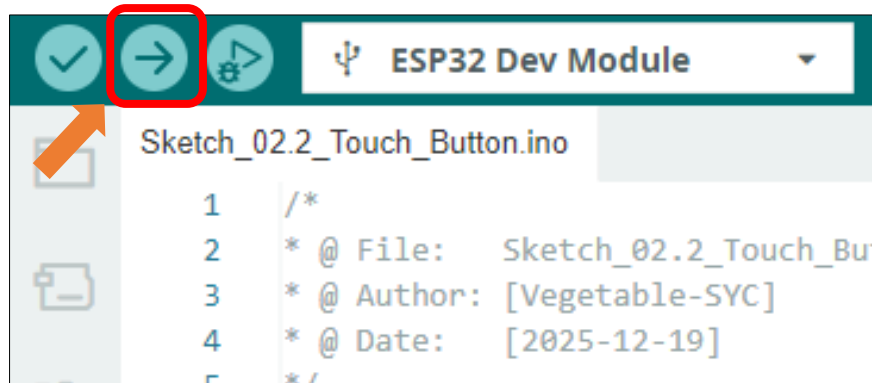
Determine if the touch duration exceeds the long-press time threshold.

```

44  if(millis() - Touch_Time > PRESS_Time)

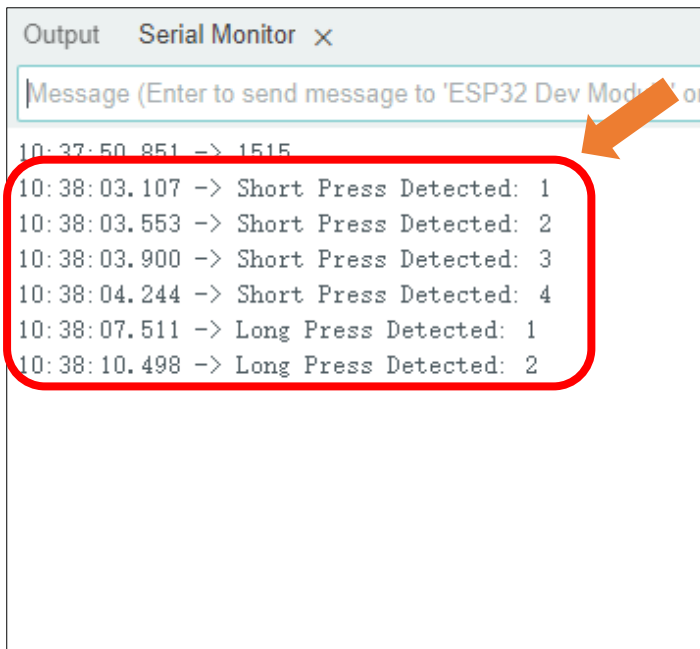
```

The purpose of this code is to display data on the serial monitor. Click “**Upload**” to upload the code to Freenove ESP32 Mini TV.



After downloading the code, open the serial port monitor, and set the baud rate to **115200**

Touch on the capacitive area of the Freenode ESP32 Mini TV, if the duration exceeds 3 seconds, it is classified as a long press; otherwise, it is considered a short press.



**Please note:** This chapter does not involve the use of the screen. After the code for this chapter, the screen may not light up, which is normal and not a hardware malfunction. If you need to verify whether the screen is functioning properly, please refer to [Chapter 6](#) for testing.

## Chapter 3 EEPROM

### Project 3.1 EEPROM

#### Component List

|                                                                                                                                                                                                                      |                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  A yellow, cube-shaped electronic device with a black screen on the front and a circular logo on top. | <p>USB cable x1</p>  A black USB cable with a standard USB-A connector on one end and a micro-USB connector on the other. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Component knowledge

##### EEPROM

The primary function of EEPROM (Electrically Erasable Programmable Read-Only Memory) is to enable non-volatile data storage, ensuring critical information remains preserved after power loss or system restart. In the ESP32 architecture, EEPROM is not a physical chip but a logical storage space emulated using a section of the internal Flash memory. Due to the limited erase-write cycles of Flash memory, frequent repeated write operations should be avoided in practical applications.

#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “Sketch\_03.1\_EEPROM” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_03.1\_EEPROM.ino”.

### Sketch\_03.1\_EEPROM

```

1  #include "EEPROM.h"
2
3  #define EEPROM_SIZE 1
4  int addr = 0;
5  void setup() {
6      // put your setup code here, to run once:
7      Serial.begin(115200);
8      EEPROM.begin(EEPROM_SIZE);
9      int val = EEPROM.read(addr) + 1;
10     delay(5);
11     EEPROM.write(addr, val);
12     EEPROM.commit();
13     delay(1000);
14     Serial.print("The data in EEPROM(0) is:");
15     Serial.println(EEPROM.read(addr));
16 }
17
18 void loop() {
19     // put your main code here, to run repeatedly:
20
21 }
```

### Code Explanation

Introduce the necessary libraries.

```
1  #include "EEPROM.h"
```

Define the size of non-volatile storage space (unit: bytes).

```
3  #define EEPROM_SIZE 1
```

Define the target starting address of the storage operation.

```
7  int addr = 0;
```

Set the baud rate to 115200.

```
3  Serial.begin(115200);
```

Initialize EEPROM and allocate buffer size

```
8  EEPROM.begin(EEPROM_SIZE);
```

Read the target address data and increment it by 1

```
9  int val = EEPROM.read(addr) + 1;
```

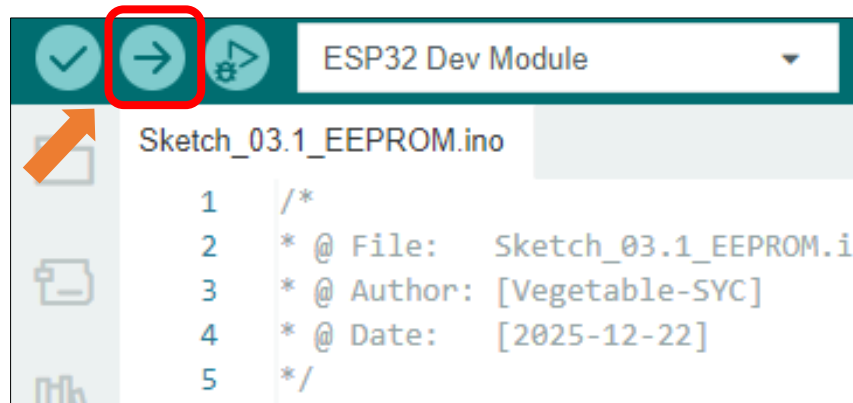
Write the new data to the memory buffer.

```
11 EEPROM.write(addr, val);
```

Update data to EEPROM.

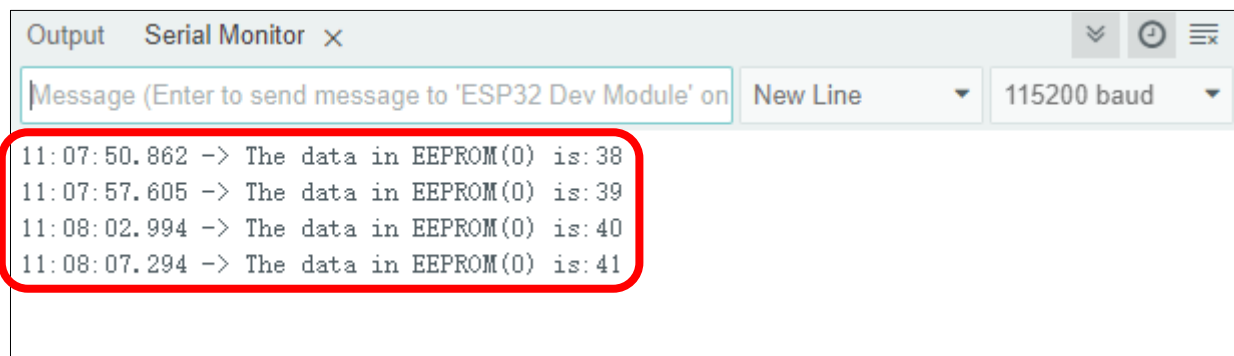
```
12 EEPROM.commit();
```

The purpose of this code is to display data on the serial monitor. Click “**Upload**” to upload the code to Freenove ESP32 Mini TV.



After downloading the code, open the serial port monitor, and set the baud rate to **115200**

After viewing the current value on the serial monitor, disconnect and then reconnect the USB cable. You will observe that the value stored in the EEPROM persists through the power cycle. Upon reinitialization after reboot, the value increments from its previously stored state.



**Please note:** This chapter does not involve the use of the screen. After the code for this chapter is uploaded, the screen may not light up, which is normal and not a hardware malfunction. If you need to verify whether the screen is functioning properly, please refer to [Chapter 6](#) for testing.

## Chapter 4 BLE

### Project 4.1 BLE USART

#### Component List

|                                                                                                                 |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  | <p>USB cable x1</p>  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|

#### Component knowledge

##### BLE (Bluetooth Low Energy)

Low Energy Bluetooth (BLE) is a new feature introduced in the Bluetooth 4.0 specification, specifically designed for low-power devices and suitable for applications involving intermittent transmission of small amounts of data.

The BLE architecture follows a client-server model and consists of the following key components:

**GATT (Generic Attribute Protocol):** The foundational protocol for BLE communication, defining a hierarchical data structure of services and characteristics.

**Service:** A collection of data that performs a specific function or feature, containing one or more characteristics.

**Characteristic:** A specific data point within a service, consisting of a value and descriptors.

**UUID:** A 128-bit universally unique identifier used to distinguish services and characteristics.

BLE device can function simultaneously as a Peripheral (providing services) and a Central (connecting to peripherals). In this section, we will learn how to configure the Freenove ESP32 Mini TV as a peripheral device to provide services.

## Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “Sketch\_04.1\_BLE\_USART” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_04.1\_BLE\_USART.ino”.

Sketch\_04.1\_BLE\_USART

The following is the program code:

```
1  #include "BLEDevice.h"
2  #include "BLEServer.h"
3  #include "BLEUtils.h"
4  #include "BLE2902.h"
5  #include "String.h"
6
7  BLECharacteristic *pCharacteristic;
8  bool deviceConnected = false;
9  uint8_t txValue = 0;
10 long lastMsg = 0;
11 String rxload="Test\n";
12
13 #define SERVICE_UUID          "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
14 #define CHARACTERISTIC_UUID_RX "6E400002-B5A3-F393-E0A9-E50E24DCCA9E"
15 #define CHARACTERISTIC_UUID_TX "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"
16
17 class MyServerCallbacks: public BLEServerCallbacks {
18     void onConnect(BLEServer* pServer) {
```

```

19     deviceConnected = true;
20 };
21 void onDisconnect(BLEServer* pServer) {
22     deviceConnected = false;
23 }
24 };
25
26 class MyCallbacks: public BLECharacteristicCallbacks {
27     void onWrite(BLECharacteristic *pCharacteristic) {
28         String rxValue = pCharacteristic->getValue();
29         if (rxValue.length() > 0) {
30             rxload="";
31             for (int i = 0; i < rxValue.length(); i++){
32                 rxload +=(char)rxValue[i];
33             }
34         }
35     }
36 };
37
38 void setupBLE(String BLEName) {
39     const char *ble_name=BLEName.c_str();
40     BLEDevice::init(ble_name);
41     BLEServer *pServer = BLEDevice::createServer();
42     pServer->setCallbacks(new MyServerCallbacks());
43     BLEService *pService = pServer->createService(SERVICE_UUID);
44     pCharacteristic=
45     pService->createCharacteristic(CHARACTERISTIC_UUID_TX, BLECharacteristic::PROPERTY_NOTIFY);
46     pCharacteristic->addDescriptor(new BLE2902());
47     BLECharacteristic *pCharacteristic =
48     pService->createCharacteristic(CHARACTERISTIC_UUID_RX, BLECharacteristic::PROPERTY_WRITE);
49     pCharacteristic->setCallbacks(new MyCallbacks());
50     pService->start();
51     pServer->getAdvertising()->start();
52     Serial.println("Waiting a client connection to notify...");
53 }
54
55 void setup() {
56     Serial.begin(115200);
57     setupBLE("ESP32_BLE");
58 }
59
60 void loop() {
61     long now = millis();
62     if (now - lastMsg > 100) {

```



```

63     if (deviceConnected&&rxload.length()>0) {
64         Serial.println(rxload);
65         rxload="";
66     }
67     if(Serial.available()>0){
68         String str=Serial.readString();
69         const char *newValue=str.c_str();
70         pCharacteristic->setValue(newValue);
71         pCharacteristic->notify();
72     }
73     lastMsg = now;
74 }
75 }

```

### Code Explanation

Include the necessary header libraries.

```

1  #include "BLEDevice.h"
2  #include "BLEServer.h"
3  #include "BLEUtils.h"
4  #include "BLE2902.h"
5  #include "String.h"

```

Define service UUID and characteristic UUID.

```

13 #define SERVICE_UUID           "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
14 #define CHARACTERISTIC_UUID_RX "6E400002-B5A3-F393-E0A9-E50E24DCCA9E"
15 #define CHARACTERISTIC_UUID_TX "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"

```

The MyServerCallbacks class handles device connection and disconnection events and updates the deviceConnected status.

```

17 class MyServerCallbacks: public BLEServerCallbacks {
18     void onConnect(BLEServer* pServer) {
19         deviceConnected = true;
20     };
21     void onDisconnect(BLEServer* pServer) {
22         deviceConnected = false;
23     }
24 };

```

The MyServerCallbacks class handles the received data and save them to the rxload string.

```

26 class MyCallbacks: public BLECharacteristicCallbacks {
27     void onWrite(BLECharacteristic *pCharacteristic) {
28         String rxValue = pCharacteristic->getValue();
29         if (rxValue.length() > 0) {
30             rxload="";
31             for (int i = 0; i < rxValue.length(); i++){
32                 rxload +=(char)rxValue[i];
33             }
34         }

```

```

35     }
36 };

```

Set the baud rate to 115200.

```

56 Serial.begin(115200);

```

BLE Device Initialization, Service Creation, and Characteristic Setup

```

57 setupBLE("ESP32_BLE");

```

The Loop function check the connection status and serial data every 100 milliseconds.

```

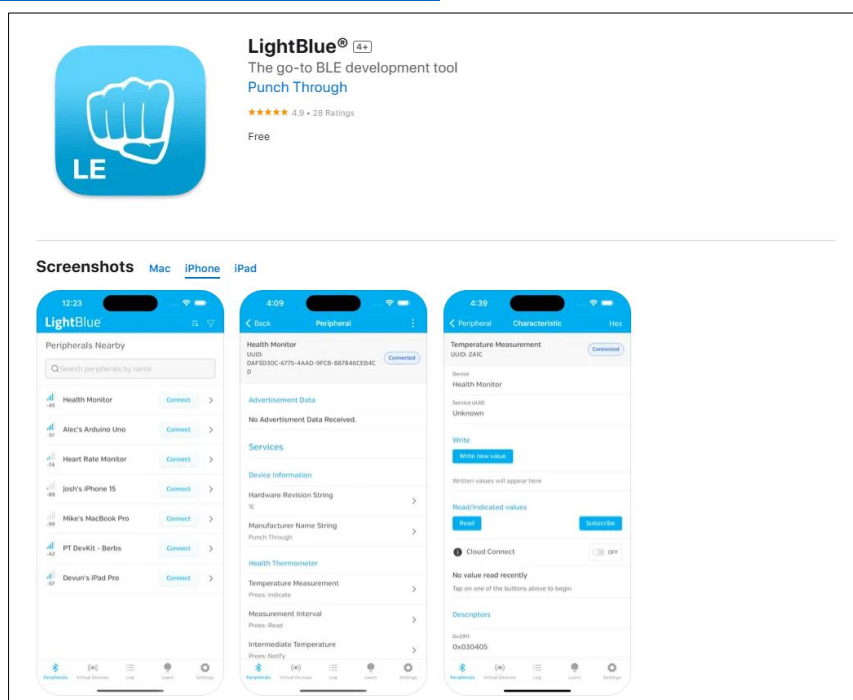
60 void loop() {
61     long now = millis();
62     if (now - lastMsg > 100) {
63         if (deviceConnected && rxload.length() > 0) {
64             Serial.println(rxload);
65             rxload="";
66         }
67         if (Serial.available() > 0) {
68             String str = Serial.readString();
69             const char *newValue = str.c_str();
70             pCharacteristic->setValue(newValue);
71             pCharacteristic->notify();
72         }
73         lastMsg = now;
74     }
75 }

```

## LightBlue

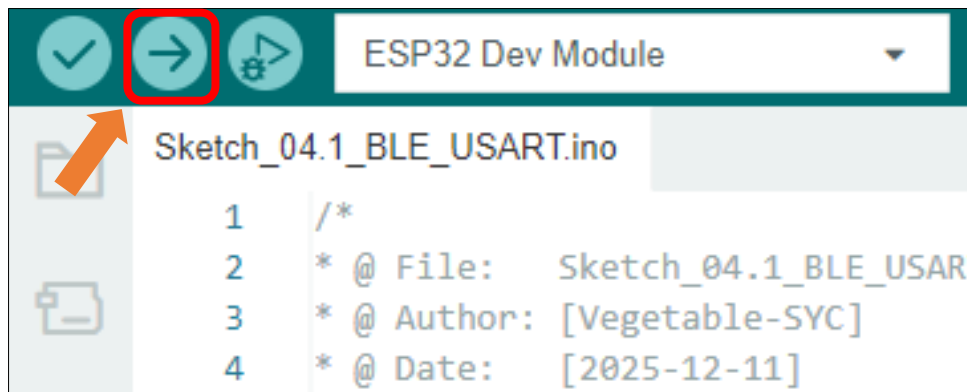
If you do not have this software installed on your phone, you can refer to this link:

<https://apps.apple.com/us/app/lightblue/id557428110>

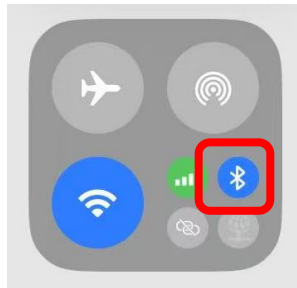


Click **“Upload”** to upload the code to Freenove ESP32 Mini TV.

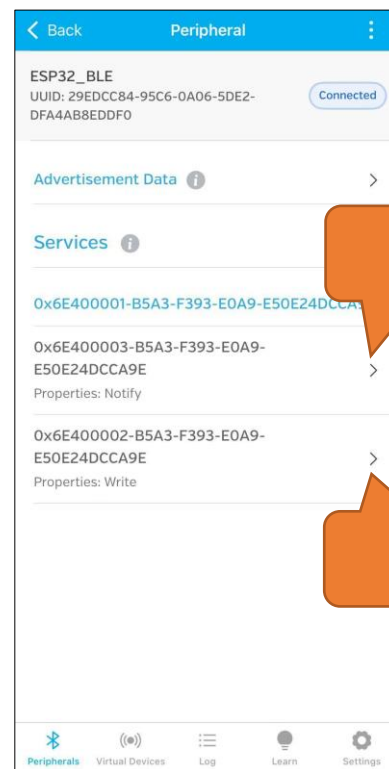
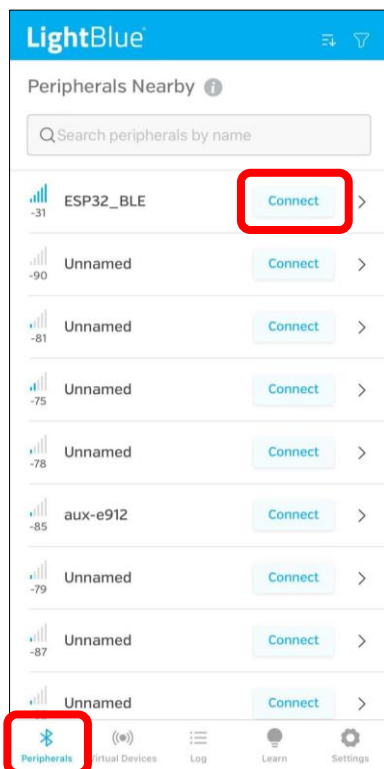
Please note: This chapter does not involve the use of the screen. After the code for this chapter, the screen may not light up, which is normal and not a hardware malfunction. If you need to verify whether the screen is functioning properly, please refer to [Chapter 6](#) for testing.



Turn ON Bluetooth on your phone, and open the LightBlue APP.

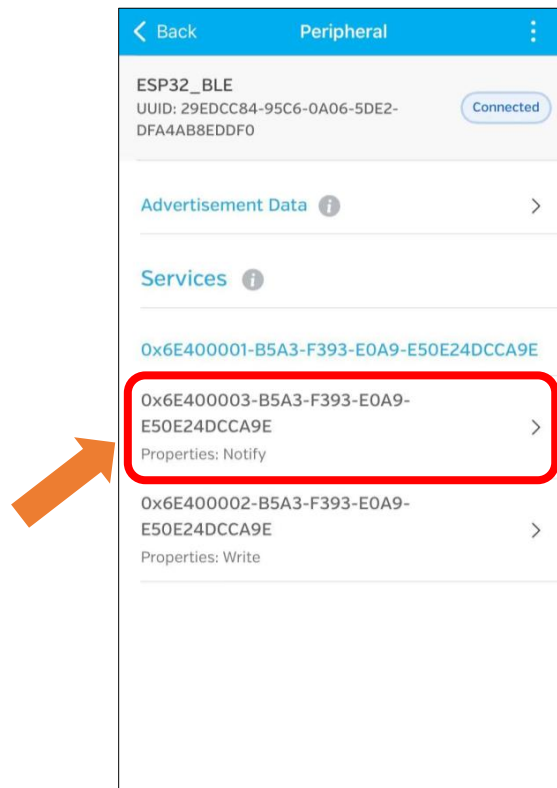


In the Scan page, swipe down to refresh the name of Bluetooth that the phone searches for. Click the Connection button of ESP32\_BLE.

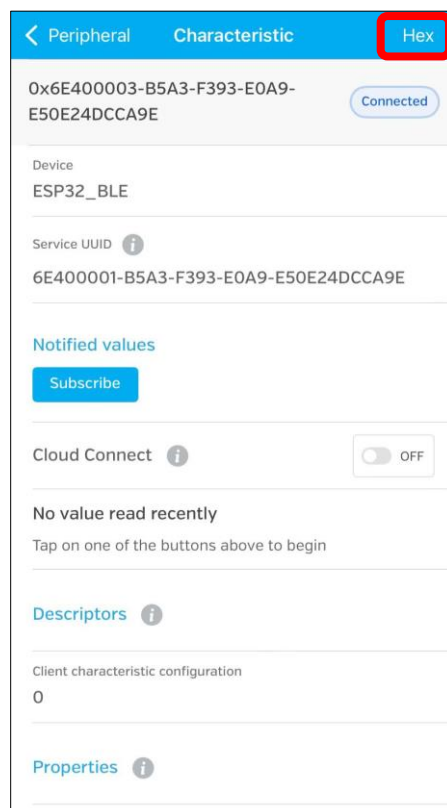


## Receiving Data

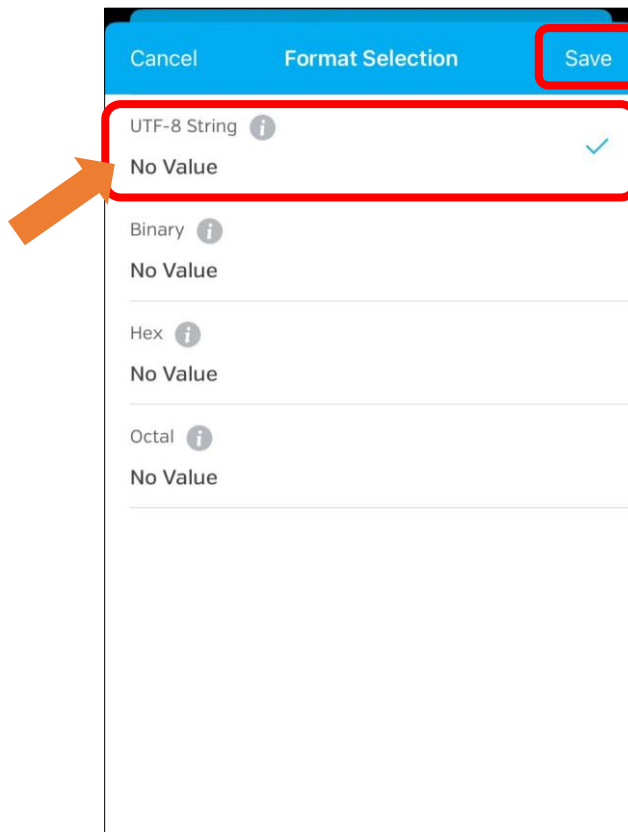
Click Notify



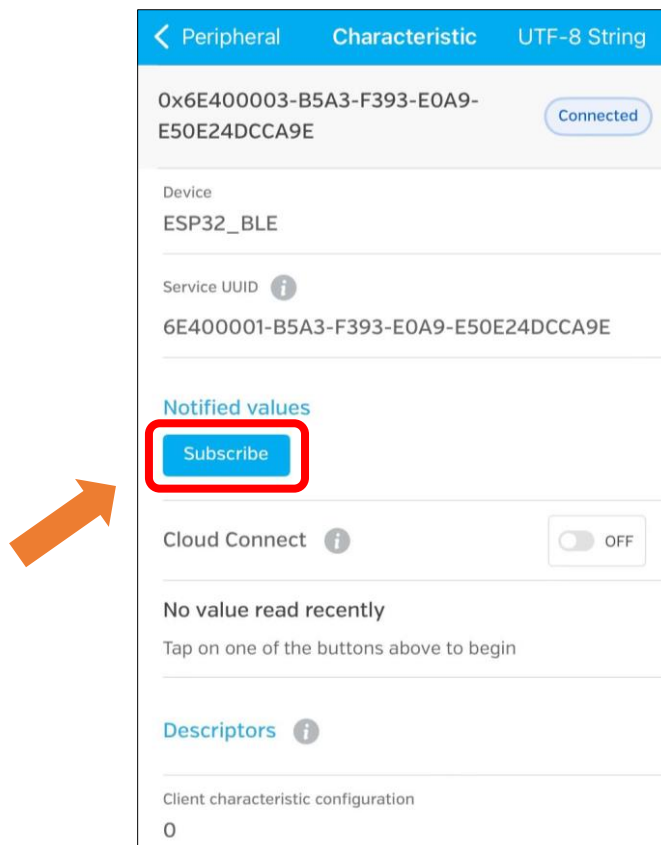
Click the top right corner to change the string type.



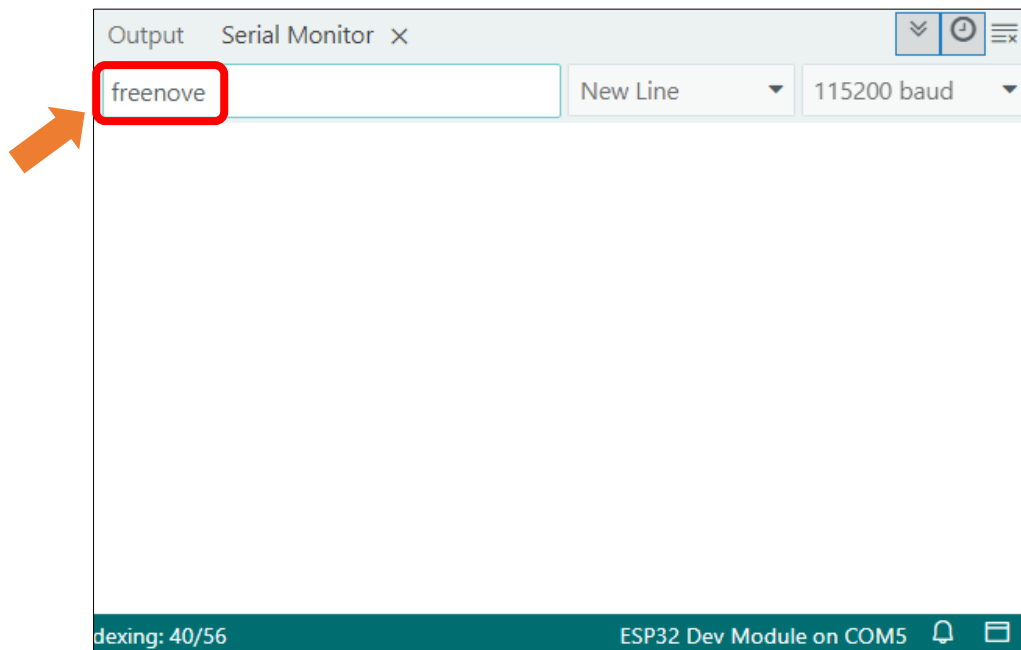
Select UTF-8 String, and click "Save".



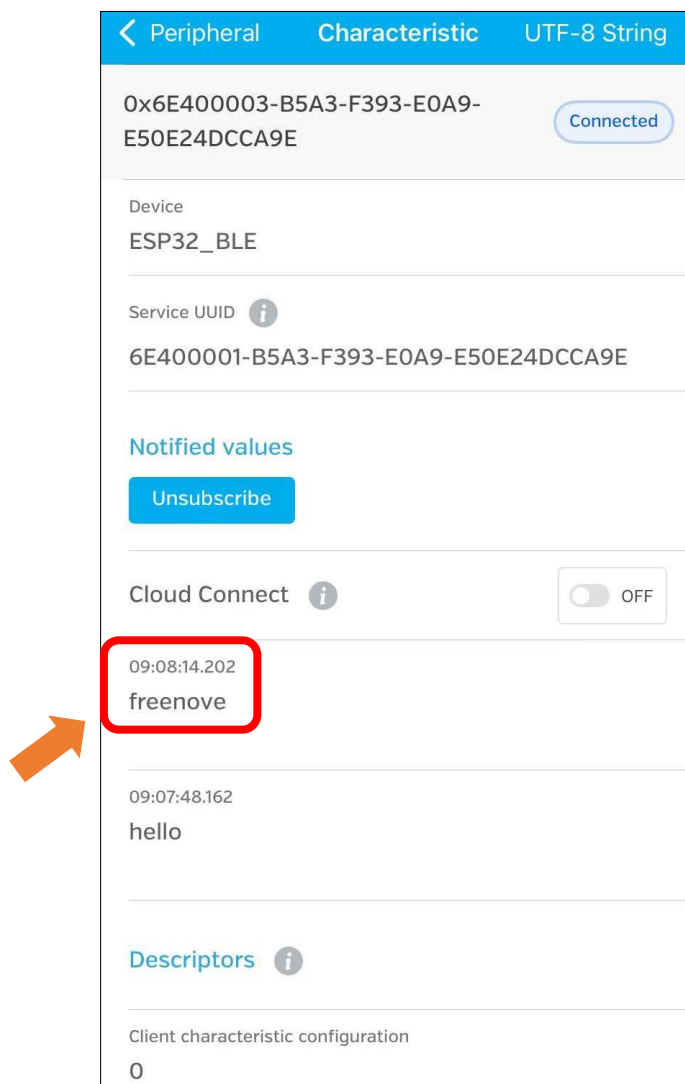
Click Subscribe



Send data on the serial monitor.

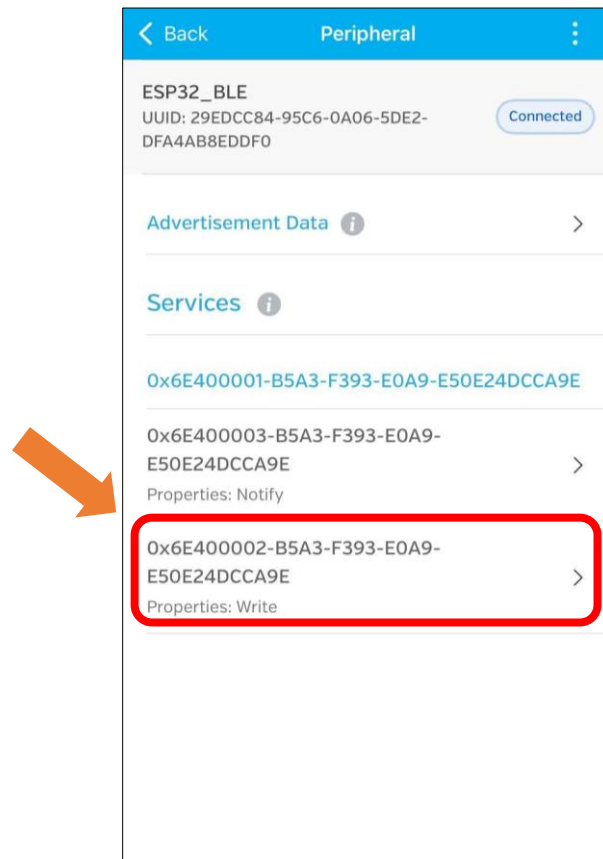


Data will be received on the LightBlue app.

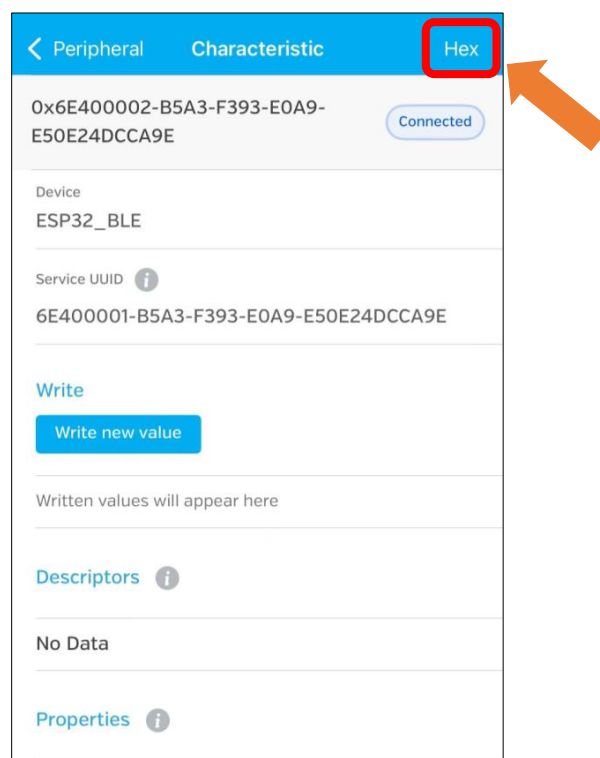


## Sending Data

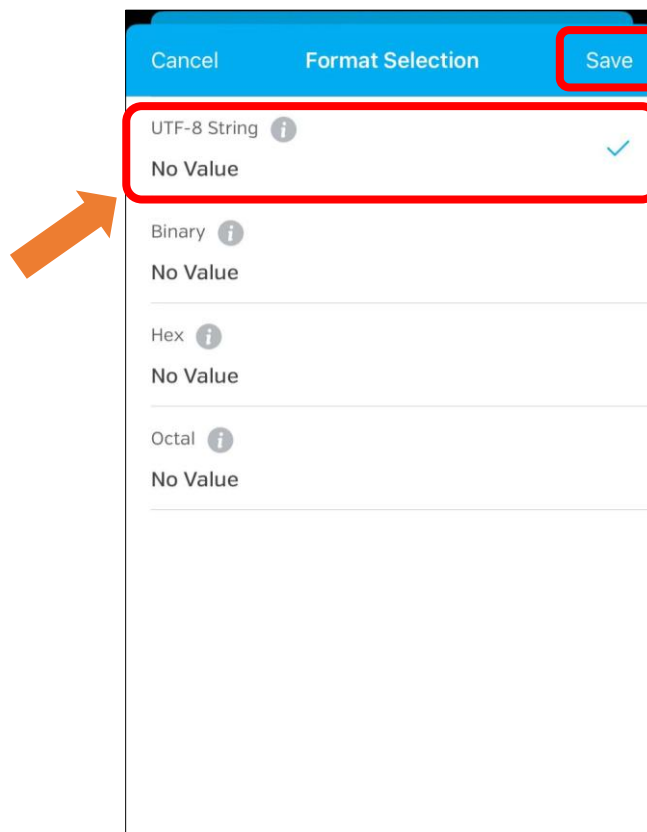
Click Write



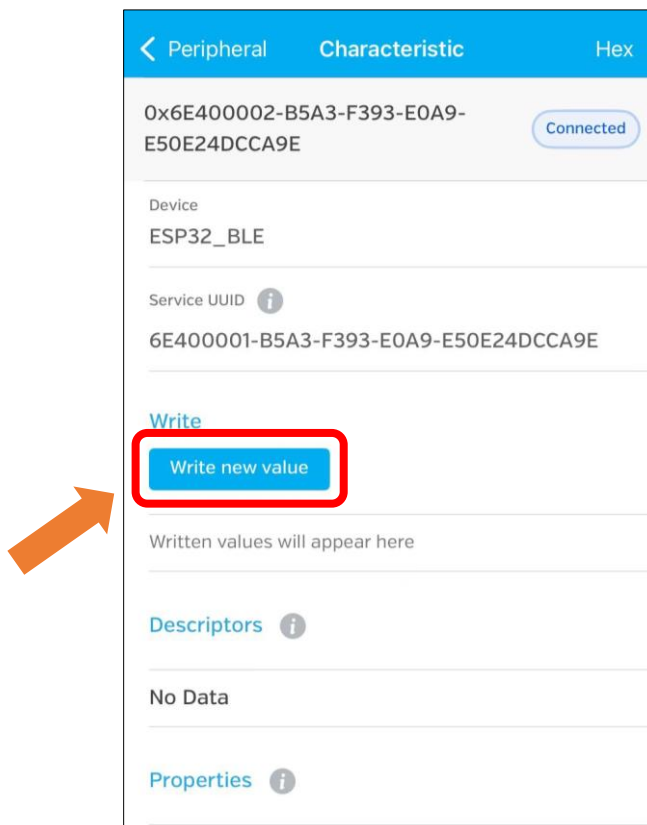
Click the top right corner to change the string type.



Select UTF-8 String and click “Save”.

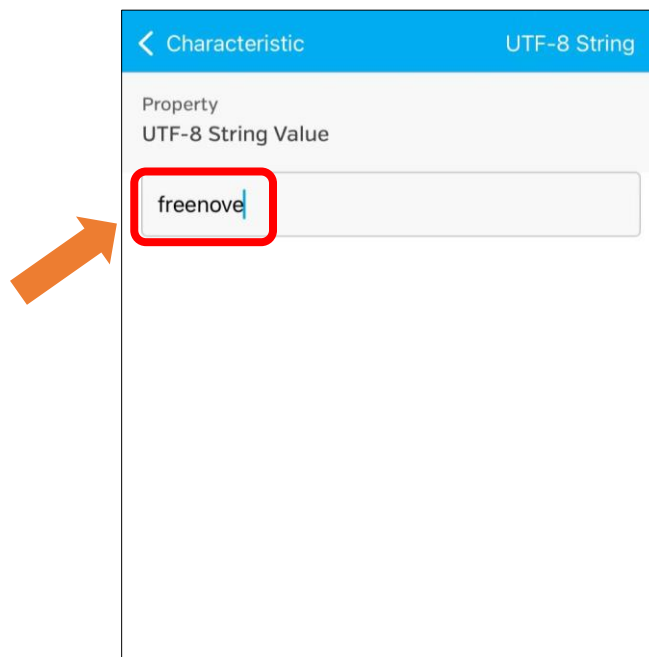


Click “Write new value”

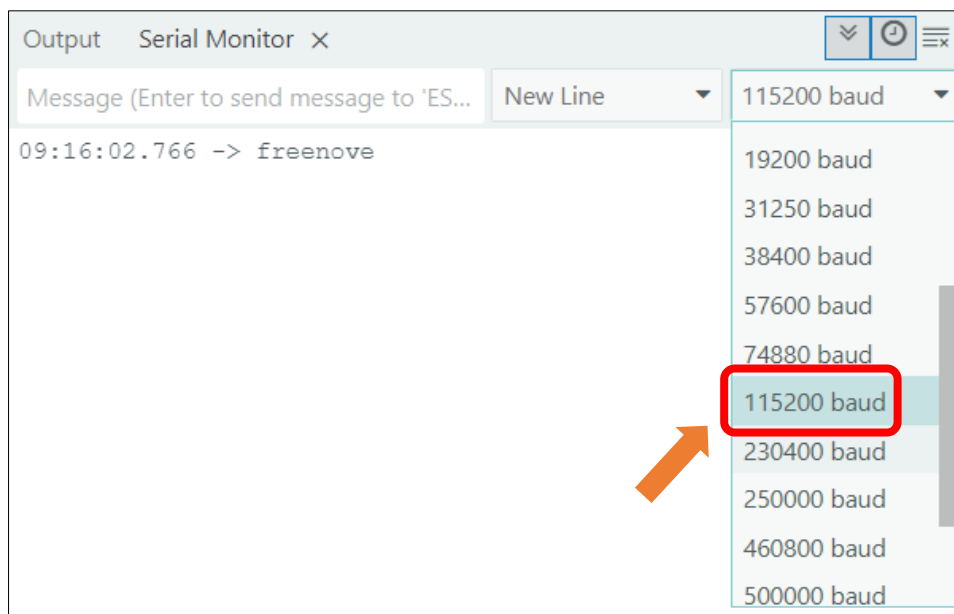




Enter the messages to send.



Set the baud rate to 115200, and the serial monitor will print the data received.



# Chapter 5 WIFI

## Project 5.1 WIFI Web Servers LED

### Component List

|                                                                                                                 |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  | <p>USB cable x1</p>  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|

### Component Knowledge

#### Wi-Fi

Wi-Fi is a wireless LAN (WLAN) technology compliant with the IEEE 802.11 standard, enabling devices to transmit data or connect to the internet via radio waves within short-range coverage (typically up to tens of meters). In embedded systems, Wi-Fi modules provide wireless communication capabilities, allowing devices to connect to routers, access cloud services, or interact with other devices.

Its core features include:

- Wireless connectivity (eliminating the need for physical cabling)
- Moderate to high-speed data transfer (depending on protocol versions such as 802.11n/ac)
- Secure communication (ensured by encryption protocols like WPA2/WPA3)

In embedded development, Wi-Fi is one of the key technologies for enabling IoT (Internet of Things) device connectivity.

#### Web

The Web (World Wide Web) is an internet-based information system that integrates global resources through hypertext links. In the embedded field, Web technology refers to devices equipped with lightweight built-in web servers, allowing users to remotely access the device interface via browsers (such as Chrome or Edge). By entering the device's IP address, users can open an interactive page to perform functions like status monitoring, parameter configuration, or firmware upgrades. Its core value lies in cross-platform compatibility (no need for dedicated software installation) and standardized protocols (HTTP/HTTPS), making it a mainstream solution for remote management of embedded devices.

#### HTML & CSS & JavaScript

- tags to define page elements, forming the foundational framework. In embedded web interfaces,

HTML describes the layout of components such as device status displays and configuration forms.

- **CSS (Cascading Style Sheets)** controls the visual presentation of web pages, including colors, fonts, spacing, and responsive layouts. Through CSS rules, developers style HTML elements into intuitive interactive interfaces. The combination of HTML and CSS enables the creation of lightweight device control pages without complex graphics libraries, reducing resource overhead in embedded systems.
- **JavaScript (a scripting language)** adds dynamic behavior and interaction logic to web pages. In embedded web interfaces, JavaScript plays a crucial role: it responds to user actions and enables asynchronous communication with the device backend through technologies like **AJAX** or **WebSocket**. This allows the webpage to fetch real-time device status data, update displays without refreshing, and send control commands or configuration parameters—delivering a smooth and efficient remote management experience.

## Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “**Sketch\_04.1\_BLE\_USART**” folder under “**Freenove\_ESP32\_Mini\_TV\Sketch**” and double-click “**Sketch\_04.1\_BLE\_USART.ino**”.

**Sketch\_04.1\_BLE\_USART**

The following is the program code:

```
1  #include <WiFi.h>
2  #include <WebServer.h>
3  #include "web.h"
4  // -----
5  // WiFi configuration
```

```
6 // -----
7 const char* ssid = "*****";
8 const char* password = "*****";
9
10 // Create a web server instance listening on HTTP port 80.
11 WebServer server(80);
12
13 // Serve an HTML page to the client.
14 void handleRoot() {
15     server.send(200, "text/html", index_html);
16 }
17
18 // Decode the POST request data.
19 void handleSubmit() {
20     if (server.hasArg("msg")) {
21         String receivedMsg = server.arg("msg");
22         Serial.println("-----");
23         Serial.print("Received message: ");
24         Serial.println(receivedMsg);
25         Serial.println("-----");
26
27         server.send(200, "text/plain", "OK");
28     } else {
29         server.send(400, "text/plain", "Bad Request");
30     }
31 }
32
33 // Process 404 (Not Found) error.
34 void handleNotFound() {
35     server.send(404, "text/plain", "Not found");
36 }
37
38 void setup() {
39     Serial.begin(115200);
40     delay(1000);
41
42     Serial.println();
43     Serial.print("Connect to WiFi: ");
44     Serial.println(ssid);
45
46     // Set up the WiFi connection.
47     WiFi.begin(ssid, password);
48
49     // Wait until WiFi connects successfully
```

```

50 while (WiFi.status() != WL_CONNECTED) {
51     delay(500);
52     Serial.print(".");
53 }
54
55 // Print the assigned local IP address
56 Serial.println("\nWiFi connected successfully!");
57 Serial.print("Access address: http://");
58 Serial.println(WiFi.localIP());
59
60 server.on("/", handleRoot);
61 server.on("/submit", handleSubmit);
62 server.onNotFound(handleNotFound);
63
64 // Start the HTTP server
65 server.begin();
66 Serial.println("HTTP Server is running");
67 }
68
69 void loop() {
70     // Continuously listen for and handle HTTP connection requests from clients
71     server.handleClient();
72 }

```

### Code Explanation

Include the necessary header files.

```

1  #include <WiFi.h>
2  #include <WebServer.h>
3  #include "web.h"

```

Define WiFi name and password.

```

7  const char* ssid = "*****";
8  const char* password = "*****";

```

Create a web server instance that listens on HTTP port 80..

```

11 WebServer server(80);

```

Connect WIFI.

```

47 WiFi.begin(ssid, password);
48
49 // Wait until WiFi connects successfully
50 while (WiFi.status() != WL_CONNECTED) {
51     delay(500);
52     Serial.print(".");
53 }

```

Continuously listen for and process HTTP connection requests from clients.

```

71 server.handleClient();

```

Input correct WiFi(2.4GHz) SSID and password.

```

9 // WiFi credentials
10 const char* ssid      = "*****";
11 const char* password = "*****";

```

Click **Upload** to upload the code to Freenove ESP32 Mini TV, and set the baudrate to 115200.



When the serial monitor prints the following information, it means the network connection is successful. Use a mobile phone or computer browser to open the IP address printed by the serial monitor.

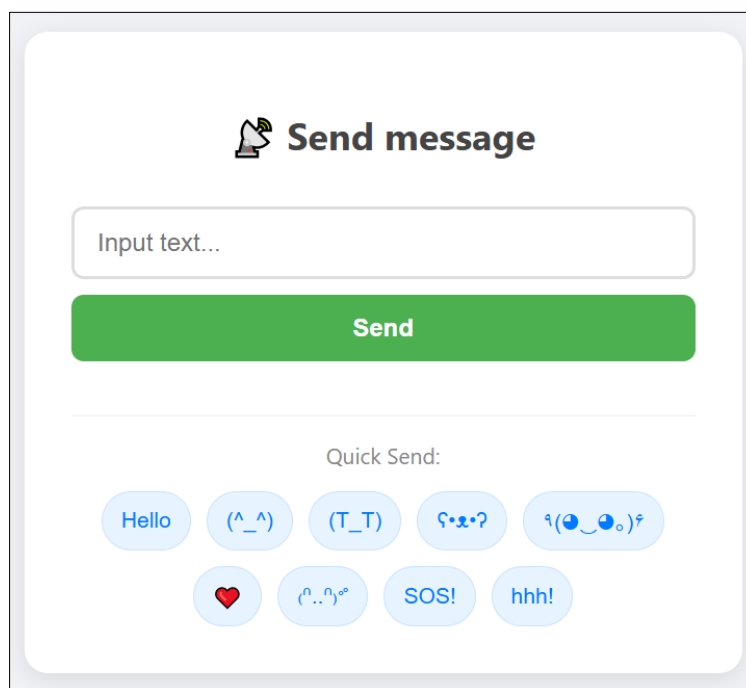
**Please note: If it keeps printing dots, please confirm whether the WiFi is in the 2.4G band.**

```

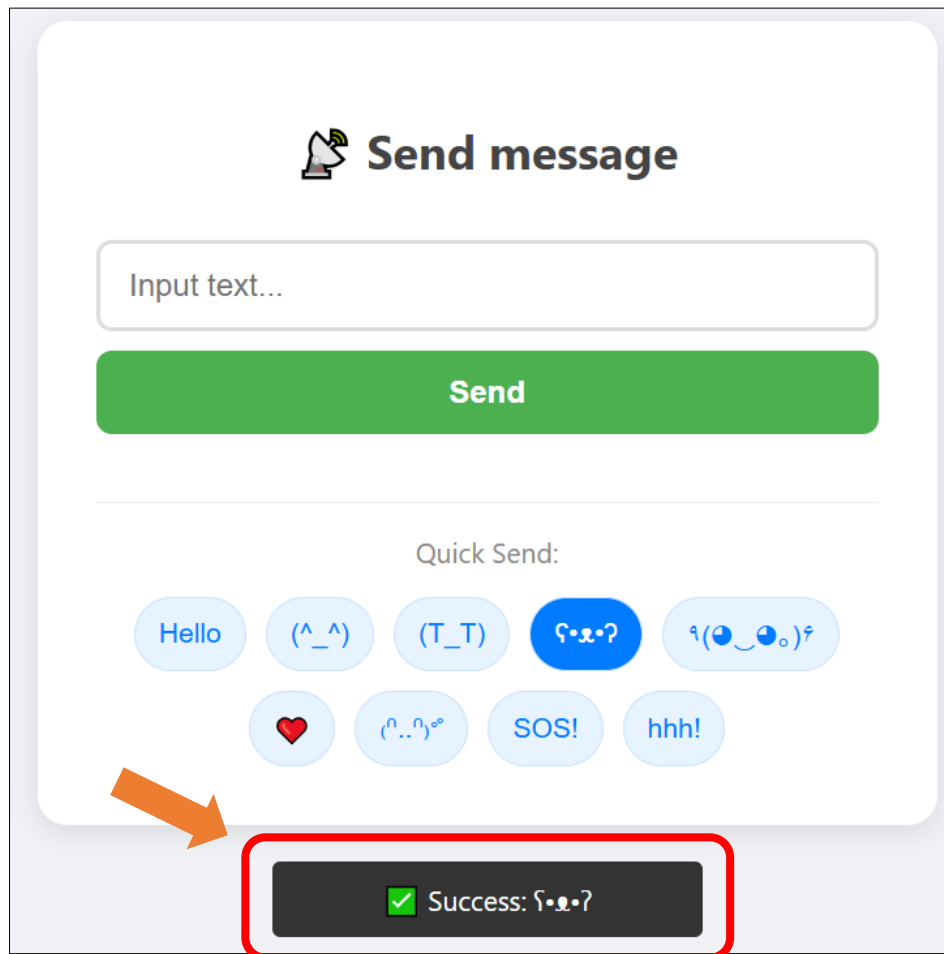
13:46:08.763 -> Connect to WiFi: FYI_2.4G
13:46:09.367 -> ...
13:46:10.332 -> WiFi connected successfully!
13:46:10.368 -> Access address http://192.168.1.26
13:46:10.368 -> HTTP Server is running

```

The browser will display the following screen. You can click the shortcut below or enter your text in the text box and click "Send".



A notification message will pop up when the message is successfully sent.



The messages you sent will print on the serial monitor.

| Output                                                           | Serial Monitor | x                              |
|------------------------------------------------------------------|----------------|--------------------------------|
| Message (Enter to send message to 'ESP32 Dev Module' on 'COM34') |                |                                |
| 14:50:30.121                                                     | ->             | -----                          |
| 14:50:30.121                                                     | ->             | Received message: Hello World! |
| 14:50:30.121                                                     | ->             | -----                          |
| 14:50:37.371                                                     | ->             | -----                          |
| 14:50:37.371                                                     | ->             | Received message: Freenove     |
| 14:50:37.371                                                     | ->             | -----                          |
| 14:50:39.626                                                     | ->             | -----                          |
| 14:50:39.626                                                     | ->             | Received message: ❤            |
| 14:50:39.626                                                     | ->             | -----                          |
| 14:50:40.484                                                     | ->             | -----                          |
| 14:50:40.484                                                     | ->             | Received message: ಠ_ಠ?         |
| 14:50:40.484                                                     | ->             | -----                          |

### Reference

#### server.begin()

This function starts the server to monitor the port.

#### server.available()

This function is used to detect and return the connected client object.

## Chapter 6 TFT

### Project 6.1 TFT\_Display

#### Component List

|                                                                                                                 |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  | <p>USB cable x1</p>  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|

#### Component knowledge

##### TFT Display

TFT (Thin Film Transistor) is an electronic component that serves as the foundation for TFT displays, the mainstream display technology in modern laptops and desktop computers. In these displays, each individual liquid crystal pixel is controlled by its own dedicated thin-film transistor embedded directly behind it. This architecture classifies TFT screens as a form of active-matrix LCD (AMLCD) technology.

As one of the finest LCD color displays available, TFT screens offer superior performance characteristics including rapid response times, exceptional brightness levels, and outstanding contrast ratios.

#### Circuit

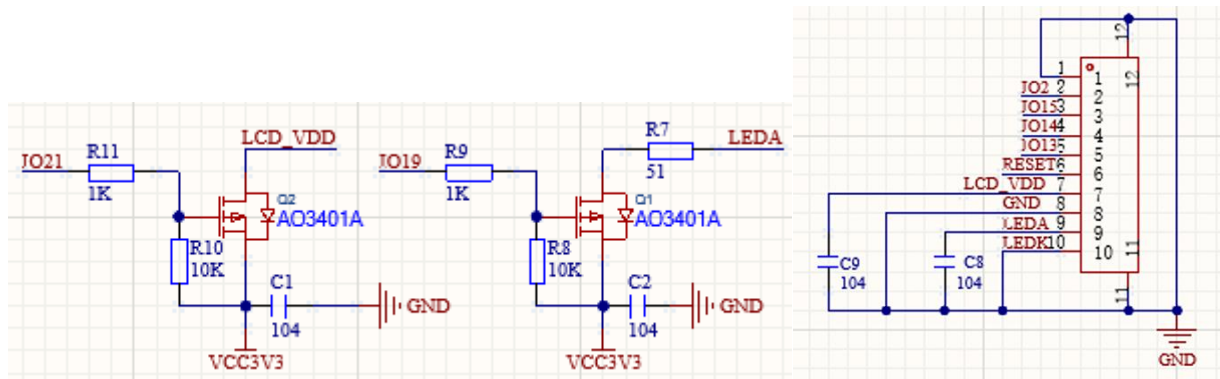
Connect Freenove ESP32 Mini TV to the computer with USB cable.





## Schematic

The schematic diagram for the TFT screen is shown below. It is important to note that the screen's power supply (LCD\_VDD) and backlight (LEDA) are driven by two independent GPIO pins, and both are active-low. The complete schematic diagram is located in the same directory as this tutorial document.



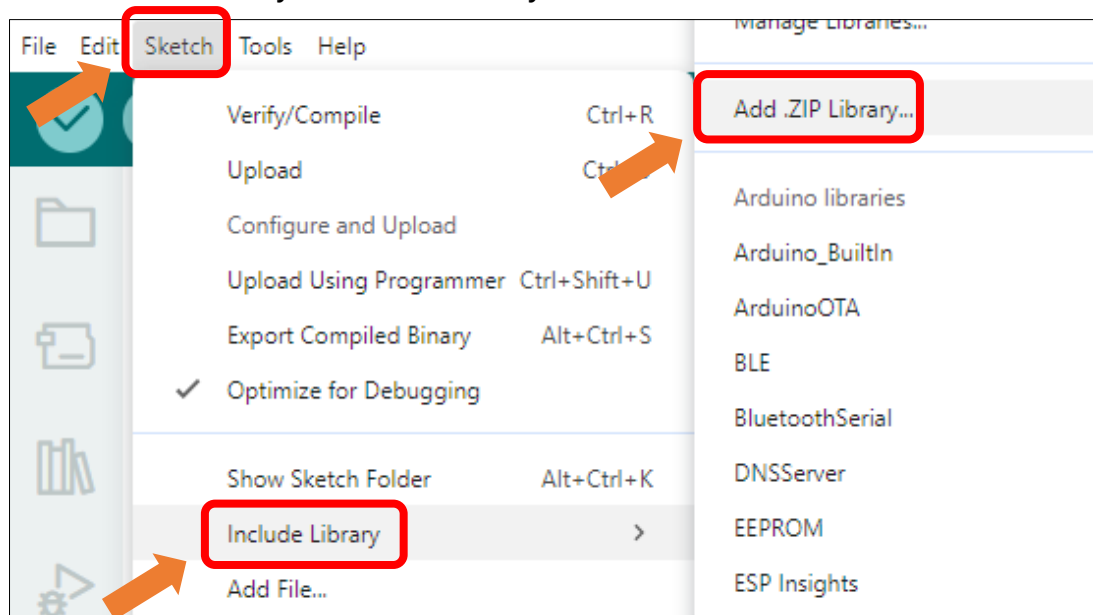
## Sketch

Open "Sketch\_06.1\_TFT\_Clock.ino" folder under "Freenove\_ESP32\_Mini\_TV\Sketches" and double-click "Sketch\_06.1\_TFT\_Clock.ino".

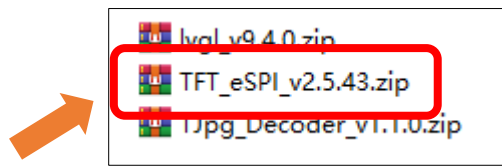
### Install Libraries

**Caution:** It is critical to follow the installation steps below precisely. Our provided library comes pre-configured for the display. Using a version downloaded online or keeping an old one will prevent the screen from working. Please ensure you completely replace any other version with this one to guarantee correct driver operation.

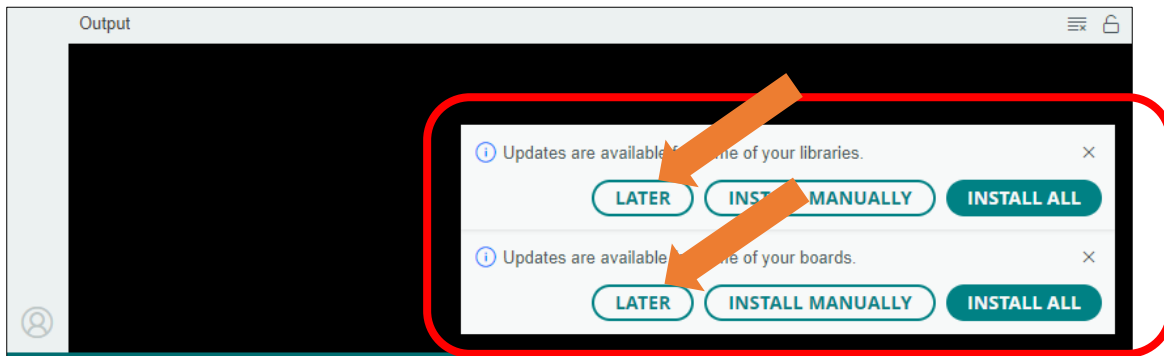
Click **Sketch** -> **Include Library** -> **Add .ZIP Library...**



Install **TFT\_eSPI\_v2.5.43.zip**



**Note:** Some libraries are not the latest version. Please do not update them even if it prompts every time you open the IDE. Just click LATER. Otherwise, it may lead to compilation failure.



#### Sketch\_06.1\_TFT\_Clock

The following is the program code:

```

1  #include <TFT_eSPI.h>
2
3  const int TFT_VCC = 21;
4
5  // Initialize TFT instance
6  TFT_eSPI tft = TFT_eSPI();
7
8  uint16_t colors[] = {
9      TFT_RED,      // Red
10     TFT_GREEN,    // Green
11     TFT_BLUE,     // Blue
12     TFT_YELLOW,   // Yellow
13     TFT_PURPLE    // Purple
14 };
15
16 int colorIndex = 0;
17
18 void setup() {
19     Serial.begin(115200);
20
21     // Power on and initialize display
22     pinMode(TFT_VCC, OUTPUT);
23     digitalWrite(TFT_VCC, LOW); // Active Low enable
24
25     tft.begin();
26     tft.fillScreen(TFT_BLACK);

```

```

27
28     Serial.println("Test Start...");
29 }
30
31 void loop() {
32     // Fill screen with the current color
33     tft.fillScreen(colors[colorIndex]);
34
35     // Configure text settings
36     tft.setTextColor(TFT_BLACK);          // Set text color
37     tft.setTextDatum(MC_DATUM);           // Set text datum to middle center
38     tft.drawString("TFT Test", 120, 100, 4); // Draw text string
39
40     // Update index
41     colorIndex++;
42     if (colorIndex >= 5) {
43         colorIndex = 0;
44     }
45
46     // Delay for 1 second
47     delay(1000);
48 }

```

#### Code Explanation

Include the necessary header files.

```
1  #include <TFT_eSPI.h>
```

Declare the TFT screen object.

```
6  TFT_eSPI tft = TFT_eSPI();
```

Enable screen power supply pin.

```
22 pinMode(TFT_VCC, OUTPUT);
23 digitalWrite(TFT_VCC, LOW);
```

Initialize the TFT screen.

```
19 tft.begin();
20 tft.fillScreen(TFT_BLACK);
```

Fill with the color corresponding to the index.

```
33 tft.fillScreen(colors[colorIndex]);
```

Set text style and draw.

```
38 tft.setTextColor(TFT_BLACK);          // Set text color
39 tft.setTextDatum(MC_DATUM);           // Set text datum to middle center
40 tft.drawString("TFT Test", 120, 100, 4); // Draw text string
```

Update the index.

```
41 colorIndex++;
42 if (colorIndex >= 5) {
43     colorIndex = 0;
44 }
```

Click “**Upload**” to upload the code to Freenove ESP32 Mini TV.



The TFT screen will change colors in the order of red -> green -> blue -> yellow -> purple, while displaying the text "TFT Test".



## Project 6.2 TFT Picture

### Component List

|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  A yellow, cube-shaped electronic device with a black rectangular screen on the front face. The top face has a circular logo with a hand pointing to a screen and the text 'ESP32' and 'TFT-2828'. | <p>USB cable x1</p>  A black USB cable with a standard USB-A connector on one end and a custom connector on the other. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Component knowledge

#### DMA

DMA, or Direct Memory Access, is a hardware feature that allows peripherals to transfer data to and from memory without needing the CPU to be directly involved, dramatically improving overall system efficiency while minimizing processor workload.

The core mechanism of DMA relies on a dedicated DMA controller taking over data transfer tasks. The CPU only needs to initialize the transfer parameters before offloading the operation, allowing computation and I/O operations to proceed in parallel.

### Circuit

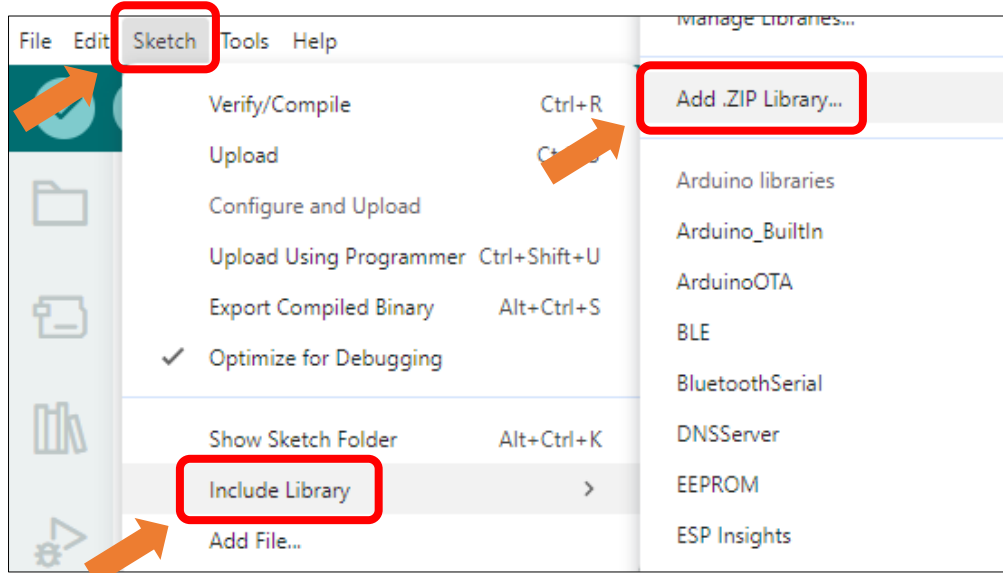
Connect Freenove ESP32 Mini TV to the computer with USB cable.



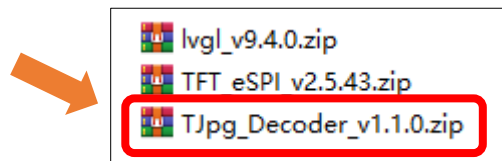
## Sketch

### Install Libraries

Click **Sketch** -> **Include Library** -> **Add .ZIP Library...**



Install **TJpg\_Decoder\_v1.1.0.zip**



Open "Sketch\_06.2\_TFT\_Picture" folder under "Freenove\_ESP32\_Mini\_TV\Sketches" and double-click "Sketch\_06.2\_TFT\_Picture.ino".

### Sketch\_06.2\_TFT\_Picture

The following is the program code:

```

1  #include <TFT_eSPI.h> // TFT display library
2  #define USE_DMA
3  #include "image.h"      // Include raw JPEG image data array
4  #include <TJpg_Decoder.h> // Include JPEG decoder library
5
6  #ifdef USE_DMA
7  uint16_t dmaBuffer1[16 * 16]; // DMA buffer for 16x16 pixel block
8  uint16_t dmaBuffer2[16 * 16]; // Second DMA buffer for toggling
9  uint16_t* dmaBufferPtr = dmaBuffer1;
10 bool dmaBufferSel = 0;
11 #endif
12
13 const int TFT_VCC = 21;
14
15 TFT_eSPI tft = TFT_eSPI(); // Create TFT object instance
16

```

```
17 // Callback function to draw decoded JPEG blocks on screen
18 bool tft_output(int16_t x, int16_t y, uint16_t w, uint16_t h, uint16_t* bitmap) {
19     if (y >= tft.height()) return 0; // Stop decoding if off-screen
20
21     #ifdef USE_DMA
22         if (dmaBufferSel) dmaBufferPtr = dmaBuffer2;
23         else dmaBufferPtr = dmaBuffer1;
24         dmaBufferSel = !dmaBufferSel;
25         tft.pushImageDMA(x, y, w, h, bitmap, dmaBufferPtr); // Start DMA transfer
26     #else
27         tft.pushImage(x, y, w, h, bitmap); // Draw without DMA
28     #endif
29
30     return 1; // Continue decoding next block
31 }
32
33 void setup() {
34     Serial.begin(115200);
35     Serial.println("\n\n Testing TJpg_Decoder library");
36     pinMode(TFT_VCC, OUTPUT);
37     digitalWrite(TFT_VCC, LOW);
38     tft.begin(); // Initialize TFT display
39     tft.setTextColor(TFT_WHITE, TFT_BLACK);
40     tft.fillScreen(TFT_BLACK); // Clear screen with black background
41
42     #ifdef USE_DMA
43         tft.initDMA(); // Setup SPI DMA engine
44     #endif
45
46     TJpgDec.setJpgScale(1); // Set scale factor (1=full size)
47     tft.setSwapBytes(true); // Match color byte order
48     TJpgDec.setCallback(tft_output); // Register drawing callback
49 }
50
51 void loop() {
52     uint16_t w = 0, h = 0;
53     TJpgDec.getJpgSize(&w, &h, img_asset, sizeof(img_asset)); // Get image dimensions
54     Serial.print("Width = ");
55     Serial.print(w);
56     Serial.print(", height = ");
57     Serial.print(h);
58
59     uint32_t dt = millis();
60     tft.startWrite();
```

```

61 TJpgDec.drawJpg(0, 0, img_asset, sizeof(img_asset)); // Draw the JPEG image
62 tft.endWrite();
63
64 dt = millis() - dt;
65 Serial.print(", dt = ");
66 Serial.print(dt);
67 Serial.println(" ms");
68
69 delay(2000); // Wait before redraw
70 }

```

### Code Explanation

Include necessary header files.

```

1  #include <TFT_eSPI.h> // TFT display library
   ...
3  #include "image.h"      // Include raw JPEG image data array
4  #include <TJpg_Decoder.h> // Include JPEG decoder library

```

Configure DMA buffer

```

7  uint16_t dmaBuffer1[16 * 16]; // DMA buffer for 16x16 pixel block
8  uint16_t dmaBuffer2[16 * 16]; // Second DMA buffer for toggling
9  uint16_t* dmaBufferPtr = dmaBuffer1;
10 bool dmaBufferSel = 0;

```

Create TFT object instance.

```

19 TFT_eSPI tft = TFT_eSPI(); // Create TFT object instance

```

JPEG decoding callback function

```

15 // Callback function to draw decoded JPEG blocks on screen
16 bool tft_output(int16_t x, int16_t y, uint16_t w, uint16_t h, uint16_t* bitmap) {
17     if (y >= tft.height()) return 0; // Stop decoding if off-screen
18
19     #ifdef USE_DMA
20         if (dmaBufferSel) dmaBufferPtr = dmaBuffer2;
21         else dmaBufferPtr = dmaBuffer1;
22         dmaBufferSel = !dmaBufferSel;
23         tft.pushImageDMA(x, y, w, h, bitmap, dmaBufferPtr); // Start DMA transfer
24     #else
25         tft.pushImage(x, y, w, h, bitmap); // Draw without DMA
26     #endif
27
28     return 1; // Continue decoding next block
29 }

```

Get JPG size.

```

46 TJpgDec.getJpgSize(&w, &h, img_asset, sizeof(img_asset)); // Get image dimensions

```

Draw images on the TFT screen.

```

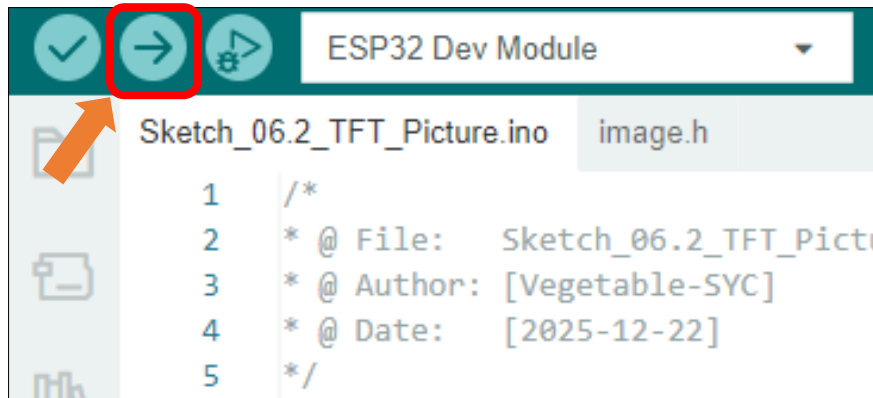
57 tft.startWrite();
58 TJpgDec.drawJpg(0, 0, img_asset, sizeof(img_asset)); // Draw the JPEG image

```



```
59 tft.endWrite();
```

Click “**Upload**” to upload the code to Freenove ESP32 Mini TV



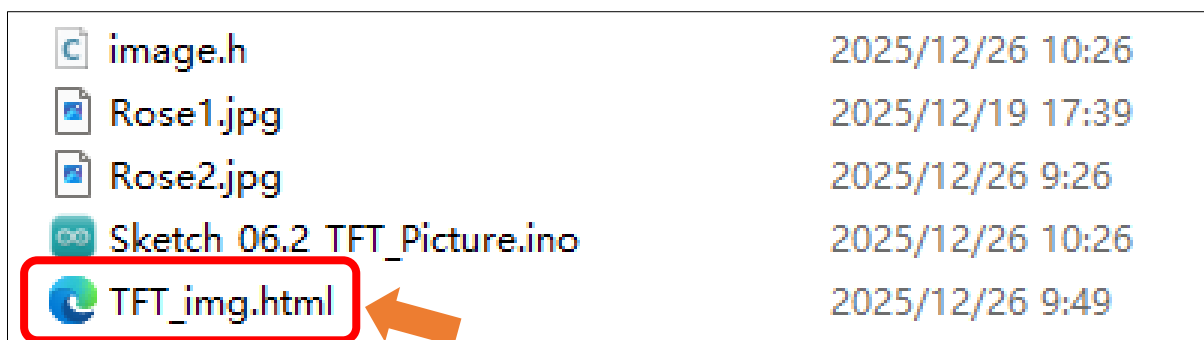
After the code is uploaded, an image will be displayed on the TFT screen



#### Custom image display

You can customize the image displayed on the display according to your personal preferences.

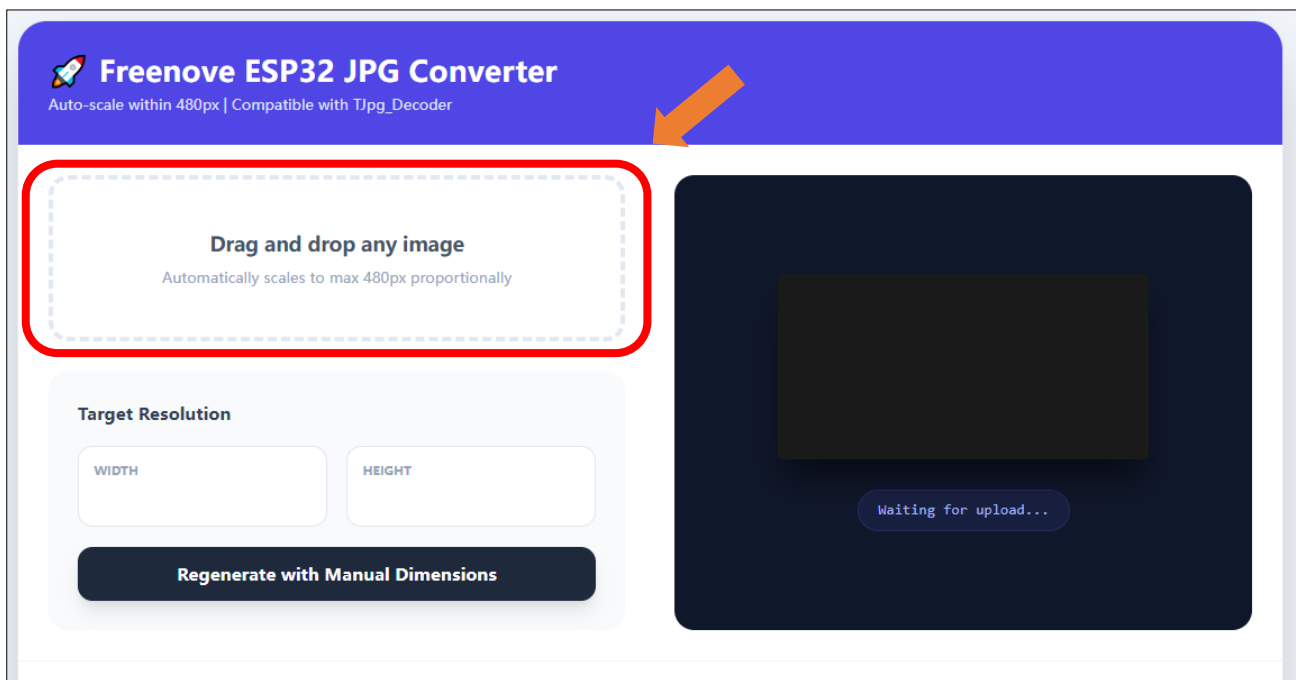
First, open **Freenove\_ESP32\_Mini\_TV \Sketch\ Sketch\_06.2\_TFT\_Picture\TFT\_img.html**



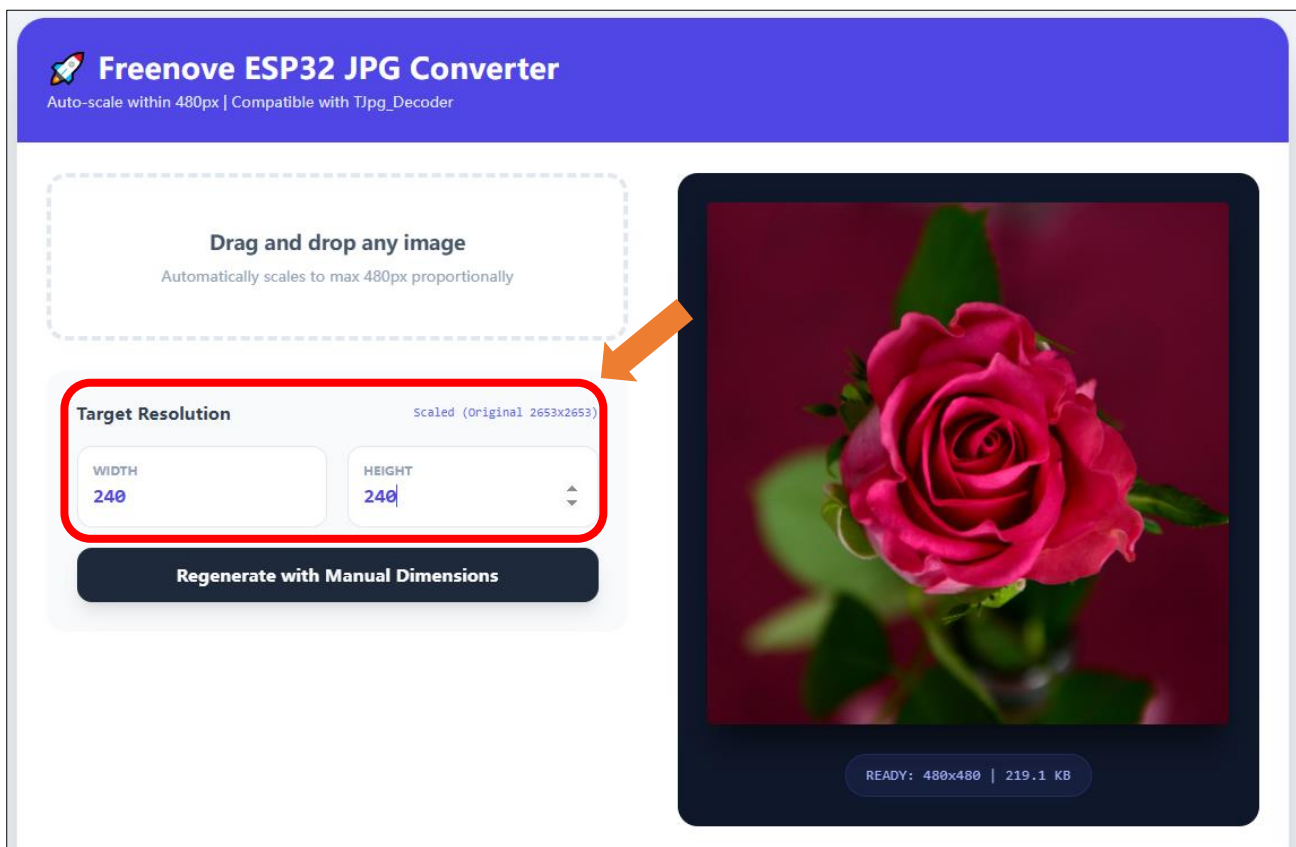
For optimal compatibility, please use Microsoft Edge or Google Chrome. You can open the file by dragging it directly into the browser window, or by right clicking it and selecting "Open with".

If you have any concerns, please feel free to contact us via [support@freenove.com](mailto:support@freenove.com)

Click the **"Drag and drop any image"** area to select an image file, or directly drag and drop a file onto it.



Set the resolution to **240x240**.



Click **"Regenerate with Manual Dimensions"** to generate arrays.

[illegible]

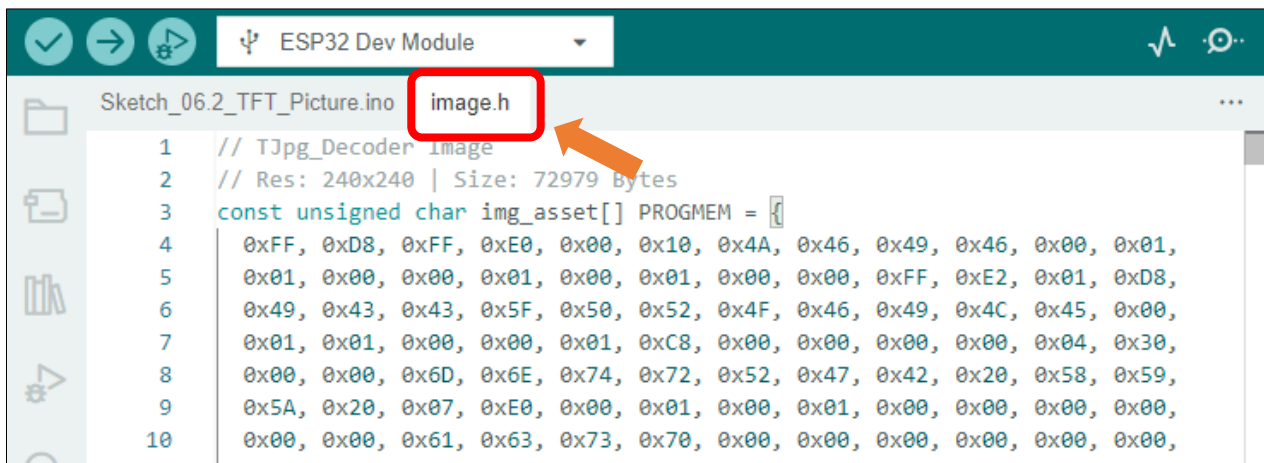
Click the **"Copy Code"** button to copy the arrays.

C Array Output

Copy Code

```
// Tjpg_Decoder Image  
// Res: 240x240 | Size: 72979 Bytes  
const unsigned char img_asset[] PROGMEM = {  
    0xFF, 0xDB, 0xFF, 0xE0, 0x00, 0x10, 0x4A, 0x46, 0x49, 0x46, 0x00, 0x01,  
    0x01, 0x00, 0x00, 0x01, 0x00, 0x01, 0x00, 0x00, 0xFF, 0xE2, 0x01, 0xD8,  
    0x49, 0x43, 0x43, 0x5F, 0x50, 0x52, 0x4F, 0x46, 0x49, 0x4C, 0x45, 0x00,  
    0x01, 0x01, 0x00, 0x00, 0x01, 0xC8, 0x00, 0x00, 0x00, 0x00, 0x04, 0x30,  
    0x00, 0x00, 0xF0, 0x6E, 0x74, 0x72, 0x52, 0x47, 0x42, 0x20, 0x58, 0x59,  
    0x5A, 0x20, 0x07, 0xE0, 0x00, 0x01, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x61, 0x63, 0x73, 0x70, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x01, 0x00, 0x00,  
    0xF6, 0xD6, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00, 0xD3, 0x2D, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,  
    0x00, 0x00, 0x64, 0x65, 0x73, 0x53, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00
```

Replace all content in **image.h** with the copied arrays.



```
1 // Tjpg_Decoder Image
2 // Res: 240x240 | Size: 72979 Bytes
3 const unsigned char img_asset[] PROGMEM = {
4   0xFF, 0xD8, 0xFF, 0xE0, 0x00, 0x10, 0x4A, 0x46, 0x49, 0x46, 0x00, 0x01,
5   0x01, 0x00, 0x00, 0x01, 0x00, 0x01, 0x00, 0x00, 0xFF, 0xE2, 0x01, 0xD8,
6   0x49, 0x43, 0x43, 0x5F, 0x50, 0x52, 0x4F, 0x46, 0x49, 0x4C, 0x45, 0x00,
7   0x01, 0x01, 0x00, 0x00, 0x01, 0xC8, 0x00, 0x00, 0x00, 0x00, 0x04, 0x30,
8   0x00, 0x00, 0x6D, 0x6E, 0x74, 0x72, 0x52, 0x47, 0x42, 0x20, 0x58, 0x59,
9   0x5A, 0x20, 0x07, 0xE0, 0x00, 0x01, 0x00, 0x01, 0x00, 0x00, 0x00, 0x00,
10  0x00, 0x00, 0x61, 0x63, 0x73, 0x70, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
```

Upload the code and the custom image will be displayed.



## Project 6.3 TFT Clock

Having explored key concepts such as EEPROM, Wi-Fi, capacitive touch, and TFT display in previous chapters, we will now synthesize this knowledge to create a practical and visually appealing classic project: a WiFi clock. Upon completing this chapter, you will be able to customize the interface to your preference.

### Component List

Freenove ESP32 Mini TV



USB cable x1



### Circuit

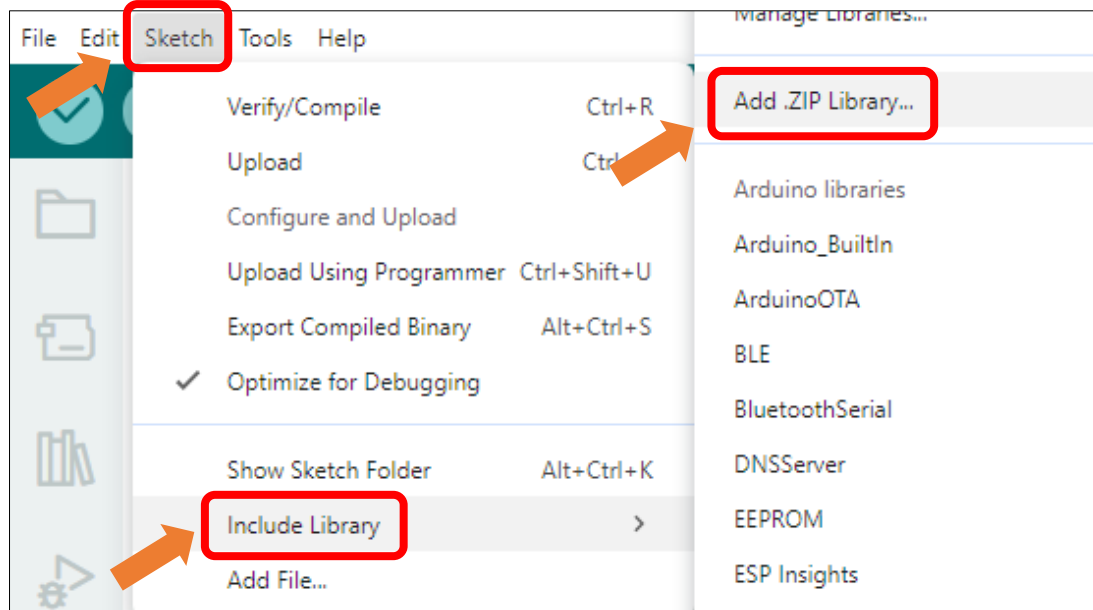
Connect Freenove ESP32 Mini TV to the computer with USB cable.



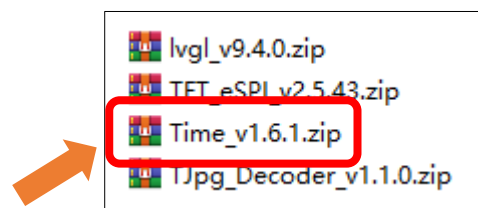
## Sketch

### Install Libraries

Click **Sketch** -> **Include Library** -> **Add .ZIP Library...**



Install **Time\_v1.6.1.zip**



Open "Sketch\_06.3\_TFT\_Clock" folder under "Freenove\_ESP32\_Mini\_TV\Sketch" and double-click "Sketch\_06.3\_TFT\_Clock.ino".

### Sketch\_06.3\_TFT\_Clock

The following is the program code:

```

1  #include <TFT_eSPI.h>
2  #include <WiFi.h>
3  #include <WiFiUdp.h>
4  #include <EEPROM.h>
5  #include <TimeLib.h>
6  #include <DNSServer.h>
7
8  /* ----- */
9  /*                               Configuration                               */
10 /* ----- */
11
12 // Pin Definitions
13 #define TFT_VCC      21

```

```
14 #define TOUCH_PIN      32
15 #define EEPROM_SIZE    1024
16 #define AP_SSID        "ESP32-Clock"
17 #define AP_IP           IPAddress(192, 168, 4, 1)
18
19 // UI Geometry
20 #define CENTER_X        120
21 #define CENTER_Y        120
22 #define RADIUS          100
23
24 #define COLOR_BG        0x0000
25 #define COLOR_SCALE     0xFFFF
26 #define COLOR_TEXT      0xFFFF
27 #define COLOR_HOUR      0xF800
28 #define COLOR_MIN       0x07E0
29 #define COLOR_SEC       0xFFE0
30 #define COLOR_CENTER    0xFFFF
31 #define COLOR_RING      0xAD55
32
33 // Global Constants
34 const char* ntpServer    = "pool.ntp.org";
35 const int timeZone       = 8;
36 const char* AUTHOR_NAME = "Freenove";
37 const int TOUCH_THRESHOLD = 80;
38 const int PRESS_TIME     = 3000;
39
40 // Object Instances
41 TFT_eSPI tft = TFT_eSPI();
42 WiFiServer server(80);
43 DNSServer;
44 WiFiUDP ntpUDP;
45
46 // State Variables
47 bool timeSynced    = false;
48 bool isAPMode      = false;
49 int currentUI      = 1;
50 int lastUI         = -1;
51 bool isFirstDraw   = true;
52 int wifiScanCount  = 0;
53 float lastHAngle=0, lastMAngle=0, lastSAngle=0;
54
55 // Touch Interaction Variables
56 uint32_t Touch_Time = 0;
57 int Touch_data = 0, Touch_old_data = 0;
```

```
58 bool LONG_PRESS_FLAG = false;
59
60 // HTML/CSS for Web Configuration Portal
61 const char* CSS_BODY = R"raw(
62 <style>
63     body {
64         background: #202020;
65         color: #fff;
66         font-family: sans-serif;
67         margin: 0;
68         padding: 0;
69     }
70     #header {
71         background: #333;
72         padding: 15px;
73         text-align: center;
74     }
75     h1 {
76         margin: 0;
77         font-size: 20px;
78     }
79     .list {
80         max-height: 400px;
81         overflow-y: auto;
82     }
83     .item {
84         padding: 15px;
85         border-bottom: 1px solid #444;
86         display: flex;
87         justify-content: space-between;
88         cursor: pointer;
89     }
90     .item:active {
91         background: #444;
92     }
93     .ssid {
94         font-size: 16px;
95         font-weight: bold;
96     }
97     .meta {
98         font-size: 12px;
99         color: #aaa;
100    }
101    .form {
```



```
102     padding: 20px;
103     background: #303030;
104     text-align: center;
105 }
106 input {
107     width: 100%;
108     padding: 12px;
109     margin: 10px 0;
110     border: none;
111     border-radius: 4px;
112     font-size: 16px !important;
113     box-sizing: border-box;
114 }
115 button {
116     width: 100%;
117     padding: 15px;
118     background: #007BFF;
119     color: #fff;
120     border: none;
121     border-radius: 4px;
122     font-size: 18px;
123     font-weight: bold;
124     cursor: pointer;
125 }
126 a {
127     color: #007BFF;
128     text-decoration: none;
129     display: block;
130     margin-top: 10px;
131     text-align: center;
132 }
133 </style>
134
135 <script>
136     function c(s) {
137         document.getElementById('s').value = s;
138         document.getElementById('p').focus();
139     }
140 </script>
141 )raw";
142
143 /* ----- */
144 /*                               Main Program                               */
145 /* ----- */
```

```

146
147 void setup() {
148     Serial.begin(115200);
149
150     // Initialize NVS storage
151     if (!EEPROM.begin(EEPROM_SIZE)) Serial.println("EEPROM Init Failed");
152
153     // Initialize Display
154     pinMode(TFT_VCC, OUTPUT);
155     digitalWrite(TFT_VCC, LOW);
156     tft.begin();
157     tft.setRotation(0);
158     tft.fillScreen(COLOR_BG);
159
160     // Initialize Touch and calibrate baseline
161     touchSetCycles(0xf000, 0x1000);
162     Touch_old_data = touchRead(TOUCH_PIN);
163
164     // Connection management
165     connectToWiFi();
166     ntpUDP.begin(123);
167 }
168
169 void loop() {
170     if (isAPMode) {
171         dnsServer.processNextRequest(); // Handle DNS queries for Captive Portal
172         handleWebServer();             // Process config portal requests
173     } else if (WiFi.isConnected()) {
174         static unsigned long lastNtpUpdate = 0;
175         // Periodic NTP synchronization
176         if (millis() - lastNtpUpdate > 3600000 || !timeSynced) {
177             syncNtpTime(timeZone);
178             lastNtpUpdate = millis();
179         }
180     }
181     checkTouchInput();
182     // Rendering logic only active after time synchronization
183     if (timeSynced) renderUI();
184     delay(5);
185 }

```

### Code Explanation

Include the header file.

```

1 #include <TFT_eSPI.h>
2 #include <WiFi.h>

```

```

3  #include <WiFiUdp.h>
4  #include <EEPROM.h>
5  #include <TimeLib.h>
6  #include <DNSServer.h>

```

Preset the required variables the time zone and author name can be modified here.

```

35  const char* ntpServer    = "pool.ntp.org";
36  const int  timeZone      = 8;
37  const char* AUTHOR_NAME  = "Freenove";
38  const int  TOUCH_THRESHOLD = 80;
39  const int  PRESS_TIME    = 3000;

```

Webpage interface design during network configuration

```

62  const char* CSS_BODY = R"raw(
63  <style>
...  ...
142 )raw";

```

Initialize the screen.

```

155  pinMode(TFT_VCC, OUTPUT);
156  digitalWrite(TFT_VCC, LOW);
157  tft.begin();
158  tft.setRotation(0);
159  tft.fillScreen(COLOR_BG);

```

Initialize the capacitive touch.

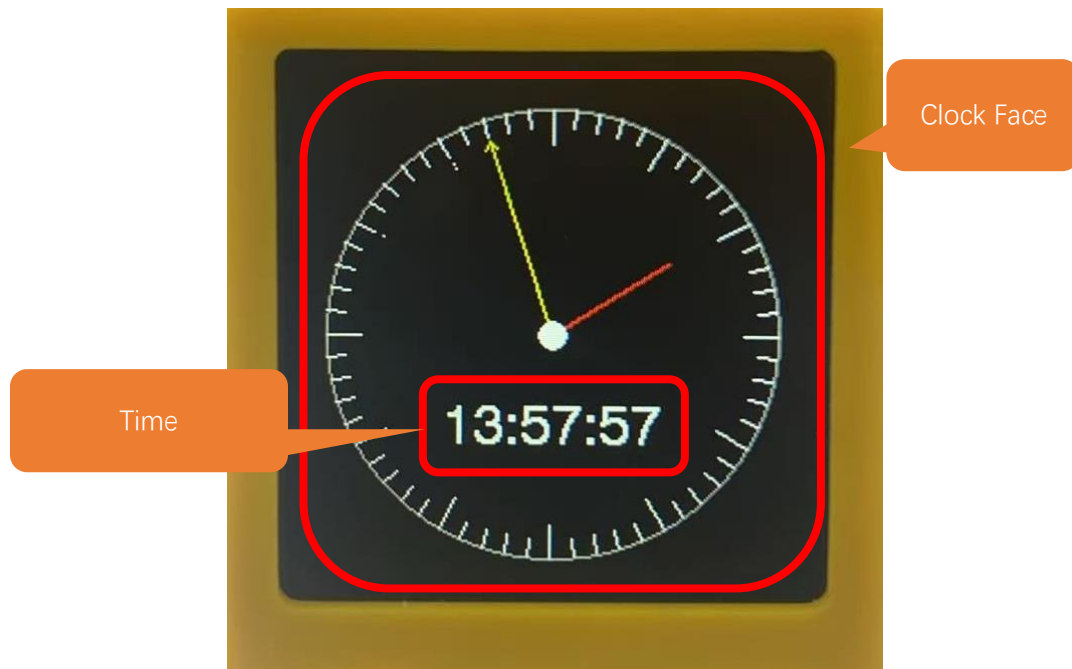
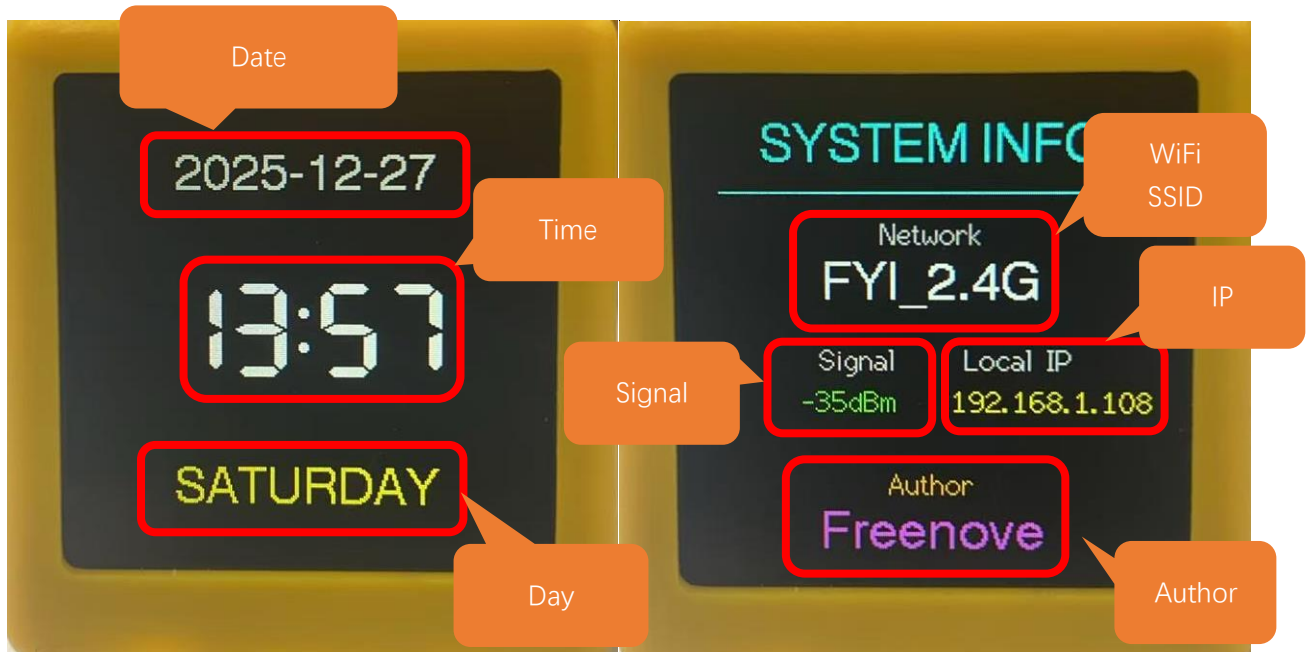
```

155  touchSetCycles(0xf000, 0x1000);
156  Touch_old_data = touchRead(TOUCH_PIN);

```

Click "Upload" to upload the code to Freenove ESP32 Mini TV





### How to Use

Upon first boot, the device requires network configuration. The WiFi credentials will be saved to EEPROM for automatic connection on subsequent startups. To reset or change the network, simply long-press the capacitive touch area for 3 seconds. This will clear the stored configuration and initiate the setup process again.

1. On first power-up, wait until the following interface appears.



2. Connect to ESP32-Clock.



- Wait for the network configuration page to appear.

**Note: If the page does not load automatically, please open a browser and navigate to <http://192.168.4.1>.**

- Select your WiFi network in the configuration page and enter the password. Click

**Important note: The WiFi network must be 2.4GHz; otherwise, the connection will fail.**



The image shows the 'ESP32 WIFI SETUP' interface. It features a list of available WiFi networks with their signal strengths. Below the list is a form with two input fields labeled 'SSID' and 'PASS', and a blue 'SAVE' button. The entire form area is highlighted with a red border.

| ESP32 WIFI SETUP |      |
|------------------|------|
| FYI_2.4G         | 100% |
| LXH_2.4G         | 100% |
| ChinaNet-y29u    | 94%  |
| FYI_2.4G_3       | 90%  |
| ChinaNet-5rTS    | 78%  |
| TP-LINK_A7BE     | 66%  |
| MT               | 62%  |
| MT               | 62%  |

Below the list, there is a configuration form with the following fields:

- SSID:
- PASS:
- SAVE:

- the following screen will be displayed. You can tap the capacitive touch area to switch between interfaces.



## Chapter 7 LVGL

### Project 7.1 LVGL

#### Component List

|                                                                                                                 |                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  | <p>USB cable x1</p>  |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|

#### Component knowledge

LVGL is a widely-used embedded GUI library that is implemented in pure C, making it highly portable and performant. It offers rich features and content, supporting both display and input devices such as touchscreens and keyboards.

|    | Features supported by LVGL                                                                                      |
|----|-----------------------------------------------------------------------------------------------------------------|
| 1  | Powerful building blocks such as buttons, charts, lists, sliders, images, and more.                             |
| 2  | Advanced graphics with animation, anti-aliasing, opacity, and smooth scrolling.                                 |
| 3  | Various input devices, such as touchpads, mice, keyboards, encoders, and more.                                  |
| 4  | Multiple languages with UTF-8 encoding.                                                                         |
| 5  | Multiple display types, including TFT and monochrome displays.                                                  |
| 6  | Fully customizable graphical elements.                                                                          |
| 7  | LVGL can be used independently of any microcontroller or display hardware.                                      |
| 8  | Highly extensible and can be configured to use very little memory (e.g. 64 kB of flash and 16 kB of RAM)        |
| 9  | It can be used with or without an operating system, and supports external memory and GPUs as optional features. |
| 10 | Single-frame buffer operation, even with advanced graphics effects.                                             |
| 11 | Written in C language to achieve maximum compatibility (compatible with C++ as well).                           |
| 12 | LVGL has a simulator that allows for embedded GUI design on a PC without any embedded hardware.                 |
| 13 | Resources to help developers quickly get started with the library, including tutorials, examples, and themes.   |
| 14 | A wide range of resources.                                                                                      |

## Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.

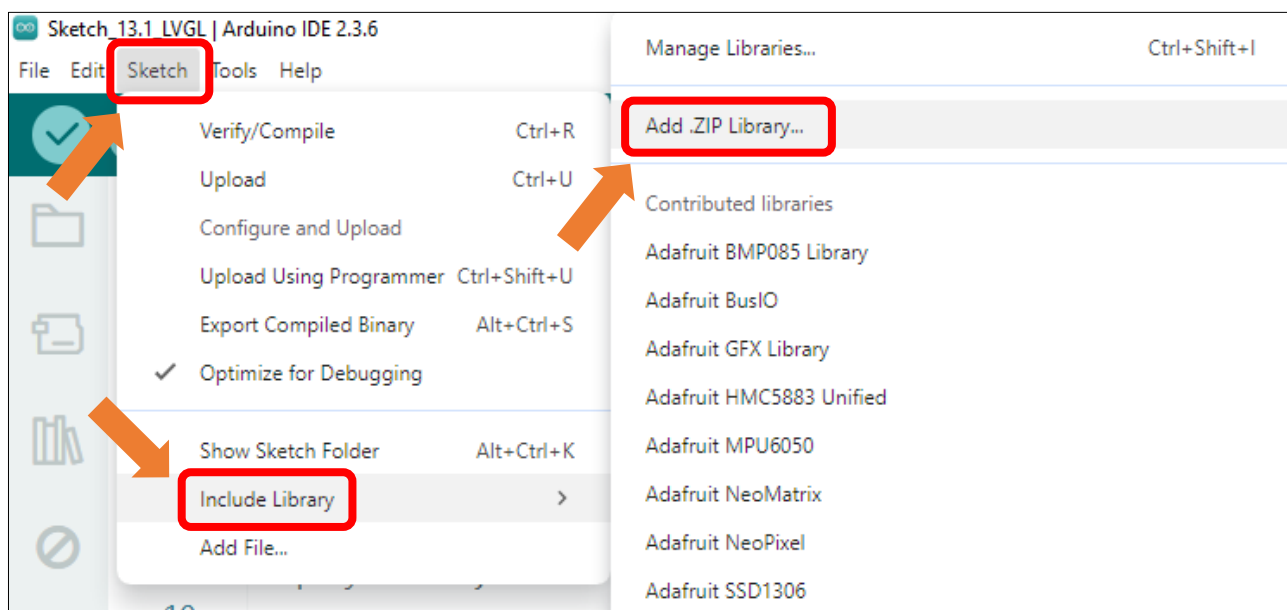


## Sketch

### Install Libraries

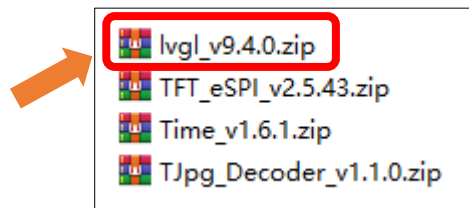
**Caution:** It is critical to follow the installation steps below precisely to install the LVGL library. Our provided library comes pre-configured for the display. Using a version downloaded online or keeping an old one will prevent the screen from working. Please ensure you completely replace any other version with this one to guarantee correct driver operation.

Click **Sketch** -> **Include Library** -> **Add .ZIP Library...**





Install **lvgl\_v9.4.0.zip**



Open “**Sketch\_07.1\_LVGL**” folder under “**Freenove\_ESP32\_Mini\_TV\Sketch**” and double-click “**Sketch\_07.1\_LVGL.ino**”.

**Sketch\_07.1\_LVGL**

The following is the program code:

```

1  #include <lvgl.h>
2  #include <TFT_eSPI.h>
3
4  /* Display resolution configuration */
5  #define SCREEN_WIDTH  240
6  #define SCREEN_HEIGHT 240
7
8  /* Hardware and driver initialization */
9  const int TFT_VCC = 21;
10  TFT_eSPI tft = TFT_eSPI();
11
12  /* LVGL display handles and draw buffers */
13  static lv_display_t * disp;
14  static uint8_t buf[SCREEN_WIDTH * SCREEN_HEIGHT / 10 * 2];
15
16  /**
17   * Flush callback: Transfers pixel data from LVGL to the TFT controller
18   */
19  void my_disp_flush(lv_display_t * disp, const lv_area_t * area, uint8_t * px_map) {
20      uint32_t w = (area->x2 - area->x1 + 1);
21      uint32_t h = (area->y2 - area->y1 + 1);
22
23      tft.startWrite();
24      tft.setAddrWindow(area->x1, area->y1, w, h);
25      tft.pushColors((uint16_t *)px_map, w * h, true);
26      tft.endWrite();
27
28      lv_display_flush_ready(disp); // Indicate flushing finished
29  }
30
31  void setup() {
32      Serial.begin(115200);
33  }

```

```

34  /* Initialize physical display */
35  tft.begin();
36
37  /* Initialize LVGL core library */
38  lv_init();
39
40  /* Setup display abstraction layer */
41  disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
42  lv_display_set_flush_cb(disp, my_disp_flush);
43  lv_display_set_buffers(disp, buf, NULL, sizeof(buf), LV_DISPLAY_RENDER_MODE_PARTIAL);
44
45  /* UI initialization and version logging */
46  String LVGL_Arduino = "Hello Arduino! ";
47  LVGL_Arduino += String('V') + lv_version_major() + "." + lv_version_minor() + "." +
48  lv_version_patch();
49
50  Serial.println(LVGL_Arduino);
51  Serial.println("System Ready: LVGL initialized");
52
53  /* Create and center-align a label widget */
54  lv_obj_t *label = lv_label_create(lv_screen_active());
55  lv_label_set_text(label, LVGL_Arduino.c_str());
56  lv_obj_align(label, LV_ALIGN_CENTER, 0, 0);
57
58  /* Enable display power */
59  pinMode(TFT_VCC, OUTPUT);
60  digitalWrite(TFT_VCC, LOW);
61 }
62
63 void loop() {
64     /* Update library tick and process pending UI tasks */
65     lv_tick_inc(5);
66     lv_timer_handler();
67     delay(5);
68 }

```

### Code Explanation

Include the header file.

```

1  #include <lvgl.h>
2  #include <TFT_eSPI.h>

```

Set the screen resolution.

```

5  #define SCREEN_WIDTH  240
6  #define SCREEN_HEIGHT 240

```

Initialize the screen.

```

35  tft.begin();

```

Initialize LVGL..

```
38 lv_init();
```

Create a text label and center it.

```
54 lv_obj_t *label = lv_label_create(lv_screen_active());
55 lv_label_set_text(label, LVGL_Arduino.c_str());
56 lv_obj_align(label, LV_ALIGN_CENTER, 0, 0);
```

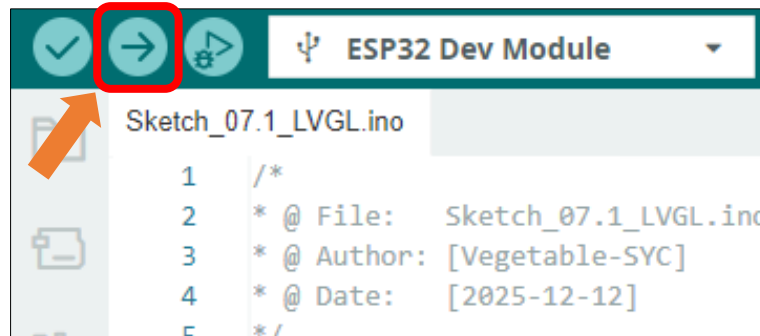
Enable the display's power supply pin.

```
59 pinMode(TFT_VCC, OUTPUT);
60 digitalWrite(TFT_VCC, LOW);
```

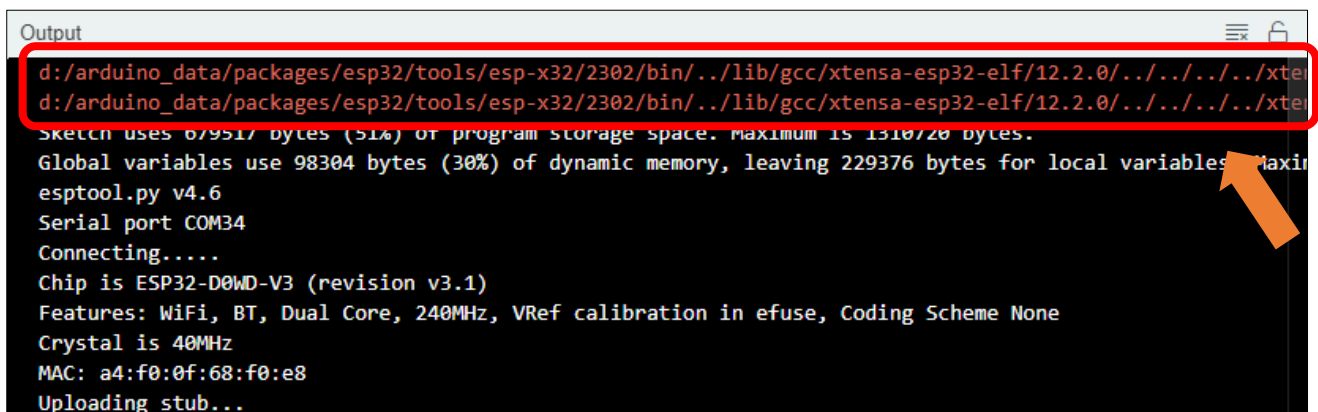
Handles UI tasks in a continuous loop.

```
65 lv_tick_inc(5);
66 lv_timer_handler();
67 delay(5);
```

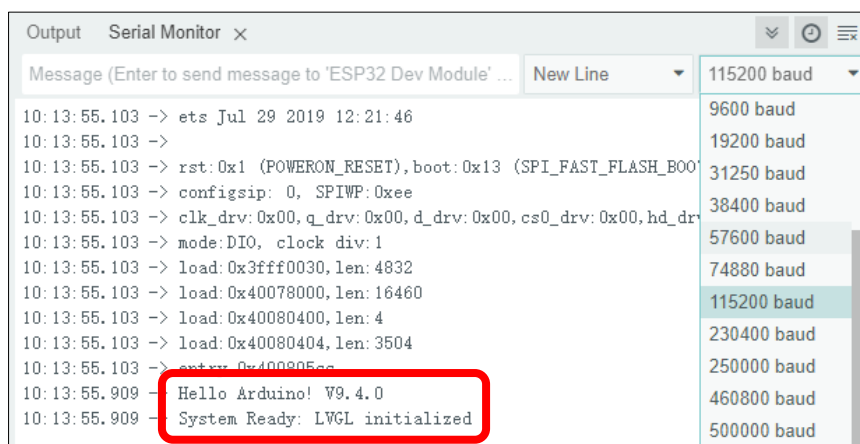
Click "Upload" to upload the code to Freenove ESP32 Mini TV. Set the baud rate to 115200.



**Note:** The following warning messages may appear in the console during compilation. They do not affect code execution and can be safely ignored.



Successful initialization information will be displayed on the serial monitor.



```

Output  Serial Monitor x
Message (Enter to send message to 'ESP32 Dev Module' ... New Line 115200 baud
10:13:55.103 -> ets Jul 29 2019 12:21:46
10:13:55.103 ->
10:13:55.103 -> rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
10:13:55.103 -> configspi: 0, SPIWP:0xee
10:13:55.103 -> clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00
10:13:55.103 -> mode:DIO, clock div:1
10:13:55.103 -> load:0x3fff0030,len:4832
10:13:55.103 -> load:0x40078000,len:16460
10:13:55.103 -> load:0x40080400,len:4
10:13:55.103 -> load:0x40080404,len:3504
10:13:55.103 -> load:0x40080500,len:3504
10:13:55.909 -> Hello Arduino! V9.4.0
10:13:55.909 -> System Ready: LVGL initialized
  
```

You will see the text reading “Hello Arduino! V9.4.0” on the display.



## Reference

```
lv_display_t * lv_display_create(int32_t hor_res, int32_t ver_res);
```

This function is designed to create a device object with specified dimensions, define the canvas size, and return a handle to the object.

```
lv_obj_t * lv_label_create(lv_obj_t * parent);
```

This function creates a text label component.

```
void lv_label_set_text(lv_obj_t * obj, const char * text);
```

This function sets the static text to be displayed by the label control.

## Chapter 8 Lvgl Touch

### Project 8.1 Lvgl Touch

Having learned about capacitive touch in previous chapters, we will now combine it with LVGL to create more intuitive interactive effects on the screen.

#### Component List

Freenove ESP32 Mini TV



USB cable x1



#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “Sketch\_08.1\_LVGL\_Touch” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_08.1\_LVGL\_Touch.ino”.

### Sketch\_08.1\_LVGL\_Touch

The following is the program code:

```

1  #include <lvgl.h>
2  #include <TFT_eSPI.h>
3
4  /* --- Configuration Parameters --- */
5  const int TOUCH_PIN = 32;          // Capacitive touch GPIO
6  const int TFT_VCC = 21;           // Power control pin
7  const int TOUCH_THRESHOLD = 80;   // Sensitivity threshold
8  const int PRESS_Time = 3000;      // Long press threshold (3000ms)
9  int touch_base_val = 0;           // Initial baseline value
10
11 #define SCREEN_WIDTH  240
12 #define SCREEN_HEIGHT 240
13
14 TFT_eSPI tft = TFT_eSPI();
15 static uint8_t buf[SCREEN_WIDTH * SCREEN_HEIGHT / 10 * 2]; // LVGL draw buffer
16
17 lv_obj_t * label_short;
18 lv_obj_t * label_long;
19 int short_press_num = 0;
20 int long_press_num = 0;
21
22 /* --- 1. Display Flush Callback --- */
23 void my_disp_flush(lv_display_t * disp, const lv_area_t * area, uint8_t * px_map) {
24     uint32_t w = (area->x2 - area->x1 + 1);
25     uint32_t h = (area->y2 - area->y1 + 1);
26     tft.startWrite();
27     tft.setAddrWindow(area->x1, area->y1, w, h);
28     tft.pushColors((uint16_t *)px_map, w * h, true);
29     tft.endWrite();
30     lv_display_flush_ready(disp); // Inform LVGL flushing is complete
31 }
32
33 /* --- 2. Input Device Read Callback --- */
34 void my_touchpad_read(lv_indev_t * indev, lv_indev_data_t * data) {
35     int val = touchRead(TOUCH_PIN);
36
37     // Determine state based on baseline comparison

```

```
38     if (touch_base_val - val > TOUCH_THRESHOLD) {
39         data->state = LV_INDEV_STATE_PRESSED;
40     } else {
41         data->state = LV_INDEV_STATE_RELEASED;
42     }
43     data->btn_id = 0; // Use the first button mapping
44 }
45
46 /* --- 3. UI Event Callback --- */
47 void btn_event_cb(lv_event_t * e) {
48     lv_event_code_t code = lv_event_get_code(e);
49
50     if(code == LV_EVENT_SHORT_CLICKED) {
51         // Triggered upon releasing after a short press
52         short_press_num++;
53         lv_label_set_text_fmt(label_short, "Short Press: %d", short_press_num);
54         Serial.println("Short Press Detected");
55     }
56     else if(code == LV_EVENT_LONG_PRESSED) {
57         // Triggered upon releasing after a long press
58         long_press_num++;
59         lv_label_set_text_fmt(label_long, "Long Press: %d", long_press_num);
60         Serial.println("Long Press Detected (3s)");
61     }
62 }
63
64 void setup() {
65     Serial.begin(115200);
66
67     // Initialize capacitive touch hardware
68     touchSetCycles(0xf000, 0x1000);
69     touch_base_val = touchRead(TOUCH_PIN);
70
71     tft.begin();
72     lv_init();
73
74     // Register LVGL display driver
75     lv_display_t * disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
76     lv_display_set_flush_cb(disp, my_disp_flush);
77     lv_display_set_buffers(disp, buf, NULL, sizeof(buf), LV_DISPLAY_RENDER_MODE_PARTIAL);
78
79     // Register Input Device (Button Type for single touch point)
80     lv_indev_t * indev = lv_indev_create();
81     lv_indev_set_type(indev, LV_INDEV_TYPE_BUTTON);
```

```
82     lv_indev_set_read_cb(indev, my_touchpad_read);
83
84     // Set LVGL long press detection threshold
85     lv_indev_set_long_press_time(indev, PRESS_Time);
86
87     // Map the hardware touch sensor to a virtual screen coordinate
88     static const lv_point_t btn_points[] = { {120, 150} }; // Coordinates within button area
89     lv_indev_set_button_points(indev, btn_points);
90
91     // Create User Interface - Interactive Button
92     lv_obj_t * btn = lv_button_create(lv_screen_active());
93     lv_obj_set_size(btn, 180, 80);
94     lv_obj_align(btn, LV_ALIGN_CENTER, 0, 30);
95
96     // Listen for all interaction events
97     lv_obj_add_event_cb(btn, btn_event_cb, LV_EVENT_ALL, NULL);
98
99     lv_obj_t * btn_label = lv_label_create(btn);
100    lv_label_set_text(btn_label, "Touch Pin 32");
101    lv_obj_center(btn_label);
102
103    // Create Feedback Labels
104    label_short = lv_label_create(lv_screen_active());
105    lv_label_set_text(label_short, "Short Press: 0");
106    lv_obj_align(label_short, LV_ALIGN_TOP_MID, 0, 40);
107
108    label_long = lv_label_create(lv_screen_active());
109    lv_label_set_text(label_long, "Long Press: 0");
110    lv_obj_align(label_long, LV_ALIGN_TOP_MID, 0, 70);
111
112    // Power up display
113    pinMode(TFT_VCC, OUTPUT);
114    digitalWrite(TFT_VCC, LOW);
115 }
116
117 void loop() {
118     // Process UI tasks
119     lv_timer_handler();
120
121     // Handle LVGL internal tick timing
122     static uint32_t last_tick = 0;
123     uint32_t current_ms = millis();
124     lv_tick_inc(current_ms - last_tick);
125     last_tick = current_ms;
```



```

126
127     delay(5);
128 }

```

### Code Explanation

Include the necessary header files.

```

1  #include <lvgl.h>
2  #include <TFT_eSPI.h>

```

Preset basic parameters.

```

5  const int TOUCH_PIN = 32;          // Capacitive touch GPIO
6  const int TFT_VCC = 21;           // Power control pin
7  const int TOUCH_THRESHOLD = 80;   // Sensitivity threshold
8  const int PRESS_Time = 3000;      // Long press threshold (3000ms)

```

Define the screen resolution.

```

11 #define SCREEN_WIDTH  240
12 #define SCREEN_HEIGHT 240

```

Initialize and calibrate the touch.

```

68 touchSetCycles(0xf000, 0x1000);
69 touch_base_val = touchRead(TOUCH_PIN);

```

Initialize the screen and LVGL.

```

71 tft.begin();
72 lv_init();

```

Draw the UI.

```

74 // Register LVGL display driver
75 lv_display_t * disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
76 lv_display_set_flush_cb(disp, my_disp_flush);
77 lv_display_set_buffers(disp, buf, NULL, sizeof(buf), LV_DISPLAY_RENDER_MODE_PARTIAL);
78
79 // Register Input Device (Button Type for single touch point)
80 lv_indev_t * indev = lv_indev_create();
81 lv_indev_set_type(indev, LV_INDEV_TYPE_BUTTON);
82 lv_indev_set_read_cb(indev, my_touchpad_read);
83
84 // Set LVGL long press detection threshold
85 lv_indev_set_long_press_time(indev, PRESS_Time);
86
87 // Map the hardware touch sensor to a virtual screen coordinate
88 static const lv_point_t btn_points[] = { {120, 150} }; // Coordinates within button area
89 lv_indev_set_button_points(indev, btn_points);
90
91 // Create User Interface - Interactive Button
92 lv_obj_t * btn = lv_button_create(lv_screen_active());
93 lv_obj_set_size(btn, 180, 80);
94 lv_obj_align(btn, LV_ALIGN_CENTER, 0, 30);
95

```

```

96 // Listen for all interaction events
97 lv_obj_add_event_cb(btn, btn_event_cb, LV_EVENT_ALL, NULL);
98
99 lv_obj_t * btn_label = lv_label_create(btn);
100 lv_label_set_text(btn_label, "Touch Pin 32");
101 lv_obj_center(btn_label);
102
103 // Create Feedback Labels
104 label_short = lv_label_create(lv_screen_active());
105 lv_label_set_text(label_short, "Short Press: 0");
106 lv_obj_align(label_short, LV_ALIGN_TOP_MID, 0, 40);
107
108 label_long = lv_label_create(lv_screen_active());
109 lv_label_set_text(label_long, "Long Press: 0");
110 lv_obj_align(label_long, LV_ALIGN_TOP_MID, 0, 70);

```

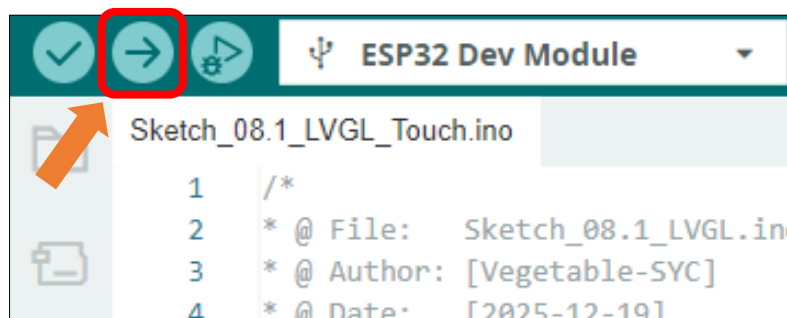
Enable the screen power supply pin.

```

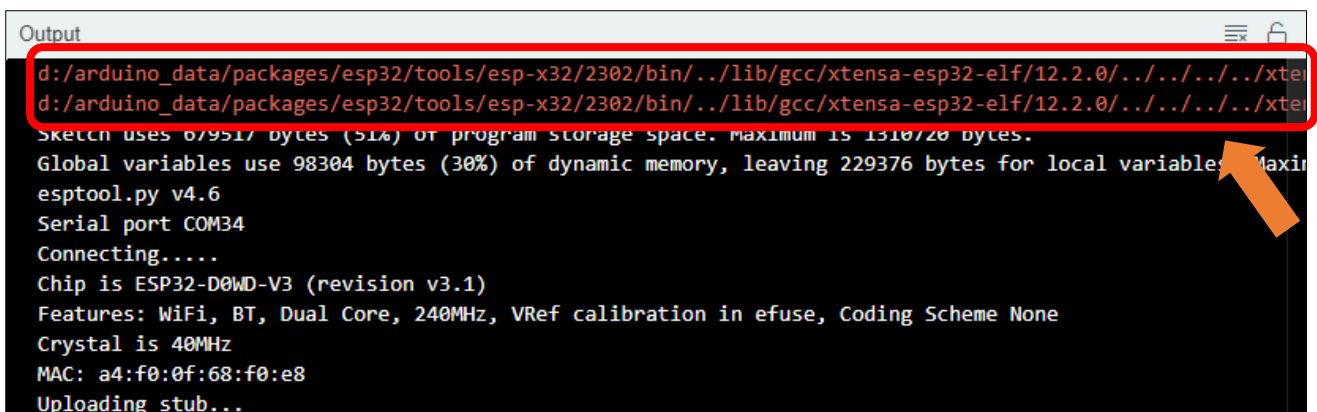
113 pinMode(TFT_VCC, OUTPUT);
114 digitalWrite(TFT_VCC, LOW);

```

Click **Upload** to upload the code to Freenove ESP32 Mini TV. Set the baud rate to 115200.



**Note:** The following warning messages may appear in the console during compilation. They do not affect code execution and can be safely ignored.



When touching the capacitive touch area of the Freenode ESP32 Mini TV with a finger, a touch event lasting longer than 3 seconds is considered a long press; otherwise, it is a short press. The on-screen button displays a click animation when pressed.



#### Reference

```
lv_obj_t * lv_button_create(lv_obj_t * parent);
```

This function is used to create a button component.

```
lv_event_dsc_t * lv_obj_add_event_cb(lv_obj_t * obj, lv_event_cb_t event_cb, lv_event_code_t  
filter, void * user_data);
```

This function is used to bind an event callback function to an object, enabling it to listen for and respond to specific interactive behaviors.

## Chapter 9 Lvgl Picture

### Project 9.1 Lvgl Picture

#### Component List

|                                                                                                             |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Freenove ESP32 Mini TV<br> | USB cable x1<br> |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|

#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



#### Sketch

Open “Sketch\_09.1\_LVGL\_Picture” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_09.1\_LVGL\_Picture.ino”.

Sketch\_09.1\_LVGL\_Picture

The following is the program code:

```
1 #include <lvgl.h>
2 #include <TFT_eSPI.h>
```

```

3  #include <image.h>
4
5  /* Display and asset dimensions */
6  #define IMG_W  240
7  #define IMG_H  240
8  #define SCREEN_WIDTH  240
9  #define SCREEN_HEIGHT 240
10
11 /* Hardware pin configuration */
12 const int TFT_VCC = 21;
13 TFT_eSPI tft = TFT_eSPI();
14
15 /* LVGL rendering buffer */
16 static uint8_t draw_buf[SCREEN_WIDTH * SCREEN_HEIGHT / 10 * 2];
17
18 /* LVGL Image Descriptor mapping the C-array asset */
19 const lv_image_dsc_t lv_img = {
20     .header = {
21         .magic = LV_IMAGE_HEADER_MAGIC,
22         .cf = LV_COLOR_FORMAT_RGB565, // Color format: 16-bit RGB
23         .flags = 0,
24         .w = IMG_W,
25         .h = IMG_H,
26         .stride = IMG_W * 2,          // Bytes per line
27         .reserved_2 = 0
28     },
29     .data_size = IMG_W * IMG_H * 2,
30     .data = img_asset,                // Pointer to external image data
31 };
32
33 /**
34  * Display flushing callback: Transfers pixels from LVGL to hardware
35  */
36 void my_disp_flush(lv_display_t * disp, const lv_area_t * area, uint8_t * px_map) {
37     uint32_t w = (area->x2 - area->x1 + 1);
38     uint32_t h = (area->y2 - area->y1 + 1);
39
40     tft.startWrite();
41     tft.setAddrWindow(area->x1, area->y1, w, h);
42     tft.pushColors((uint16_t *)px_map, w * h, true);
43     tft.endWrite();
44
45     lv_display_flush_ready(disp); // Inform library flushing is complete
46 }

```

```
47
48 void setup() {
49     Serial.begin(115200);
50
51     /* Hardware and library initialization */
52     tft.begin();
53     lv_init();
54
55     /* Create and configure display abstraction layer */
56     lv_display_t * disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
57     lv_display_set_flush_cb(disp, my_disp_flush);
58     lv_display_set_buffers(disp, draw_buf, NULL, sizeof(draw_buf),
59 LV_DISPLAY_RENDER_MODE_PARTIAL);
60
61     /* Initialize UI components: Image display */
62     lv_obj_t * img_obj = lv_image_create(lv_screen_active());
63     lv_image_set_src(img_obj, &lv_img);
64     lv_obj_center(img_obj);
65
66     /* Status label configuration */
67     lv_obj_t * label = lv_label_create(lv_screen_active());
68     lv_label_set_text(label, "LVGL v9 Image Display");
69     lv_obj_align(label, LV_ALIGN_BOTTOM_MID, 0, -20);
70     lv_obj_set_style_text_color(label, lv_color_white(), 0);
71
72     /* Enable display power */
73     pinMode(TFT_VCC, OUTPUT);
74     digitalWrite(TFT_VCC, LOW);
75
76     Serial.println("System initialized: LVGL asset display ready.");
77 }
78
79 void loop() {
80     /* Process timer events and system task handlers */
81     lv_timer_handler();
82
83     /* Advance internal library tick counts */
84     static uint32_t last_tick = 0;
85     uint32_t current_ms = millis();
86     lv_tick_inc(current_ms - last_tick);
87     last_tick = current_ms;
88
89     delay(5);
90 }
```

### Code Explanation

Include the necessary header files

```
1  #include <lvgl.h>
2  #include <TFT_eSPI.h>
3  #include <image.h>
```

Define image size

```
6  #define IMG_W  240
7  #define IMG_H  240
```

Define screen resolution

```
8  #define SCREEN_WIDTH  240
9  #define SCREEN_HEIGHT 240
```

Set the baudrate to 115200

```
8  Serial.begin(115200);
```

Drawing pictures

```
61  /* Initialize UI components: Image display */
62  lv_obj_t * img_obj = lv_image_create(lv_screen_active());
63  lv_image_set_src(img_obj, &lv_img);
64  lv_obj_center(img_obj);
65
66  /* Status label configuration */
67  lv_obj_t * label = lv_label_create(lv_screen_active());
68  lv_label_set_text(label, "LVGL v9 Image Display");
69  lv_obj_align(label, LV_ALIGN_BOTTOM_MID, 0, -20);
70  lv_obj_set_style_text_color(label, lv_color_white(), 0);
```

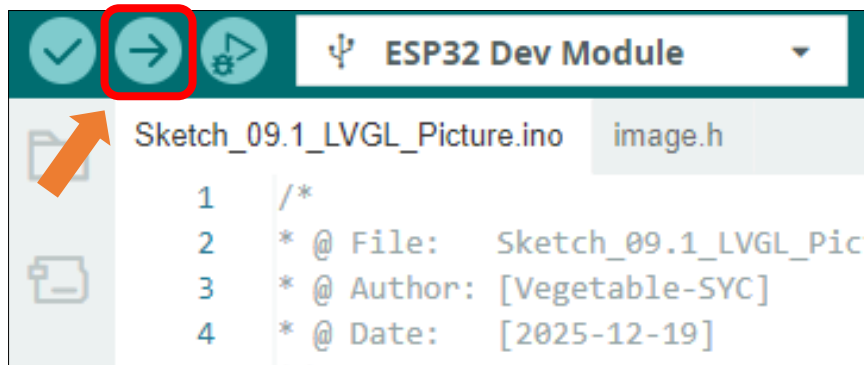
Enable screen power supply pin

```
72  /* Enable display power */
73  pinMode(TFT_VCC, OUTPUT);
74  digitalWrite(TFT_VCC, LOW);
```

Loop through UI tasks

```
80  /* Process timer events and system task handlers */
81  lv_timer_handler();
82
83  /* Advance internal library tick counts */
84  static uint32_t last_tick = 0;
85  uint32_t current_ms = millis();
86  lv_tick_inc(current_ms - last_tick);
87  last_tick = current_ms;
88
89  delay(5);
```

Click “**Upload**” to upload the code to Freenove ESP32 Mini TV



After the code is uploaded, an image will be displayed on the TFT screen



#### Reference

```
lv_obj_t * lv_image_create(lv_obj_t * parent);
```

This function is used to create an image component.

```
void lv_image_set_src(lv_obj_t * obj, const void * src);
```

This function is used to provide a data source for the image component, thereby displaying the specific image content.

#### Custom image display

You can customize the image displayed on the display according to your personal preferences.

First, open **Freenove\_ESP32\_Mini\_TV\Sketch\Sketch\_09.1\_LVGL\_Picture\LVGL\_img.html**



For optimal compatibility, please use Microsoft Edge or Google Chrome. You can open the file by dragging it directly into the browser window, or by right clicking it and selecting "Open with".

If you have any concerns, please feel free to contact us via [support@freenove.com](mailto:support@freenove.com)



Click the **"Drag and drop any image"** area to select an image file, or directly drag and drop a file onto it.

**Freenove ESP32 LVGL v9** v1.0  
Drag & Drop Support | Auto-Scaling | Compatible with ESP32 & Arduino

**1. Drag & Drop or Select Image**  
**Choose File**  
Suggested: 240x240 or adjust based on screen

**2. Target Dimensions (Resizing)**  
WIDTH: 240 HEIGHT: 240  
☒ Keep Aspect Ratio

**3. Color Format**  
RGB565

Source: -  
Output: -  
Data: - Bytes

**Convert & Generate Code**

**C Array Output** **Copy Code**  
Waiting for image import...

With the default settings, click **"Convert & Generate Code"** to generate the array.

**Freenove ESP32 LVGL v9** v1.0  
Drag & Drop Support | Auto-Scaling | Compatible with ESP32 & Arduino

**1. Drag & Drop or Select Image**  
**Choose File**  
Suggested: 240x240 or adjust based on screen

**2. Target Dimensions (Resizing)**  
WIDTH: 240 HEIGHT: 240  
☒ Keep Aspect Ratio

**3. Color Format**  
RGB565

Source: 265x265  
Output: 240x240  
Data: 115,200 Bytes

**Convert & Generate Code**

**C Array Output** **Copy Code**  
Waiting for image import...

Click “Copy Code” to copy the arrays.

### Freenove ESP32 LVGL v9 v1.0

Drag & Drop Support | Auto-Scaling | Compatible with ESP32 & Arduino

1. Drag & Drop or Select Image

[Choose File](#)

Suggested: 240x240 or adjust based on screen

2. Target Dimensions (Resizing)

WIDTH:  HEIGHT:

☒ Keep Aspect Ratio

3. Color Format



Source: 2653x2653  
Output: 240x240  
Data: 115,200 Bytes

[Success! Convert Again](#)

C Array Output

[Copy Code](#)

```

/*
 * @ File: [image.h]
 * @ Type: [RGB565]
 * @ Author: [Vegetable-SYC]
 */

const unsigned char img_asset[] PROGMEM = {
  0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58,
  0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58,
  0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58,
  0x44, 0x60, 0x44, 0x58, 0x44, 0x58, 0x24, 0x58, 0x44, 0x58, 0x24, 0x58, 0x44, 0x58, 0x24, 0x58,
  0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x44, 0x58, 0x45, 0x58, 0x65, 0x58, 0x64, 0x58, 0x65, 0x58, 0x64, 0x58, 0x65, 0x58, 0x64, 0x58,
  0x65, 0x58, 0x65, 0x58, 0x65, 0x58, 0x65, 0x58, 0x64, 0x58, 0x65, 0x58, 0x64, 0x58, 0x65, 0x58,
  0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x23, 0x58, 0x23, 0x58, 0x23, 0x58,
  0x23, 0x58, 0x23, 0x58, 0x43, 0x48, 0x23, 0x48, 0x23, 0x58, 0x23, 0x48, 0x23, 0x48, 0x44, 0x48,
  0x64, 0x48, 0x64, 0x48, 0x64, 0x58, 0x64, 0x48, 0x84, 0x58, 0x84, 0x58, 0x84, 0x48, 0x84, 0x48,
  0xc5, 0x48, 0xa4, 0x48, 0xa4, 0x48, 0xc4, 0x48, 0xc4, 0x48, 0xe5, 0x48, 0xe5, 0x50, 0xc5, 0x50,
  0x24, 0x49, 0x42, 0x41, 0x42, 0x41, 0x42, 0x41, 0x62, 0x41, 0x42, 0x49, 0x62, 0x49, 0x51, 0x45,
  0x69, 0x71, 0x48, 0x71, 0x28, 0x69, 0x48, 0x69, 0x28, 0x61, 0x48, 0x69, 0x48, 0x61, 0x48, 0x61,
  0x48, 0x69, 0x48, 0x61, 0x48, 0x61, 0x48, 0x59, 0x47, 0x59, 0x68, 0x59, 0x67, 0x59, 0x26, 0x51,
  0x85, 0x49, 0x85, 0x49, 0xe5, 0x48, 0x85, 0x49, 0x85, 0x49, 0x85, 0x49, 0x85, 0x49, 0x85, 0x49,
  0x85, 0x49, 0x25, 0x49, 0x25, 0x49, 0x25, 0x49, 0x25, 0x49, 0x46, 0x49, 0x46, 0x49, 0x46, 0x49,
  0x46, 0x51, 0x46, 0x49, 0x46, 0x49, 0x46, 0x51, 0x46, 0x51, 0x47, 0x51, 0x47, 0x51, 0x27, 0x59,
  0x87, 0x59, 0x87, 0x61, 0xe7, 0x60, 0xe6, 0x60, 0xe6, 0x60, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6,
  0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6, 0xc6,
  0xa6, 0x60, 0x86, 0x60, 0x85, 0x60, 0x85, 0x60, 0x85, 0x60, 0x85, 0x60, 0x85, 0x60, 0x85, 0x60,
  0xe5, 0x60, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58,
  0x45, 0x58, 0x45, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x44, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x58,
  0x44, 0x58, 0x45, 0x58, 0x45, 0x58, 0x45, 0x60, 0x45, 0x60, 0x45, 0x60, 0x45, 0x60, 0x45, 0x60,
  0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x45, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58, 0x44, 0x58,
  0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58,
  0x24, 0x58, 0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x45, 0x58, 0x45, 0x58,
  0x24, 0x58, 0x24, 0x58, 0x44, 0x58, 0x44, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58, 0x24, 0x58,

```

Replace all the content in the **image.h** file with the copied arrays.

✓

→

⚙️

ESP32 Dev Module

Sketch\_09.1\_LVGL\_Picture.ino

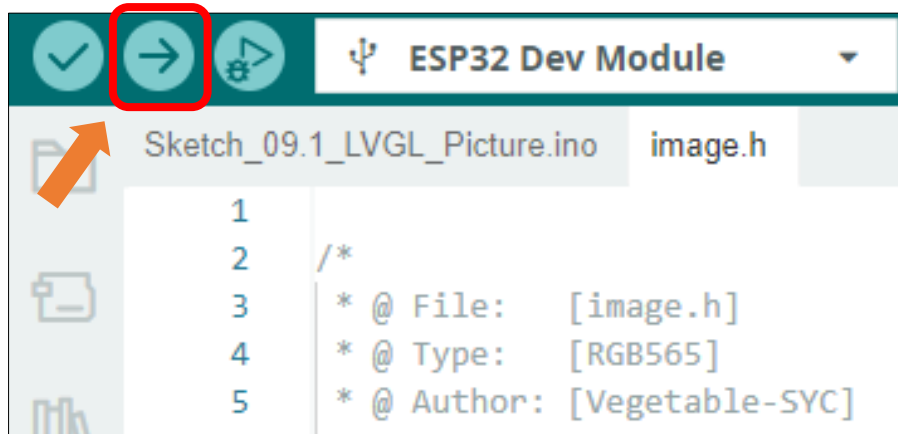
**image.h**

```

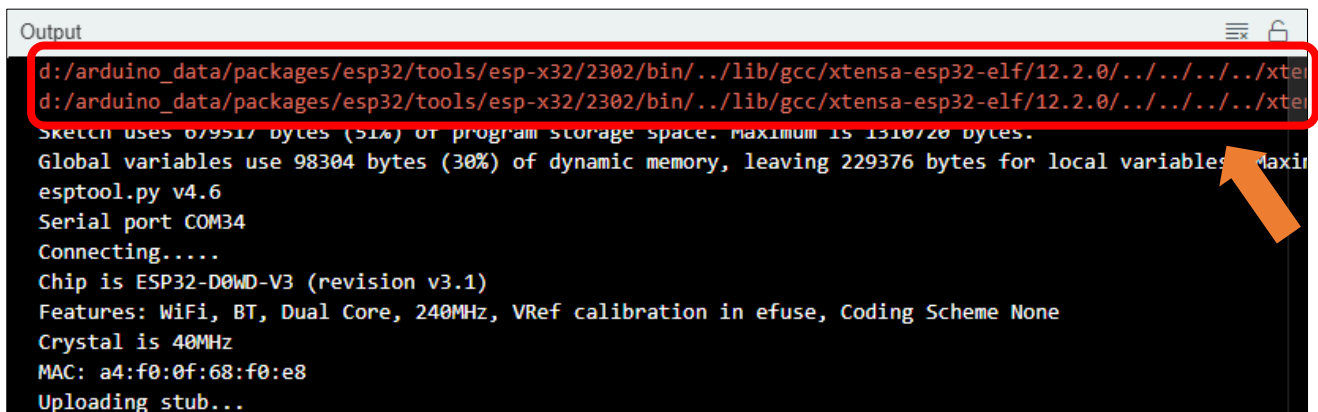
1
2  /*
3   * @ File: [image.h]
4   * @ Type: [RGB565]
5   * @ Author: [Vegetable-SYC]
6   */
7
8  const unsigned char img_asset[] PROGMEM = {
9      0x49, 0x4a, 0x69, 0x4a, 0x49, 0x4a, 0x28, 0x42,
10     0x49, 0x4a, 0x69, 0x4a, 0x49, 0x4a, 0x49, 0x4a,
11     0x49, 0x4a, 0x49, 0x4a, 0x49, 0x4a, 0x49, 0x4a,

```

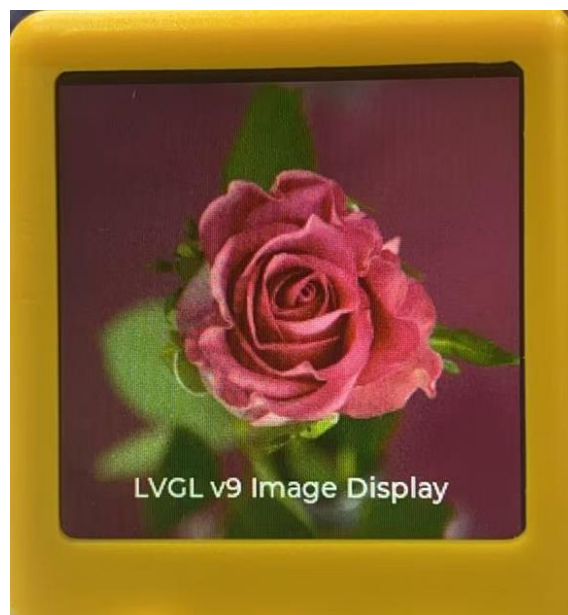
Click “**Upload**” to upload the code to Freenove ESP32 Mini TV. Set the baud rate to 115200.



**Note:** The following warning messages may appear in the console during compilation. They do not affect code execution and can be safely ignored.



After uploading the code, the custom image will be displayed on the screen.



## Chapter 10 Lvgl Timer

### Project 10.1 Lvgl Timer

#### Component List

|                                                                                                             |                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Freenove ESP32 Mini TV<br> | USB cable x1<br> |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|

#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



#### Sketch

Open “Sketch\_10.1\_LVGL\_Timer” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_10.1\_LVGL\_Timer.ino”.

Sketch\_10.1\_LVGL\_Timer

The following is the program code:

```
1 #include <lvgl.h>
2 #include <TFT_eSPI.h>
```

```

3
4  /* Hardware mapping & Constants */
5  const int TOUCH_PIN = 32;          // Capacitive touch GPIO
6  const int TFT_VCC = 21;           // Display power control
7  const int TOUCH_THRESHOLD = 80;   // Sensitivity calibration
8  const int PRESS_TIME = 3000;      // Reset threshold (ms)
9  int touch_base_val = 0;           // Auto-calibrated baseline
10
11 /* Display specifications */
12 #define SCREEN_WIDTH 240
13 #define SCREEN_HEIGHT 240
14
15 TFT_eSPI tft = TFT_eSPI();
16
17 /* Memory-aligned draw buffer for optimized DMA/rendering */
18 static uint32_t buf_aligned[ (SCREEN_WIDTH * SCREEN_HEIGHT / 10 * 2) / 4 + 1 ];
19 uint8_t * draw_buf_ptr = (uint8_t *)buf_aligned;
20
21 /* Global state machine & variables */
22 bool is_running = false;
23 uint32_t elapsed_time = 0;
24 uint32_t last_millis = 0;
25
26 /* UI Object handles */
27 lv_obj_t * label_time;
28 lv_obj_t * label_status;
29 lv_obj_t * arc_deco;
30
31 /**
32  * Flush callback: Bridges LVGL software rendering to hardware SPI
33  */
34 void my_disp_flush(lv_display_t * disp, const lv_area_t * area, uint8_t * px_map) {
35     uint32_t w = (area->x2 - area->x1 + 1);
36     uint32_t h = (area->y2 - area->y1 + 1);
37     tft.startWrite();
38     tft.setAddrWindow(area->x1, area->y1, w, h);
39     tft.pushColors((uint16_t *)px_map, w * h, true);
40     tft.endWrite();
41     lv_display_flush_ready(disp);
42 }
43
44 /**
45  * Input device callback: Logic for capacitive touch state acquisition
46  */

```

```

47 void my_touchpad_read(lv_indev_t * indev, lv_indev_data_t * data) {
48     int val = touchRead(TOUCH_PIN);
49     if (touch_base_val - val > TOUCH_THRESHOLD) {
50         data->state = LV_INDEV_STATE_PRESSED;
51     } else {
52         data->state = LV_INDEV_STATE_RELEASED;
53     }
54     data->btn_id = 0;
55 }
56
57 /**
58  * Timer callback: Manages chronological calculations and visual effects
59  */
60 void timer_refresh_cb(lv_timer_t * timer) {
61     if (is_running) {
62         uint32_t now = millis();
63         elapsed_time += (now - last_millis);
64         last_millis = now;
65
66         /* Calculate clock components */
67         uint32_t sec = (elapsed_time / 1000) % 60;
68         uint32_t min = (elapsed_time / 60000) % 60;
69         uint32_t ms = (elapsed_time % 1000) / 100;
70         lv_label_set_text_fmt(label_time, "%02d:%02d.%d", min, sec, ms);
71
72         /* Procedural arc animation */
73         static uint16_t angle = 0;
74         angle = (angle + 4) % 360;
75         lv_arc_set_rotation(arc_deco, angle);
76     }
77 }
78
79 /**
80  * Interaction event manager: Handles toggle and reset logic
81  */
82 void btn_event_cb(lv_event_t * e) {
83     lv_event_code_t code = lv_event_get_code(e);
84
85     if(code == LV_EVENT_SHORT_CLICKED) {
86         is_running = !is_running;
87         if(is_running) {
88             last_millis = millis();
89             lv_label_set_text(label_status, "RUNNING");
90             lv_obj_set_style_text_color(label_status, lv_palette_main(LV_PALETTE_GREEN), 0);

```

```

91         lv_obj_set_style_arc_color(arc_deco, lv_palette_main(LV_PALETTE_GREEN),
92 LV_PART_INDICATOR);
93     } else {
94         lv_label_set_text(label_status, "PAUSED");
95         lv_obj_set_style_text_color(label_status, lv_palette_main(LV_PALETTE_ORANGE), 0);
96         lv_obj_set_style_arc_color(arc_deco, lv_palette_main(LV_PALETTE_ORANGE),
97 LV_PART_INDICATOR);
98     }
99 }
100 else if(code == LV_EVENT_LONG_PRESSED) {
101     /* System Reset Sequence */
102     is_running = false;
103     elapsed_time = 0;
104     lv_label_set_text(label_time, "00:00.0");
105     lv_label_set_text(label_status, "READY");
106     lv_obj_set_style_text_color(label_status, lv_palette_main(LV_PALETTE_CYAN), 0);
107     lv_obj_set_style_arc_color(arc_deco, lv_palette_main(LV_PALETTE_CYAN),
108 LV_PART_INDICATOR);
109     Serial.println("System Reset Successful");
110 }
111 }
112
113 void setup() {
114     Serial.begin(115200);
115
116     /* Hardware Layer Init */
117     tft.begin();
118     tft.fillScreen(TFT_BLACK);
119     touchSetCycles(0xf000, 0x1000);
120     touch_base_val = touchRead(TOUCH_PIN);
121
122     /* Framework Layer Init */
123     lv_init();
124
125     /* Display Registration */
126     lv_display_t * disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
127     lv_display_set_flush_cb(disp, my_disp_flush);
128     lv_display_set_buffers(disp, draw_buf_ptr, NULL, (SCREEN_WIDTH * SCREEN_HEIGHT / 10) * 2,
129 LV_DISPLAY_RENDER_MODE_PARTIAL);
130
131     /* Input Device Configuration */
132     lv_indev_t * indev = lv_indev_create();
133     lv_indev_set_type(indev, LV_INDEV_TYPE_BUTTON);
134     lv_indev_set_read_cb(indev, my_touchpad_read);

```

```

135     lv_indev_set_long_press_time(indev, PRESS_TIME);
136
137     /* Map hardware sensor to logical button area */
138     static const lv_point_t btn_points[] = { {120, 120} };
139     lv_indev_set_button_points(indev, btn_points);
140
141     /* UI Theme & Orchestration */
142     lv_obj_t * scr = lv_screen_active();
143     lv_obj_set_style_bg_color(scr, lv_color_hex(0x000000), 0);
144
145     /* Stylized Decorative Arc */
146     arc_deco = lv_arc_create(scr);
147     lv_obj_set_size(arc_deco, 220, 220);
148     lv_arc_set_range(arc_deco, 0, 100);
149     lv_arc_set_value(arc_deco, 15);
150     lv_arc_set_bg_angles(arc_deco, 0, 360);
151     lv_obj_set_style_arc_width(arc_deco, 6, LV_PART_MAIN);
152     lv_obj_set_style_arc_width(arc_deco, 6, LV_PART_INDICATOR);
153     lv_obj_set_style_arc_color(arc_deco, lv_color_hex(0x1A1A1A), LV_PART_MAIN);
154     lv_obj_set_style_arc_color(arc_deco, lv_palette_main(LV_PALETTE_CYAN),
155 LV_PART_INDICATOR);
156     lv_obj_remove_style(arc_deco, NULL, LV_PART_KNOB);
157     lv_obj_center(arc_deco);
158
159     /* Button Interaction */
160     lv_obj_t * btn_full = lv_button_create(scr);
161     lv_obj_set_size(btn_full, SCREEN_WIDTH, SCREEN_HEIGHT);
162     lv_obj_set_style_bg_opa(btn_full, 0, 0);
163     lv_obj_set_style_border_opa(btn_full, 0, 0);
164     lv_obj_set_style_shadow_opa(btn_full, 0, 0);
165     lv_obj_add_event_cb(btn_full, btn_event_cb, LV_EVENT_ALL, NULL);
166
167     /* Chrono Readout Setup */
168     label_time = lv_label_create(scr);
169     lv_label_set_text(label_time, "00:00.0");
170     lv_obj_set_style_text_font(label_time, &lv_font_montserrat_40, 0);
171     lv_obj_set_style_text_color(label_time, lv_color_white(), 0);
172     lv_obj_align(label_time, LV_ALIGN_TOP_LEFT, 50, 95);
173     lv_obj_set_style_text_align(label_time, LV_TEXT_ALIGN_LEFT, 0);
174
175     /* Secondary Information Readout */
176     label_status = lv_label_create(scr);
177     lv_label_set_text(label_status, "READY");
178     lv_obj_set_style_text_font(label_status, &lv_font_montserrat_14, 0);

```



```

179     lv_obj_set_style_text_color(label_status, lv_palette_main(LV_PALETTE_CYAN), 0);
180     lv_obj_align(label_status, LV_ALIGN_CENTER, 0, 45);
181
182     /* Design separation line */
183     lv_obj_t * line = lv_obj_create(scr);
184     lv_obj_set_size(line, 80, 2);
185     lv_obj_set_style_bg_color(line, lv_color_hex(0x333333), 0);
186     lv_obj_set_style_border_opa(line, 0, 0);
187     lv_obj_align(line, LV_ALIGN_CENTER, 0, 20);
188
189     /* Primary Application Task Initiation */
190     lv_timer_create(timer_refresh_cb, 50, NULL);
191
192     /* TFT Power Engagement */
193     pinMode(TFT_VCC, OUTPUT);
194     digitalWrite(TFT_VCC, LOW);
195 }
196
197 void loop() {
198     /* Handle pending framework tasks and signal management */
199     lv_timer_handler();
200
201     /* System synchronization - synchronize library clock to hardware millis */
202     static uint32_t last_tick = 0;
203     uint32_t current_ms = millis();
204     lv_tick_inc(current_ms - last_tick);
205     last_tick = current_ms;
206
207     delay(5);
208 }

```

### Code Explanation

Include the necessary header files

```

1  #include <lvgl.h>
2  #include <TFT_eSPI.h>

```

Preset capacitive touch related parameters

```

5  const int TOUCH_PIN = 32;           // Capacitive touch GPIO
6  const int TFT_VCC = 21;             // Display power control
7  const int TOUCH_THRESHOLD = 80;     // Sensitivity calibration
8  const int PRESS_TIME = 3000;        // Reset threshold (ms)

```

Define screen resolution

```

12 #define SCREEN_WIDTH  240
13 #define SCREEN_HEIGHT 240

```

Draw the outer ring of the timer

```

146 /* Stylized Decorative Arc */

```

```

147 arc_deco = lv_arc_create(scr);
148 lv_obj_set_size(arc_deco, 220, 220);
149 lv_arc_set_range(arc_deco, 0, 100);
150 lv_arc_set_value(arc_deco, 15);
151 lv_arc_set_bg_angles(arc_deco, 0, 360);
152 lv_obj_set_style_arc_width(arc_deco, 6, LV_PART_MAIN);
153 lv_obj_set_style_arc_width(arc_deco, 6, LV_PART_INDICATOR);
154 lv_obj_set_style_arc_color(arc_deco, lv_color_hex(0x1A1A1A), LV_PART_MAIN);
155 lv_obj_set_style_arc_color(arc_deco, lv_palette_main(LV_PALETTE_CYAN), LV_PART_INDICATOR);
156 lv_obj_remove_style(arc_deco, NULL, LV_PART_KNOB);
157 lv_obj_center(arc_deco);
158
159 /* Button Interaction */
160 lv_obj_t * btn_full = lv_button_create(scr);
161 lv_obj_set_size(btn_full, SCREEN_WIDTH, SCREEN_HEIGHT);
162 lv_obj_set_style_bg_opa(btn_full, 0, 0);
163 lv_obj_set_style_border_opa(btn_full, 0, 0);
164 lv_obj_set_style_shadow_opa(btn_full, 0, 0);
165 lv_obj_add_event_cb(btn_full, btn_event_cb, LV_EVENT_ALL, NULL);
166
167 /* Chrono Readout Setup */
168 label_time = lv_label_create(scr);
169 lv_label_set_text(label_time, "00:00.0");
170 lv_obj_set_style_text_font(label_time, &lv_font_montserrat_40, 0);
171 lv_obj_set_style_text_color(label_time, lv_color_white(), 0);
172 lv_obj_align(label_time, LV_ALIGN_TOP_LEFT, 50, 95);
173 lv_obj_set_style_text_align(label_time, LV_TEXT_ALIGN_LEFT, 0);
174
175 /* Secondary Information Readout */
176 label_status = lv_label_create(scr);
177 lv_label_set_text(label_status, "READY");
178 lv_obj_set_style_text_font(label_status, &lv_font_montserrat_14, 0);
179 lv_obj_set_style_text_color(label_status, lv_palette_main(LV_PALETTE_CYAN), 0);
180 lv_obj_align(label_status, LV_ALIGN_CENTER, 0, 45);
181
182 /* Design separation line */
183 lv_obj_t * line = lv_obj_create(scr);
184 lv_obj_set_size(line, 80, 2);
185 lv_obj_set_style_bg_color(line, lv_color_hex(0x333333), 0);
186 lv_obj_set_style_border_opa(line, 0, 0);
187 lv_obj_align(line, LV_ALIGN_CENTER, 0, 20);

```

Create a timer task

```

190 lv_timer_create(timer_refresh_cb, 50, NULL);

```

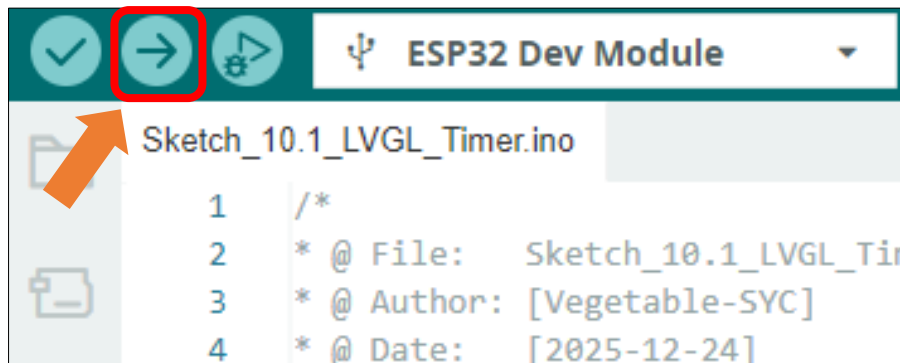
Enable screen power supply pin

```
193  /* TFT Power Engagement */
194  pinMode(TFT_VCC, OUTPUT);
195  digitalWrite(TFT_VCC, LOW);
```

Loop through UI tasks

```
198  /* Handle pending framework tasks and signal management */
199  lv_timer_handler();
200
201  /* System synchronization - synchronize library clock to hardware millis */
202  static uint32_t last_tick = 0;
203  uint32_t current_ms = millis();
204  lv_tick_inc(current_ms - last_tick);
205  last_tick = current_ms;
206
207  delay(5);
```

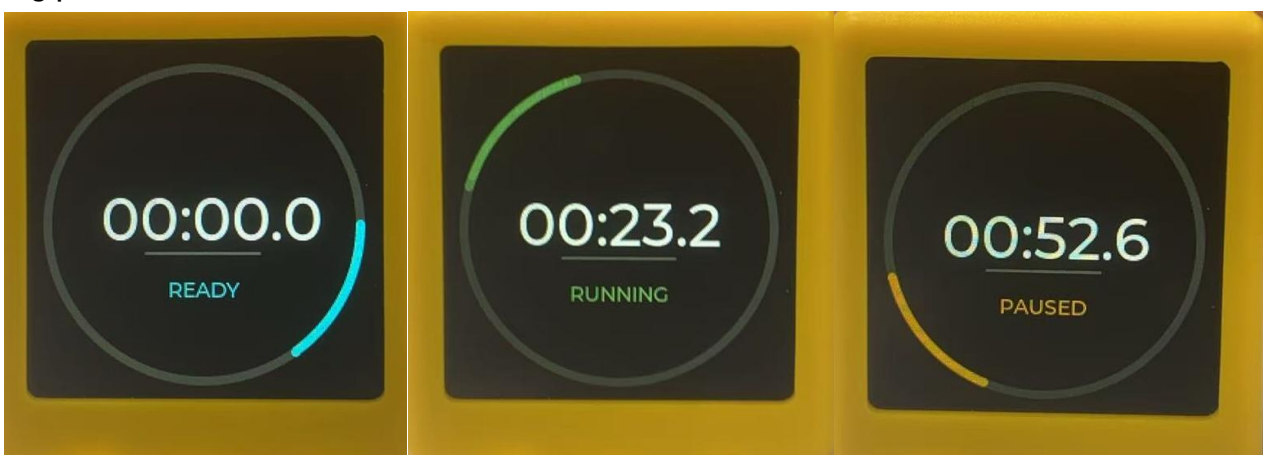
Click "Upload" to upload the code to Freenove ESP32 Mini TV



Touch the capacitive touch area of the Freenove ESP32 Mini TV with your finger:

**Short press:** Start/Pause timer;

**Long press:** Reset timer.



## Chapter 11 LVGL Game

Having explored key concepts like EEPROM, capacitive touch, TFT display, and LVGL, we will now synthesize this knowledge to create a classic and entertaining mini-game: Flappy Bird.

### Project 11.1 Lvgl Game

#### Component List

|                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  A yellow, cube-shaped electronic device with a black rectangular screen on the front face. The top face features a circular logo with a hand icon and the text 'ESP32' and 'TOUCH'. | <p>USB cable x1</p>  A black USB cable with a standard USB-A connector on one end and a micro-USB connector on the other. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.



## Sketch

Open “**Sketch\_11.1\_LVGL\_Game**” folder under “**Freenove\_ESP32\_Mini\_TV\Sketch**” and double-click “**Sketch\_11.1\_LVGL\_Game.ino**”.

### Sketch\_11.1\_LVGL\_Game

The following is the program code:

```
1  #include <lvgl.h>
2  #include <TFT_eSPI.h>
3  #include "image.h"
4  #include <EEPROM.h>
5
6  /* --- Hardware Configuration --- */
7  const int TOUCH_PIN = 32;
8  const int TFT_VCC = 21;
9  const int TOUCH_THRESHOLD = 80;
10 #define SCREEN_WIDTH 240
11 #define SCREEN_HEIGHT 240
12 #define EEPROM_SIZE 4
13
14 /* --- Graphics Instances --- */
15 TFT_eSPI tft = TFT_eSPI();
16 static uint32_t buf_aligned[ (SCREEN_WIDTH * SCREEN_HEIGHT / 10 * 2) / 4 + 4 ];
17
18 /* --- LVGL Image Descriptor --- */
19 const lv_image_dsc_t img_bg = {
20     .header = {
21         .magic = LV_IMAGE_HEADER_MAGIC,
22         .cf = LV_COLOR_FORMAT_RGB565,
23         .flags = 0,
24         .w = 240,
25         .h = 240,
26         .stride = 240 * 2,
27         .reserved_2 = 0,
28     },
29     .data_size = sizeof(img_bg_data),
30     .data = img_bg_data,
31     .reserved = NULL,
32 };
33
34 /* --- Game Physics & Constants --- */
35 const int GRAVITY = 1;
36 const int JUMP_FORCE = -9;
37 const int PIPE_SPEED = 4;
```

```
38  const int PIPE_GAP = 100;
39  const int PIPE_WIDTH = 40;
40
41  /* --- Game State Variables --- */
42  int bird_y = 120;
43  int bird_vel = 0;
44  int pipe_x = 240;
45  int pipe_gap_y = 100;
46  int score = 0;
47  bool game_running = false;
48  int touch_base_val = 0;
49  int best_score = 0;
50
51  /* --- UI Object References --- */
52  lv_obj_t * bird = NULL;
53  lv_obj_t * pipe_top = NULL;
54  lv_obj_t * pipe_bottom = NULL;
55  lv_obj_t * label_score = NULL;
56  lv_obj_t * label_msg = NULL;
57
58  /**
59   * @brief Load best score from EEPROM
60   */
61  void load_best_score() {
62      EEPROM.begin(EEPROM_SIZE);
63      EEPROM.get(0, best_score);
64      if (best_score < 0 || best_score > 9999) {
65          best_score = 0;
66      }
67  }
68
69  /**
70   * @brief Handle game termination logic
71   */
72  void game_over_action() {
73      game_running = false;
74
75      bool is_new_record = false;
76      if (score > best_score) {
77          best_score = score;
78          EEPROM.put(0, best_score);
79          EEPROM.commit();
80          is_new_record = true;
81      }
```

```

82
83     lv_label_set_text_fmt(label_msg,
84         "%s\nSCORE: %d\nBEST: %d",
85         is_new_record ? "#008000 NEW RECORD!#" : "#FF0000 GAME OVER#",
86         score,
87         best_score);
88
89     lv_label_set_recolor(label_msg, true);
90     lv_obj_remove_flag(label_msg, LV_OBJ_FLAG_HIDDEN);
91 }
92
93 /**
94  * @brief LVGL Display flushing callback
95  */
96 void my_disp_flush(lv_display_t * disp, const lv_area_t * area, uint8_t * px_map) {
97     uint32_t w = (area->x2 - area->x1 + 1);
98     uint32_t h = (area->y2 - area->y1 + 1);
99     tft.startWrite();
100    tft.setAddrWindow(area->x1, area->y1, w, h);
101    tft.pushColors((uint16_t *)px_map, w * h, true);
102    tft.endWrite();
103    lv_display_flush_ready(disp);
104 }
105
106 /**
107  * @brief Touch input reading callback
108  */
109 void my_touchpad_read(lv_indev_t * indev, lv_indev_data_t * data) {
110     int val = touchRead(TOUCH_PIN);
111     if (touch_base_val - val > TOUCH_THRESHOLD) {
112         data->state = LV_INDEV_STATE_PRESSED;
113     } else {
114         data->state = LV_INDEV_STATE_RELEASED;
115     }
116 }
117
118 /**
119  * @brief Main game loop timer callback
120  */
121 void game_loop_cb(lv_timer_t * timer) {
122     if (!game_running || bird == NULL) return;
123
124     // Apply gravity and update bird position
125     bird_vel += GRAVITY;

```

```

126     bird_y += bird_vel;
127     lv_obj_set_y(bird, bird_y);
128
129     // Update pipe position and handle scrolling
130     pipe_x -= PIPE_SPEED;
131     if (pipe_x < -PIPE_WIDTH) {
132         pipe_x = SCREEN_WIDTH;
133         pipe_gap_y = 30 + rand() % 80;
134         score++;
135         lv_label_set_text_fmt(label_score, "%d", score);
136     }
137
138     // Refresh pipe UI elements
139     if(pipe_top && pipe_bottom) {
140         lv_obj_set_x(pipe_top, pipe_x);
141         lv_obj_set_x(pipe_bottom, pipe_x);
142         lv_obj_set_height(pipe_top, pipe_gap_y);
143         lv_obj_set_y(pipe_bottom, pipe_gap_y + PIPE_GAP);
144         lv_obj_set_height(pipe_bottom, SCREEN_HEIGHT - (pipe_gap_y + PIPE_GAP));
145     }
146
147     // Boundary collision detection
148     if (bird_y < 0 || bird_y > (SCREEN_HEIGHT - 20)) {
149         game_running = false;
150         lv_obj_remove_flag(label_msg, LV_OBJ_FLAG_HIDDEN);
151         lv_label_set_text(label_msg, "CRASHED!\nTap to Restart");
152         game_over_action();
153     }
154
155     // Pipe collision detection logic
156     if (pipe_x < 60 && (pipe_x + PIPE_WIDTH) > 40) {
157         if (bird_y < pipe_gap_y || (bird_y + 18) > (pipe_gap_y + PIPE_GAP)) {
158             game_running = false;
159             lv_obj_remove_flag(label_msg, LV_OBJ_FLAG_HIDDEN);
160             lv_label_set_text(label_msg, "HIT PIPE!\nTap to Restart");
161             game_over_action();
162         }
163     }
164 }
165
166 /**
167  * @brief UI event handler for touch interactions
168  */
169 void event_handler(lv_event_t * e) {

```



```
170     if (lv_event_get_code(e) == LV_EVENT_PRESSED) {
171         if (!game_running) {
172             // Reset game state on restart
173             bird_y = 100; bird_vel = 0; pipe_x = 240; score = 0;
174             lv_label_set_text(label_score, "0");
175             lv_obj_add_flag(label_msg, LV_OBJ_FLAG_HIDDEN);
176             game_running = true;
177         } else {
178             bird_vel = -7.5;
179         }
180     }
181 }
182
183 void setup() {
184     Serial.begin(115200);
185     delay(1000);
186
187     load_best_score();
188     Serial.printf("Best score loaded: %d\n", best_score);
189
190     // Hardware Pin Setup
191     pinMode(TFT_VCC, OUTPUT);
192     digitalWrite(TFT_VCC, LOW);
193     delay(100);
194
195     // Display Driver Setup
196     tft.begin();
197     tft.fillScreen(TFT_BLACK);
198
199     // Touch Calibration
200     touchSetCycles(0xf000, 0x1000);
201     touch_base_val = touchRead(TOUCH_PIN);
202
203     // LVGL Framework Initialization
204     lv_init();
205     Serial.println("LVGL Init OK");
206
207     // Display Interface Configuration
208     lv_display_t * disp = lv_display_create(SCREEN_WIDTH, SCREEN_HEIGHT);
209     if(!disp) return;
210     lv_display_set_flush_cb(disp, my_disp_flush);
211     lv_display_set_buffers(disp, (uint8_t *)buf_aligned, NULL, sizeof(buf_aligned),
212 LV_DISPLAY_RENDER_MODE_PARTIAL);
213 }
```

```

214 // Input Device Registration
215 lv_indev_t * indev = lv_indev_create();
216 lv_indev_set_type(indev, LV_INDEV_TYPE_BUTTON);
217 lv_indev_set_read_cb(indev, my_touchpad_read);
218 static const lv_point_t btn_pts[] = {{120, 120}};
219 lv_indev_set_button_points(indev, btn_pts);
220
221 // UI Layout - Background
222 lv_obj_set_style_bg_color(lv_screen_active(), lv_color_hex(0x050510), 0);
223 lv_obj_t * bg_img = lv_image_create(lv_screen_active());
224 lv_image_set_src(bg_img, &img_bg);
225 lv_obj_center(bg_img);
226
227 // Button
228 lv_obj_t * bg_btn = lv_button_create(lv_screen_active());
229 lv_obj_set_size(bg_btn, SCREEN_WIDTH, SCREEN_HEIGHT);
230 lv_obj_set_style_bg_opa(bg_btn, 0, 0);
231 lv_obj_set_style_border_opa(bg_btn, 0, 0);
232 lv_obj_add_event_cb(bg_btn, event_handler, LV_EVENT_ALL, NULL);
233
234 // UI Layout - Pipes
235 pipe_top = lv_obj_create(lv_screen_active());
236 lv_obj_set_size(pipe_top, PIPE_WIDTH, 100);
237 lv_obj_set_style_bg_color(pipe_top, lv_palette_main(LV_PALETTE_GREEN), 0);
238 lv_obj_set_style_border_color(pipe_top, lv_palette_main(LV_PALETTE_GREEN), 0);
239 lv_obj_set_style_border_width(pipe_top, 2, 0);
240
241 pipe_bottom = lv_obj_create(lv_screen_active());
242 lv_obj_set_size(pipe_bottom, PIPE_WIDTH, 100);
243 lv_obj_set_style_bg_color(pipe_bottom, lv_palette_main(LV_PALETTE_GREEN), 0);
244 lv_obj_set_style_border_color(pipe_bottom, lv_palette_main(LV_PALETTE_GREEN), 0);
245 lv_obj_set_style_border_width(pipe_bottom, 2, 0);
246
247 // UI Layout - Character (Bird)
248 bird = lv_obj_create(lv_screen_active());
249 lv_obj_set_size(bird, 18, 18);
250 lv_obj_set_style_radius(bird, LV_RADIUS_CIRCLE, 0);
251 lv_obj_set_style_bg_color(bird, lv_palette_main(LV_PALETTE_PINK), 0);
252 lv_obj_set_x(bird, 40);
253 lv_obj_set_scrollbar_mode(bird, LV_SCROLLBAR_MODE_OFF);
254
255 // UI Layout - Score
256 label_score = lv_label_create(lv_screen_active());
257 lv_label_set_text(label_score, "0");

```

```

258 lv_obj_set_style_text_font(label_score, &lv_font_montserrat_40, 0);
259 lv_obj_set_style_text_color(label_score, lv_color_white(), 0);
260 lv_obj_set_style_text_opa(label_score, 255, 0);
261 lv_obj_align(label_score, LV_ALIGN_TOP_MID, 0, 20);
262
263 // UI Layout - Information
264 label_msg = lv_label_create(lv_screen_active());
265 lv_label_set_recolor(label_msg, true);
266 lv_obj_set_style_text_align(label_msg, LV_TEXT_ALIGN_CENTER, 0);
267 lv_label_set_text_fmt(label_msg,
268     "#FF3399 CYBER BIRD#\n\n#FF0000 BEST SCORE: %d#\n#000000 Touch to Fly#",
269     best_score);
270
271 lv_obj_set_style_text_color(label_msg, lv_color_white(), 0);
272 lv_obj_center(label_msg);
273
274 // Start timer
275 lv_timer_create(game_loop_cb, 30, NULL);
276 Serial.println("Setup Complete");
277 }
278
279 void loop() {
280     // Refresh LVGL tasks
281     lv_timer_handler();
282     static uint32_t last_tick = 0;
283     uint32_t current_ms = millis();
284     lv_tick_inc(current_ms - last_tick);
285     last_tick = current_ms;
286     delay(5);
287 }

```

Click **Upload** to upload the code to Freenove ESP32 Mini TV



### Game Rules

To start the game, tap the capacitive touch area after powering on the device.

Control the ball by briefly tapping the touch zone to make it rise; release to let it fall. Navigate through obstacle pillars to score points—each successful pass earns one point. The game ends if the ball hits any obstacle or the screen edges.

A new high score is automatically saved to EEPROM and persists after restart, preserving your best achievement.



## Chapter 12 LVGL Weather Clock

### Project 12.1 Weather Clock

#### Component List

|                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Freenove ESP32 Mini TV</p>  A yellow, cube-shaped electronic device with a black rectangular screen on the front face. The top face has a circular logo with a stylized 'P' and 'F' and the text 'ESP32' below it. | <p>USB cable x1</p>  A black USB cable with a standard USB-A connector on one end and a micro-USB connector on the other. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Circuit

Connect Freenove ESP32 Mini TV to the computer with USB cable.

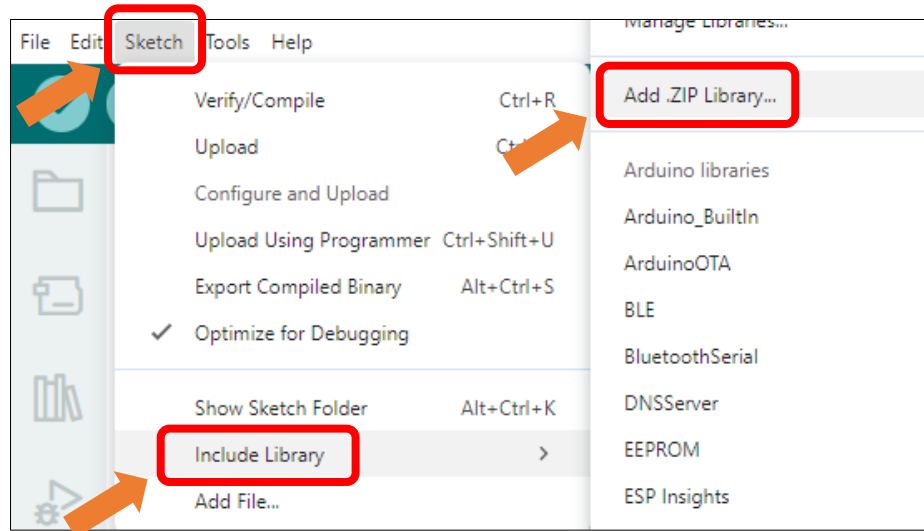


## Sketch

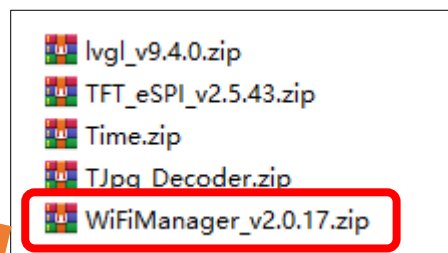
Open “Sketch\_12.1\_LVGL\_Weather\_Clock” folder under “Freenove\_ESP32\_Mini\_TV\Sketch” and double-click “Sketch\_12.1\_LVGL\_Weather\_Clock.ino”.

Install the needed libraries.

Click **Sketch** -> **Include Library** -> **Add .ZIP Library...**



Select **WiFiManager\_v2.0.17.zip**



Sketch\_12.1\_LVGL\_Weather\_Clock

The following is the program code:

```

1  #include "debug_log.h"
2  #include "lvgl_driver.h"
3  #include "wifi_setup.h"
4  #include "config_store.h"
5  #include "ntp_time.h"
6  #include "weather.h"
7  #include "ui_clock.h"
8  #include "touch_button.h"
9
10 #ifndef WEATHER_REFRESH_MS
11 #define WEATHER_REFRESH_MS (10UL * 60UL * 1000UL)
12 #endif
13
14 static uint32_t last_time_ms = 0;
```

```
15 static uint32_t last_weather_ms = 0;
16 static bool ui_ready = false;
17
18 static void boot_status_live(const char *text) {
19     ui_clock_show_boot(text);
20 }
21
22 /** Show a boot page */
23 static void boot_page(BootPage page, const char *detail, uint32_t hold_ms = 450) {
24     ui_clock_show_boot_page(page, detail);
25     uint32_t start = millis();
26     while (millis() - start < hold_ms) {
27         lvgl_driver_handler();
28         delay(5);
29     }
30 }
31
32 /** Fetch weather */
33 static bool refresh_weather(void) {
34     if (!ui_ready) {
35         boot_page(BOOT_PAGE_WEATHER, "Locating city\nby public IP...", 300);
36     }
37
38     bool ok = weather_update_auto();
39
40     if (ok && ui_ready) {
41         ui_clock_update_weather(weather_get());
42     } else if (!ok && !ui_ready) {
43         boot_page(BOOT_PAGE_WEATHER, "Weather failed\nWill retry later", 600);
44     }
45
46     last_weather_ms = millis();
47     return ok;
48 }
49
50 static void on_theme_long_press(void) {
51     uint8_t next = ui_clock_toggle_theme();
52     config_store_save_theme(next);
53     DBG_PRINTF("[UI] Theme -> %s\n", next == THEME_LIGHT ? "light" : "dark");
54 }
55
56 static void on_face_long_press(void) {
57     uint8_t next = ui_clock_toggle_face();
58     config_store_save_ui_face(next);
```

```

59     const char *name = "weather";
60     if (next == UI_FACE_ANALOG) {
61         name = "analog";
62     } else if (next == UI_FACE_XL) {
63         name = "xl";
64     }
65     DBG_PRINTF("[UI] Face -> %s\n", name);
66 }
67
68 void setup() {
69     DBG_BEGIN(115200);
70     delay(200);
71
72     lvgl_driver_init();
73     touch_button_begin();
74     config_store_begin();
75     ui_clock_set_theme(config_store_get().theme);
76
77     boot_page(BOOT_PAGE_START, "Weather Clock\nStarting system...", 500);
78     DBG_PRINTLN("System Ready");
79
80     boot_page(BOOT_PAGE_WIFI, "Preparing WiFi...", 350);
81     if (!wifi_setup_begin(boot_status_live)) {
82         boot_page(BOOT_PAGE_WIFI, "WiFi setup failed\nReset to retry", 0);
83         return;
84     }
85     boot_page(BOOT_PAGE_WIFI, "WiFi connected", 400);
86
87     boot_page(BOOT_PAGE_NTP, "Syncing time\nvia NTP...", 350);
88     if (!ntp_time_begin(config_store_get().timezone)) {
89         boot_page(BOOT_PAGE_NTP, "NTP sync failed\nClock may be wrong", 800);
90     } else {
91         boot_page(BOOT_PAGE_NTP, "Time synced OK", 400);
92     }
93
94     boot_page(BOOT_PAGE_WEATHER, "Fetching weather...", 300);
95     refresh_weather();
96     if (weather_get().valid) {
97         const WeatherInfo &w = weather_get();
98         char msg[96];
99         snprintf(msg, sizeof(msg), "Got weather\n%s  %.0fC",
100                 w.resolved_name[0] ? w.resolved_name : "OK",
101                 w.temperature_c);
102         boot_page(BOOT_PAGE_WEATHER, msg, 500);

```



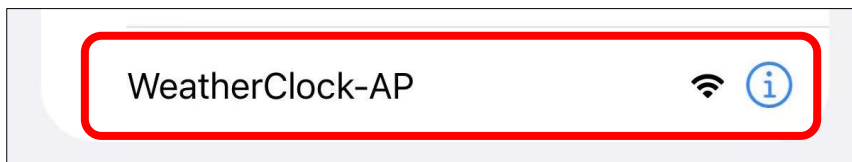
```
103     }
104
105     boot_page(BOOT_PAGE_READY, "Building UI...", 350);
106     ui_clock_create();
107     ui_clock_set_face(config_store_get().ui_face);
108     ui_ready = true;
109     ui_clock_update_time();
110     if (weather_get().valid) {
111         ui_clock_update_weather(weather_get());
112     }
113 }
114
115 void loop() {
116     lvgl_driver_handler();
117
118     TouchEvent ev = touch_button_poll();
119     if (ev == TOUCH_EVENT_THEME) {
120         on_theme_long_press();
121     } else if (ev == TOUCH_EVENT_FACE) {
122         on_face_long_press();
123     }
124
125     uint32_t now = millis();
126
127     if (ui_ready && (now - last_time_ms >= 1000)) {
128         last_time_ms = now;
129         ui_clock_update_time();
130         ntp_time_print();
131     }
132
133     if (ui_ready && (now - last_weather_ms >= WEATHER_REFRESH_MS)) {
134         refresh_weather();
135     }
136
137     delay(5);
138 }
```

Click “**Upload**” to upload the code to the Freenove ESP32 Mini TV

After the code is uploaded successfully, the Freenove ESP32 Mini TV will automatically enter WiFi configuration mode, as shown below:



Connect your phone or computer to the WiFi hotspot “WeatherClock-AP”. No password is required for this hotspot.



After connecting, the following configuration page will automatically appear. If it does not appear, open a browser and visit 192.168.4.1.

Click **Configure WiFi**.



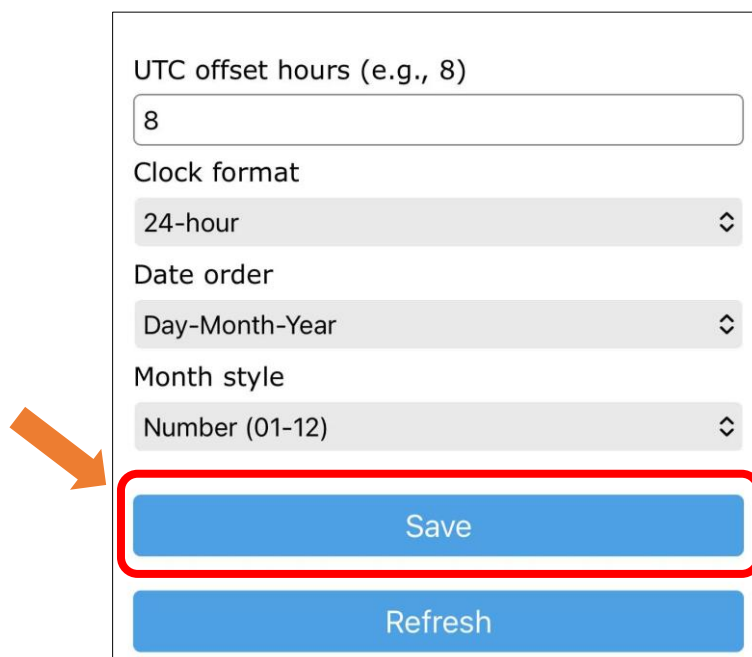
Configure the following parameters based on your actual settings:

The screenshot shows the configuration page with several settings highlighted by red boxes and orange callouts:

- SSID:** FYI\_2.4G (highlighted by a red box, with an orange callout 'WiFi SSID and password' pointing to it).
- Password:** (highlighted by a red box, with an orange callout 'WiFi SSID and password' pointing to it).
- Show Password:** (checkbox, not highlighted).
- UTC offset hours (e.g., 8):** 8 (highlighted by a red box, with an orange callout 'Time zone' pointing to it).
- Clock format:** 24-hour (highlighted by a red box, with an orange callout 'Clock format' pointing to it).
- Date order:** Day-Month-Year (highlighted by a red box, with an orange callout 'Date format' pointing to it).
- Month style:** Number (01-12) (highlighted by a red box, with an orange callout 'Month display style' pointing to it).

At the bottom of the form, there are two blue buttons: 'Save' and 'Refresh'.

Click Save to save the settings.



UTC offset hours (e.g., 8)

8

Clock format

24-hour

Date order

Day-Month-Year

Month style

Number (01-12)

Save

Refresh

After the configuration is completed, the device will automatically connect to the Internet and retrieve weather information. **Please wait patiently during this process.**

The Freenove Weather Clock features three built-in clock faces and two themes (black and white), as shown below:



Tap the touch button to switch between clock faces. Press and hold the touch button for 3 seconds to switch between themes.

Press and hold the touch button for 6 seconds to bring up a prompt interface. Continue holding for another 10 seconds to clear the network configuration and re-enter pairing mode.

